



GE VERNOVA

ADMS

16.25.1

Distribution Operator Training Simulator (DOTS) User Guide

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Change History

Date	Change Description
Apr 8, 2023	ADMS 3.16: • Updated Injecting a Scada Measurement Event. • Updated Creating a Status Type Measurement Event. • Added Creating a Multi-Phase Status Type Measurement Event. • Editorial and formatting updates.
Oct 20, 2023	ADMS 16.1.0: • The product version numbering has changed. The prefix '3' is no longer included in the version numbering for ADMS.
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1. Distribution Operator Training Simulator Overview

The Distribution Operator Training Simulator (DOTS) provides an integrated system utilizing the same software components, interfaces, and user interfaces as the real-time network management system, as shown in the figure below.

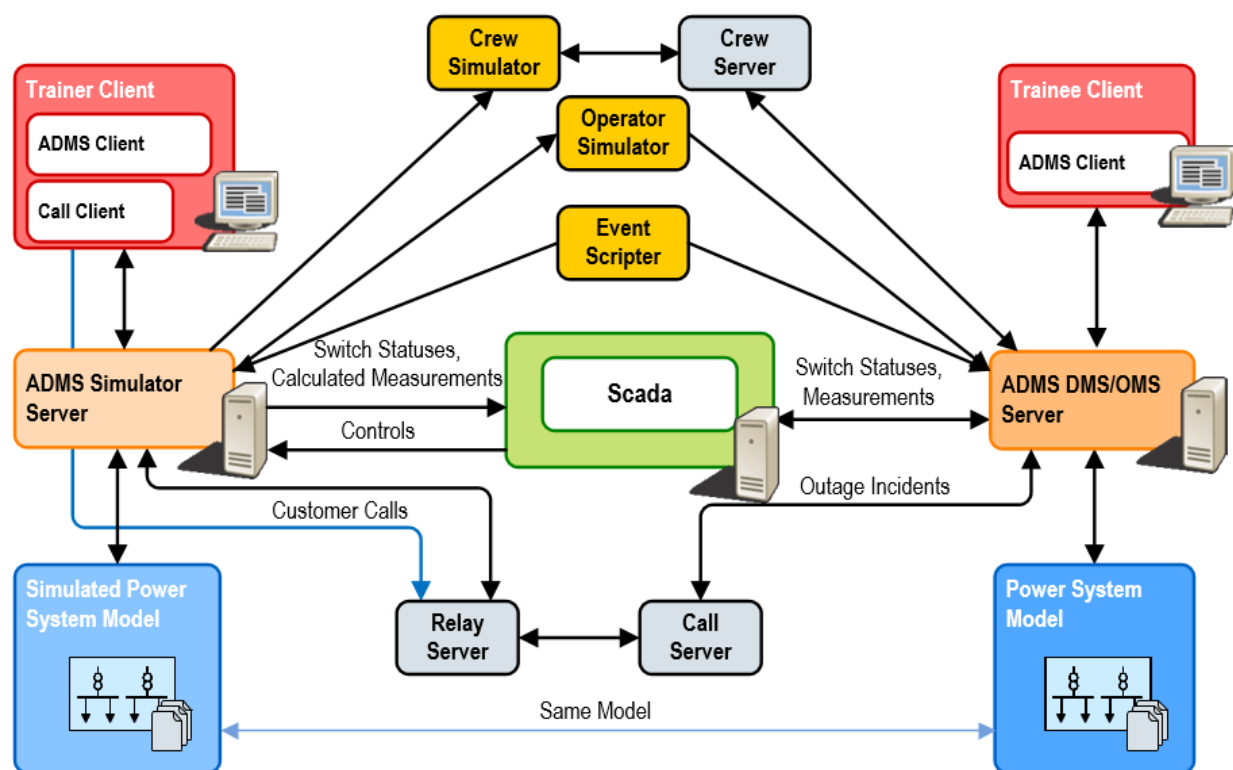


Figure 1. Distribution Operator Training Simulator Overview

The training simulator includes the following main parts:

- One instance is the NetSrv process running in real-time mode, identical to the normal control room environment.
- The other instance is the Simulator server process SimSrv, which simulates the response of the physical distribution network.

To support simulating the Outage Management System (OMS), the NetServer process needs to be run alongside NetSrv, and NetServerSim needs to be run instead of SimSrv.

The NetSrv and SimSrv (or NetServerSim) processes are also joined by a replication of the normal Scada system. The only simulation required for the Scada system is the front-end processors.

The simulation instance of the network model is automatically created as part of the standard modeling update process, meaning that no additional work is required to create the simulation, and the training simulator is always up to date with the real-time network model.

The display background color distinguishes the simulator view of the network from the real-time and study views (see figure below).

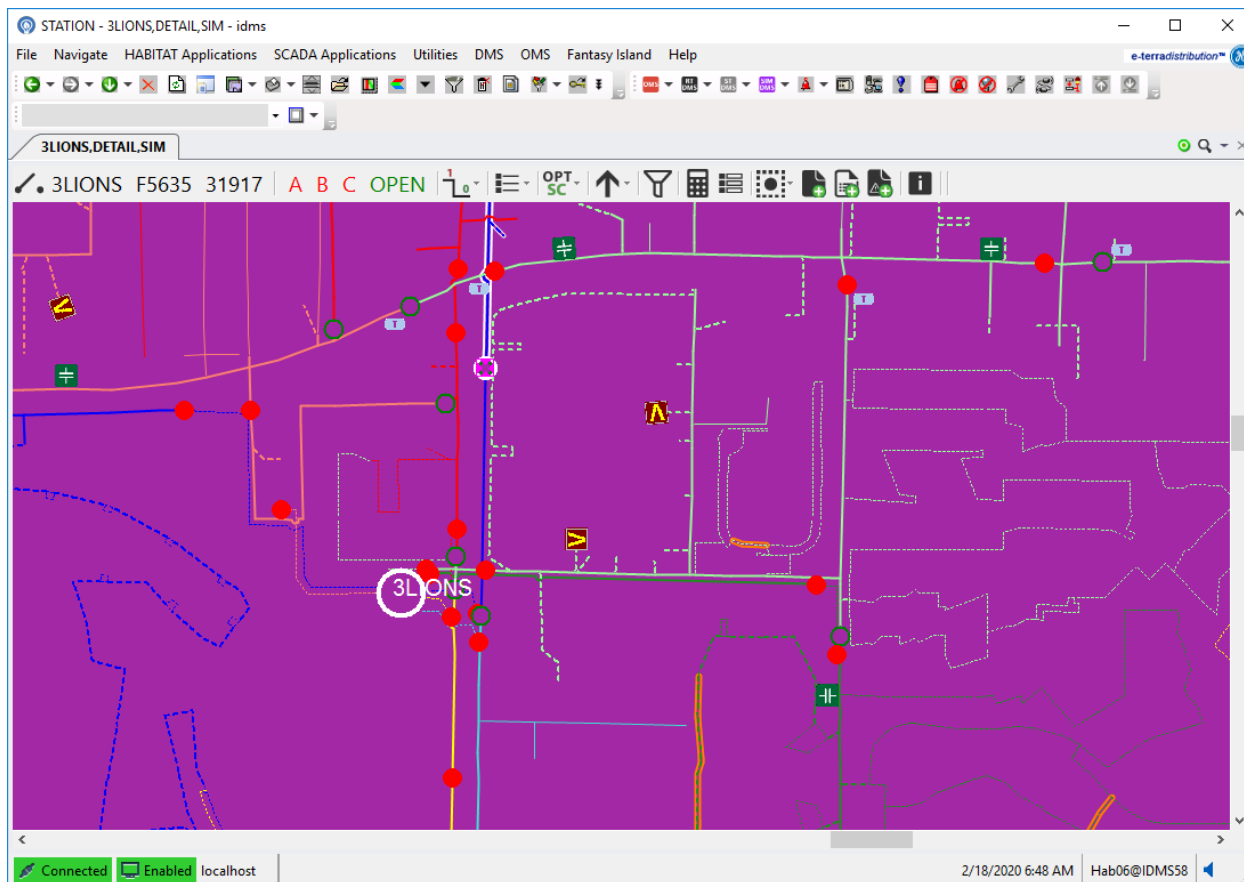


Figure 2. Simulator View of the Network

A simulated power system model generates the required network responses. The Event Scripter application allows complex training scenarios to be developed.

1.1. Training and Testing Possibilities

The Distribution Operator Training Simulator (DOTS) creates an effective training and testing environment to operate with the actual network model, data, and functions without the risk of disturbing the real-world system. Dispatchers can be trained in both routine and emergency operations that accurately represent the behavior and response of the real system. In addition, the new and existing DMS functions can be safely tested.

The training simulator supports the complete ADMS user interface. The simulation engine can exchange data with a Scada simulation, providing an environment for the following types of training and testing:

Distribution operators training and testing:

- Study mode and real-time operations:

- User interface presentation and navigation
- Simulation of actual electric system operation by scripting of events
- Snapshot of real-time environment
- Switching operations (manual and supervisory)
- Annotations, tagging, and topology processing, including feeder coloring and tracing
- Scada operations and alarms
- Network analysis and optimization
- Crew management training (RoboTools)
- Event forensics
 - Post event analysis
 - Forensic event re-enactments
 - Human performance improvement (HPI)
- Training
 - New employee training
 - Employee development (skills, deficiencies, etc.)
 - Scenario training (emergency and day-to-day operations)
- Testing and certification
 - Initial evaluation and testing of potential operators
 - Operator certifications (initial and on-going)
 - Operator testing

Operating environment training and testing:

- Processes, procedures, policies, and configurations
 - Develop, revise, update, and test procedures and processes
 - Demo and study new features and functionality
 - Simulate and test operational process design
 - Simulate and test new smart equipment
- DMS implementation and deployment support
 - Software regression testing
 - Simulation and testing of software functionality:

Volt/VAR Optimization (VVO), Fault Location (FL), Fault Isolation and System Reconfiguration (FISR), switching, DMS interfaces to other live, in-production systems (such as OMS)
 - Provides a tool for software and device deployment to simulate operation, determine optimal system configurations and settings, and simulate performance
- Emergency, emergency restoration planning and simulation

- Storm and weather events, load shed, black start, peak load response, cold load
- Automated system restoration solutions
 - Simulation and testing of closed loop FISR functionality before implementing an actual pilot
- Event forensics
 - Post event root cause analysis
 - Forensic switching error simulation and analysis
- Day-to-day operations
 - Distributed generation impact simulations
 - Allows full featured simulations to be run for virtually any operational scenario to evaluate the best course of action
 - Facilitate study and analysis of system operations
 - Switching verification

1.2. Network Analysis and Optimization Functions

The simulation engine includes a power flow solution and event script capability. This gives the simulation the ability to feed power system data points to the simulated Scada, and to support training and testing of the network analysis and optimization functions.

The simulator view allows the instructor to control the operation of the "physical network", independent of the operator's displays but with the same style of interface and the same range of tools.

When a fault is placed on the simulator view of the network, the simulator traces upstream to the first protective device (fuse, circuit breaker, etc.) and causes it to operate. The customers (and/or their AMI meters) that are de-energized generate trouble calls in accordance with call-response parameters that have been set. If the outage has affected any Scada controlled devices, the appropriate messages are sent to the Scada simulation. In this way, a truly realistic representation of network events is provided to the operators.

1.3. Outage Management

The simulation engine provides a full OMS simulation. This functionality includes:

- Trouble order (customer calls or AMI observations) message simulation based on the location of faults placed by event scripts
- Fault location prediction
- Switching operations for restoration



Note

Replicas of external systems such as IVR/VRU, CIS, WMS etc., are not included in the standard training simulator. The functionality of these systems may be addressed in the simulator by developing tools to allow the instructor to simulate the responses of these systems.

1.4. Event Scripter

The Event Scripter provides the means of creating and playing scripts that inject events into the simulated and real-time networks. The scripts are a list of network events, each with a time tag giving the number of seconds after the start of the script at which the event is executed (see figure below). Script events can consist of:

- Faults on lines and buses
- Faults on distribution transformers, power transformers, autotransformers, and regulators.
- Customer calls
- Modification of call rate (OMS) parameters
- Status and parameter changes for switching devices, such as switches, fuses, reclosers, and breakers
- Status changes for fault indicators
- Modification of device parameters (stations, transformers, feeders, and capacitors)
- Pause events (pauses execution of the script, allowing the instructor to resume execution at an appropriate point)
- Scada measurement simulation (value and quality changes)
- Advanced Metering Infrastructure (AMI) observations

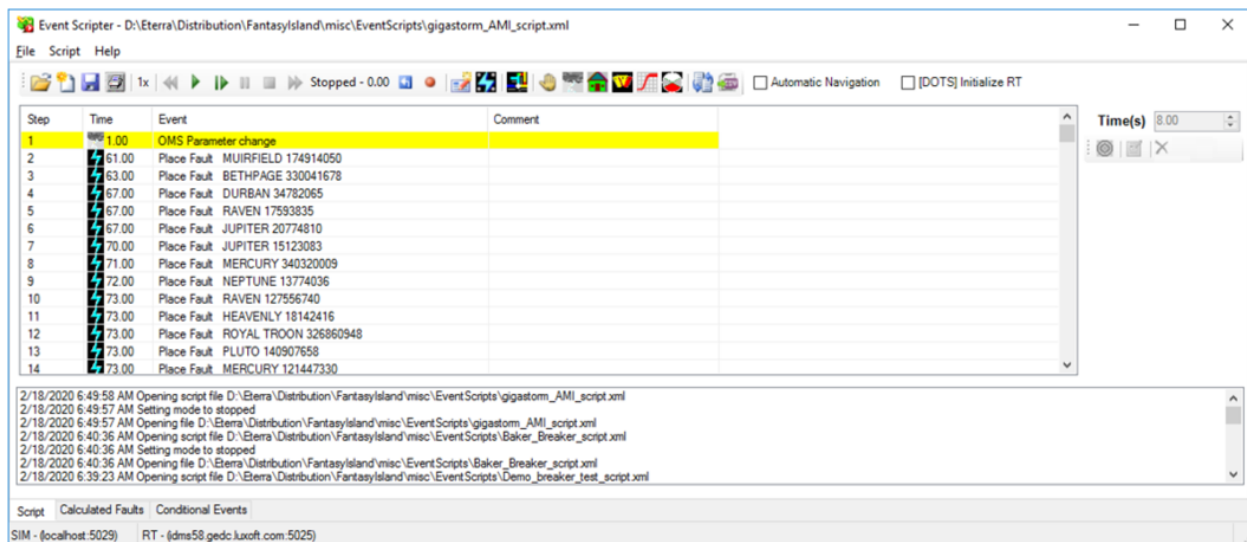


Figure 3. Event Scripter Tool

The Event Scripter provides a quick and easy method of adding events to a script. The Event

Scripter's geographic network view function is used to point and click on the selected device, and then the type of event is selected. Events within the script can be edited, deleted, or executed individually. The time sequence of events can be easily changed.

The Calculated Faults function of the Event Scripter can be used to create scripts with large number of faults to simulate storms and other high activity scenarios. The scenario is built up by defining the probabilities of different types of events causing faults on lines, transformers, and customers. The stations to be included and a time scale for the scenario are selected. The system then calculates a list of faults. The fault list can be saved as a script or merged into an existing script. The scenarios can also be saved separately so that they can be used with different scripts.

An example of the scenario matrix is shown in figure below.

Reason	Reason Code	OH Line All phases	OH Line one phase	OH Line Temporary	OH Transformer	UG Line	UG Transformer	Single Customer
UNNR	0	100	0	0	0	0	0	500
ANNR	1	0	0	0	0	0	0	0
CFNR	2	0	0	0	0	0	0	0
CONR	3	0	0	0	0	0	0	0
EFNR	4	0	0	0	0	0	0	0
ERNR	5	0	0	0	0	0	0	0
HFNR	6	0	0	0	0	0	0	0
IRNR	7	0	0	0	0	0	0	0
LINR	8	1000	300	0	0	0	0	0
OTNR	9	0	0	0	0	0	0	0
CUPR	10	0	0	0	0	0	0	0

Figure 4. Event Scripter Scenario Matrix

The Conditional Events function of the Event Scripter can be used to create scripts with events that are triggered depending on the current network conditions.

The overall execution timing of a script can be compressed or expanded as required. The execution time for each event is rescaled proportionately.

Scripts can be saved and recalled or shared with other users using the File Updater tool. For more information, refer to the *ADMS File Updater Software Installation and Maintenance Guide*.

1.5. Tools

The simulation environment includes several simulation utility applications:

- **Crew Simulator (RoboCrew) (OMS Only):** Simulates behavior of the field crew. Works together with the crew server and the operator simulator for automatic changing of crew status, crew position, and fault removal. For more information about the RoboCrew tool, refer to [Running the Crew Simulator \(RoboCrew\)](#).

- **Operator Simulator (RoboDispatcher) (OMS Only):** Simulates behavior of the operator, automatically assigns crew to the incident, keeps track of crew work, and clears incidents. For more information about the RoboDispatcher tool, refer to [Running the Operator Simulator \(RoboDispatcher\)](#).
- **Event Scripter:** Allows you to create and run special scripts that simulate a number of customer trouble calls, device switchings, AMI outages, unassociated calls, or device faults. For more information about the Event Scripter, refer to [Event Scripter](#).

2. Using the Training Simulator

This chapter provides instructions on how to run the training simulator system.

2.1. Introduction

In the training environment, the trainee uses the same displays that are available in real time. The instructor uses the simulator to simulate events on the trainee's "real-time" system. The instructor looks at either the real-time displays or the simulator network view. For example, the instructor can place a fault on a line, to check how the trainee works with customer calls and line de-energization.

i Note

The fault placed from the simulator is not displayed on the "real-time" view. It can only be processed by fault indicators and analysis functions, such as Fault Location.

Real-time and simulation displays are differentiated by the background color, as shown in figure below. The background color is configurable by the display administrator.

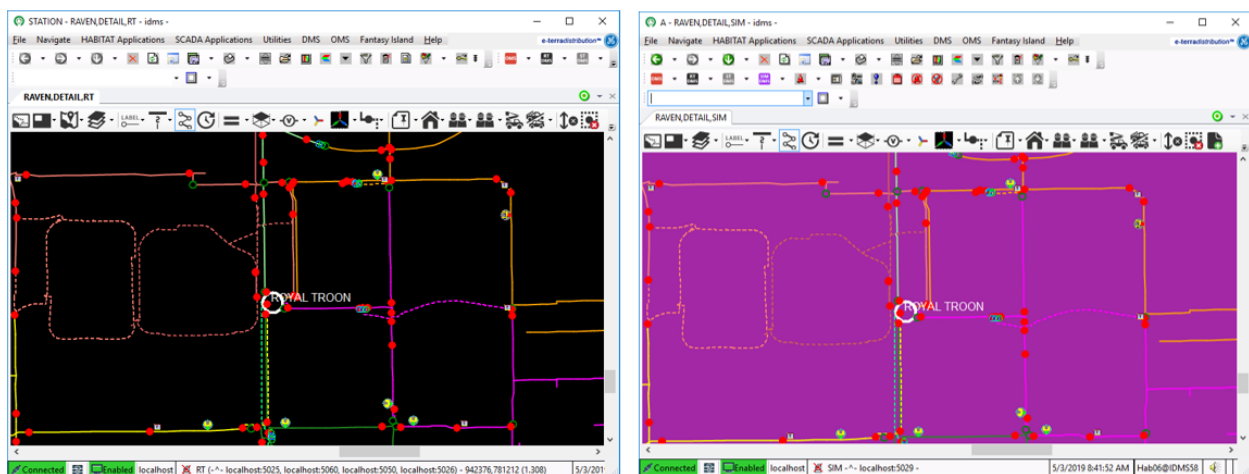


Figure 5. Real-Time vs. Simulation Background Colors

All displays that are available in the real-time environment are available in simulation mode. However, you can operate only those displays for which functions are available in the simulation network server mode. For example, optimized network functions in the simulation network server are limited.

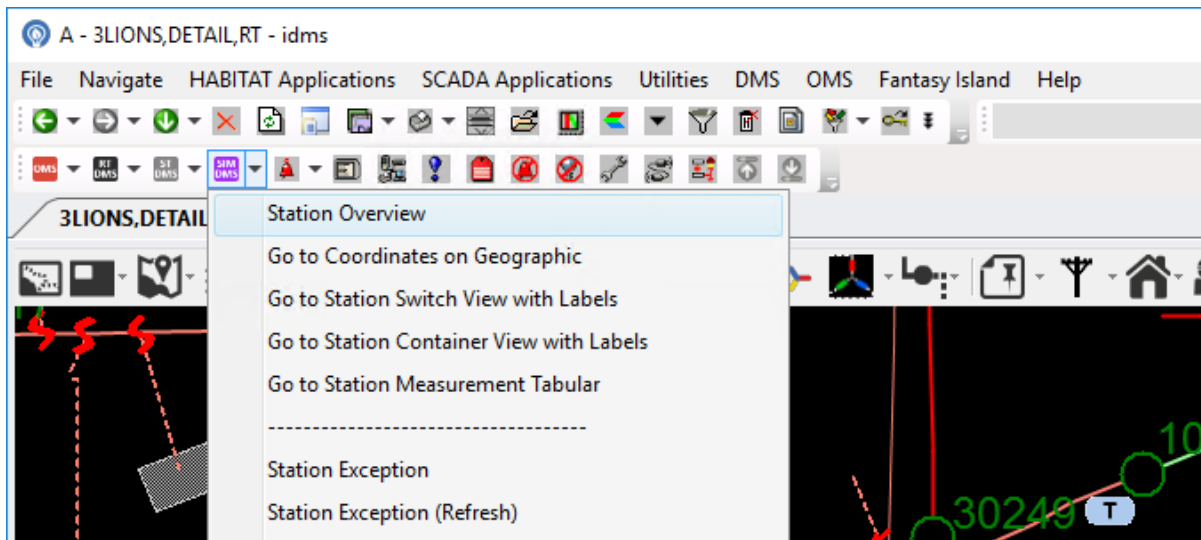
Some displays are only available in the simulation environment — for example, displays to operate the time-based event scripts.

The simulation displays are only intended for use by the instructor. In some scenarios, the instructor or the instructor's assistant can play the role of a lineman to mimic operations from the field on the simulator. This provides effective training for communicating and coordinating crews in the field.

2.2. Accessing the Simulator View

To open the simulator view:

1. Start the geographic network view.
2. On the Main toolbar click the SIM DMS button.



The simulator view appears.

3. Do one of the following:
 - Enter the substation name in the navigation field on the toolbar, and click the SIM DMS  button. The new viewport is opened, directed at the station of interest.
 - Select the pull-down menu to the right of the SIM DMS button, and select SIM Station Overview. Then click the desired area, to open a new viewport that is directed at the station or area of interest.
 - Enter the display call-up URL in the navigation field on the toolbar, and push Enter. The "dynamics" object in the URL specifies the server to connect to. For example:
 - NETVIEW://SIM/VIEW/DETAIL/RAVEN calls up the display with the SIM dynamics object and server.
 - NETVIEW://RT/VIEW/DETAIL/RAVEN calls up the geographic view for the RAVEN substation with the RT dynamics object and server.

The dynamics are configured in the `netconf.xml` file using the Net Dynamics Configuration Editor and are used by the client at startup time.

2.3. Interactive Operation of the Simulator

In the simulator's normal operating mode, the instructor initiates events and power system

responses to create the required training scenario.

The functions available to the instructor in the simulation mode are:

- Performing switching operations
- Manually applying faults
- Scaling station loads
- Changing feeder loading
- Changing network configuration (temporary modifications)
- Changing device parameters via manual entry displays

2.3.1. Performing Switching Operations

In the simulation mode, the instructor can locate and change the state of the desired switch or other electrical equipment in the same manner as in the real-time view. The simulator view updates the topology to show the effect on the network.

Note

The simulator view represents what is happening on the physical network so the changes may not appear on the trainee's real-time view.

After the instructor completes a switching operation:

- Scada-controlled switches and capacitors are updated in the real-time view.
- Manually operated switches and capacitors are not updated in the real-time view, but the simulator generates customer calls for affected customers, if appropriate.
- The OMS system may issue incorrect predictions after unplanned operations since many switches in the network are not protective devices. The OMS assumes that the outage resulting in customer calls was caused by a protective device operation.

2.3.2. Manually Applying Faults

In the simulation mode, the instructor can locate a desired section of the network and place a fault in the same manner as in the real-time view.

To place the fault, the instructor selects a transformer or a desired point on a line, and clicks the Fault button on the toolbar. The faults can be placed from the geographic, station internals, and station automated views. As opposed to the real-time view, the line must not be de-energized before placing a fault.

Placing fault from the simulator's or Event Scripter's network view invokes the Fault pop-up display, where the instructor modifies options as needed and clicks OK to place the fault.

The options include:

- **Name:** Name of the fault.
- **Station:** Name of the associated substation.
- **Line:** Name of the associated line.
- **Phases:** Changes the phasing of the fault.
- **Temporary:** Creates a temporary fault with a defined duration.
 - **Set Momentary Change Detection:** Creates a momentary fault. The duration of a momentary fault is too short to be displayed on the Simulation Network View. The protective devices operate as during the regular fault event.
 - **Set Time To Last in Sec:** Defines the duration for the fault.
- **Trip Additional Upstream Device:** Causes the fault to trip the first two upstream protective devices. For example, if the fault is placed on a fused lateral, only the fuse would normally operate. If Trip Upstream Device is selected, the fuse and the next upstream device (recloser or feeder breaker) will operate.

i Note

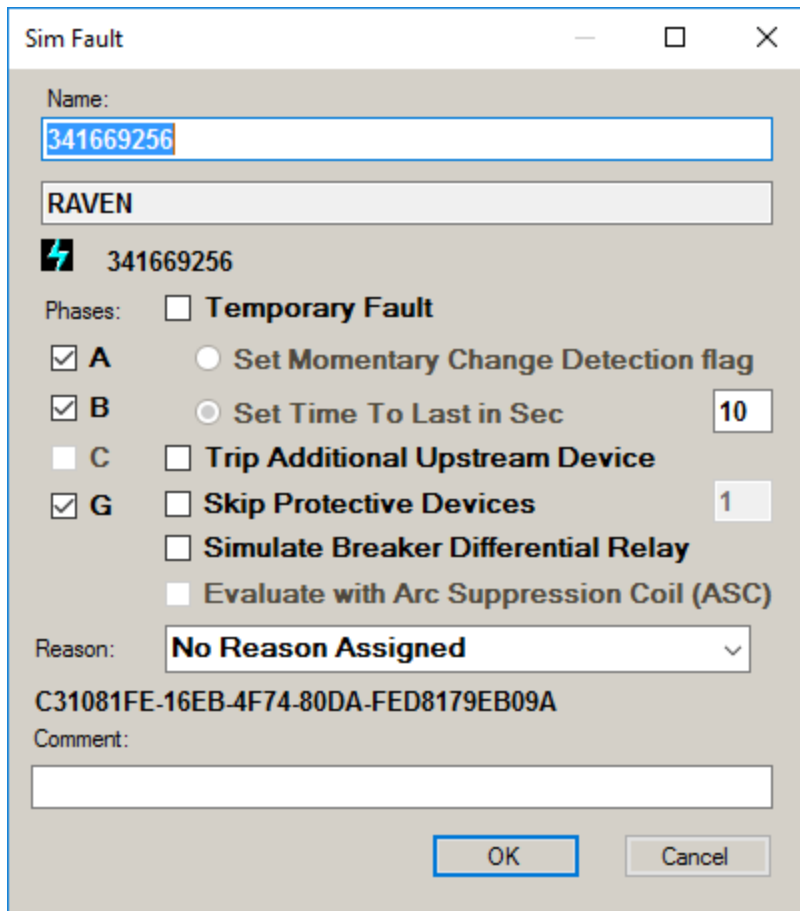
If the first device to trip is a recloser or breaker, no other upstream devices will be opened.

- **Skip Protective Devices:** This option is similar to Trip Additional Upstream Device, but it allows skipping a specified number of protective devices. For example, if the fault is placed on a fused lateral, the fuse would normally operate. If Skip Protective Device "1" is selected, the fuse does not operate and the next upstream device (recloser or feeder breaker) operates. This option allows simulating more severe faults or faulty protective devices.
- **Simulate Breaker Differential Relay:** This option simulates a differential relay scheme between breakers.

In a real world scenario, the input and exiting currents of a primary power transformer would be monitored for differences. If a discrepancy were detected, the upstream and downstream breakers would trip in order to protect the TF from a faulted or back flow condition.

When this option is chosen in the simulator and a fault is placed on a primary transformer in the station internals, the immediate upstream breaker and all immediate, normally connected downstream breakers will open.

- **Evaluate with Arc Suppression Coil (ASC):** This option simulates a station transformer arc suppression coil (ASC) fault indicator event. This option is only available for lines that are associated with a power transformer with an arc suppression coil fault indicator connected to the secondary side.
- **Reason:** Select the reason of the fault. This reason is only shown on the Sim Fault dialog and not passed to the OMS.
- **Comment:** Contains additional comments on the fault. The explanation is visible to the instructor only.



The image shows a 'Sim Fault' dialog box with the following fields and options:

- Name:** A text box containing '341669256'.
- RAVEN:** A text box containing 'RAVEN'.
- Phases:** A section with a lightning bolt icon and the ID '341669256'. It includes:
 - ☐ **Temporary Fault**
 - ☒ **A** ☐ **Set Momentary Change Detection flag**
 - ☒ **B** ☐ **Set Time To Last in Sec** (with a value box of '10')
 - ☐ **C** ☐ **Trip Additional Upstream Device**
 - ☒ **G** ☐ **Skip Protective Devices** (with a value box of '1')
 - ☐ **Simulate Breaker Differential Relay**
 - ☐ **Evaluate with Arc Suppression Coil (ASC)**
- Reason:** A dropdown menu showing 'No Reason Assigned'.
- UUID:** A text box containing 'C31081FE-16EB-4F74-80DA-FED8179EB09A'.
- Comment:** An empty text box.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

Figure 6. Place SIM Fault Pop-Up Display

When a fault is placed, the Simulation NetServer traces upstream to the first protective device (i.e., fuse, recloser or breaker) and causes it to operate. At this moment, the simulation network view shows the de-energized sections of the network, while the real-time network view shows only those updates that are passed by Scada. The Simulation NetServer also determines which customers are de-energized and starts generating trouble calls. In addition, in the simulation network view, manual fault indicators that are upstream of the fault become set.

2.3.3. Scaling Station Loads

In the simulation mode, the instructor can apply a scaling factor to alter the loading of specific stations. Scaling the station loads changes the total station load by a fixed proportion, but still allows it to vary within the defined load curves.

To set the station load scaling factor, the instructor navigates to the Manual Input Parameter Entry display for the simulator mode (see figure below).

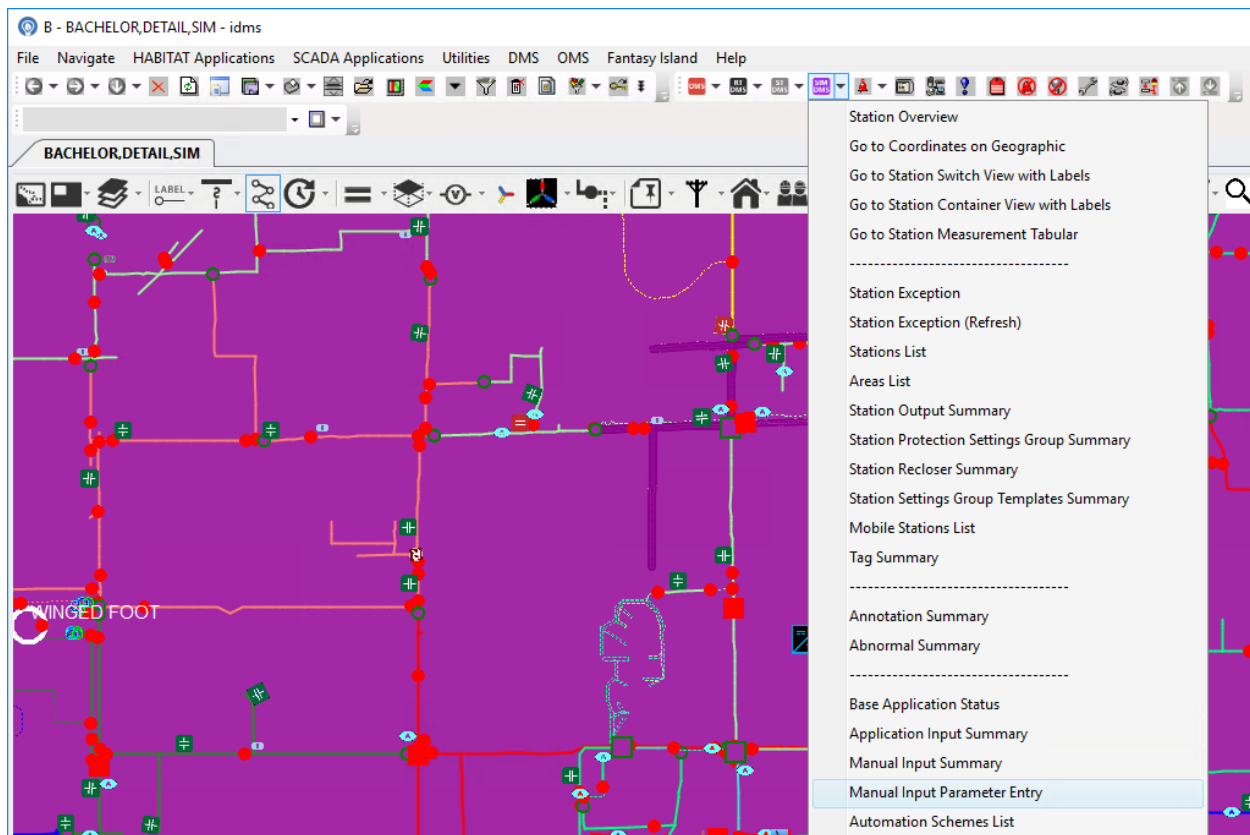


Figure 7. Simulator Manual Input Parameter Entry Display Call Up

This opens the Manual Input Parameter Entry display (figure below).

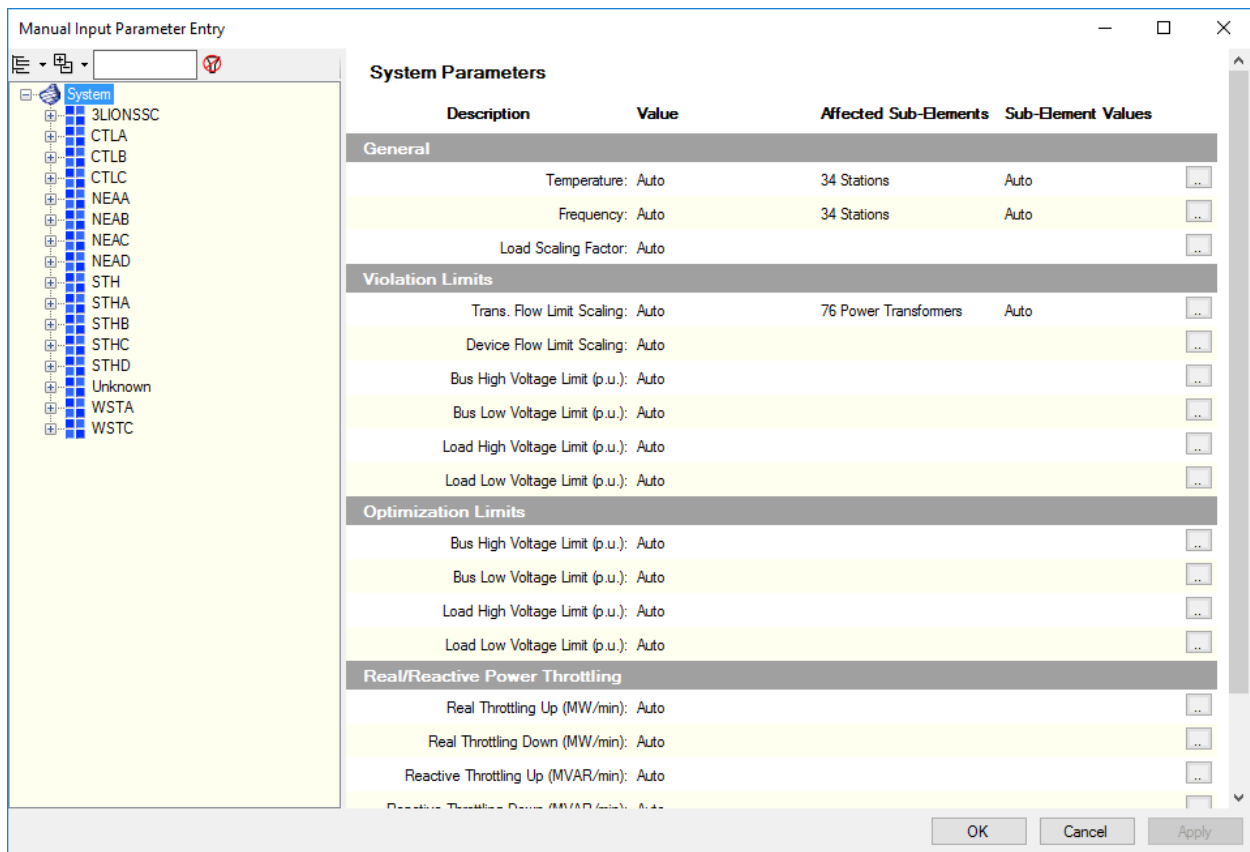
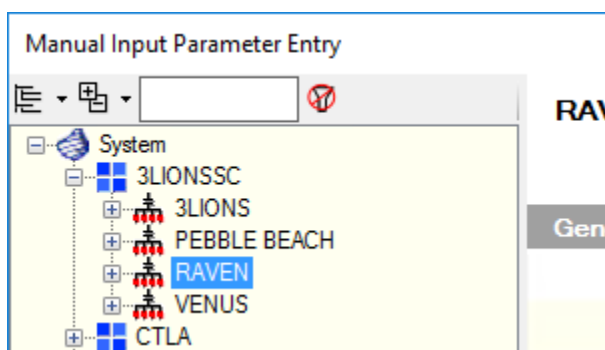


Figure 8. Manual Input Parameter Entry Display

The default load scaling factor is 1.0.

To increase the station load by 25 percent:

1. Select the station from the tree structure.



2. Click the ellipses  button for the Load Scaling Factor.

RAVEN Station Parameters			
Description	Value	Affected Sub-Elements	Sub-Element Values
General			
Temperature:	Auto (Inherited)		..
Frequency:	Auto		..
Load Scaling Factor:	Auto (Inherited)	5 Feeders	Auto

This opens the Edit Parameter dialog box.

3. In the Edit Parameter dialog box, select the "Manual – Set manual override" option.
4. In the Manual Override field, enter a new scaling value of 1.25.

Edit Parameter

Modify the value of Load Scaling Factor for Station RAVEN

☐ Inherit - Get value from element System
☐ Auto - Clear manual override
☒ Manual - Set manual override:

☐ Modify the parameter value for sub-elements

Sub-elements to Modify:

☐ Stations ☒ Transformers ☒ Feeders

How to modify sub-elements:

☒ Inherit - Get value from current element station RAVEN
☐ Auto - Clear manual override
☐ Manual - Set manual override:

OK Cancel

By default, station sub-elements (power transformers and feeders) inherit the station load scaling factor.

5. Click OK to save the new value.
6. Click Apply on the Manual Input Parameter Entry display to save changes.

RAVEN Station Parameters			
Description	Value	Affected Sub-Elements	Sub-Element Values
General			
Temperature:	Auto (Inherited)		..
Frequency:	Auto		..
Load Scaling Factor:	1.25	5 Feeders	1.25

The new station load is applied the next time the distribution power flow is run for the simulation power system (every 15 minutes by default).

Note

Similarly, the instructor can set load scaling factors for station sub-elements (power transformers and feeders).

To apply the load change immediately, the instructor must run the distribution power flow calculation process manually:

1. Locate the required station on the simulation geographic network view.
2. Select the station, open the DNAF Solution display, and run Distribution Power Flow.
 - The new station load values appear as changes in the feeder current on the real-time substation one-line (assuming that the station has telemetry on the feeders).
 - The scaling factor stays in effect until a new value is entered or the system is re-initialized.

2.3.4. Changing Feeder Loading

In the simulation mode, the instructor can set the loading of individual feeders to specific Amp values.

To change the feeder loading, the instructor navigates to the Station Feeder Input display (SIM DMS menu > Application Input Summary > Station Feeder Input) and then clicks the Feeder Input icon.

Station Feeder Input									
	Station	Feeder Input	Nominal Load Scaling Factor	Manual Load Scaling Factor	Manual LVM Maximum Power Factor Target Setting	Block All Regulating Transformers in the Station	Enable Voltage Reduction for all Transformers in the Station	Current Temperature	Manual LVM Minimum Power Factor Target Setting
	3LIONS		1.00	1.25	-0.90	☒	☐	--	0.90

Figure 9. Station Feeder Input Display

This invokes the Feeder Input tabular display shown in figure below.

Feeder Input for Station 3LIONS										
Feeder	Enable Manual Feeder Header Flow	Manual Feeder Head Flow Phase A (Amps)	Manual Feeder Head Flow Phase B (Amps)	Manual Feeder Head Flow Phase C (Amps)	Manual Feeder Head Flow Power Factor	Manual Load Scaling Setting	Manual Flow Limit Multiplier Setting	Manual Feeder LVM Maximum Power Factor Target Setting	Manual Feeder LVM Minimum Power Factor Target Setting	Manual Bus High Voltage Violation Limit (p.u.)
	△	△	△	△	△		△	△	△	△
F5631	☒	250.00	250.00	250.00	0.97	1.25	--	-0.90	0.90	--
F5632	☒	250.00	250.00	250.00	0.97	1.25	--	-0.90	0.90	--
F5633	☒	250.00	250.00	250.00	0.97	1.25	--	-0.90	0.90	--
F5634	☐	0.00	0.00	0.00	0.97	1.25	--	-0.90	0.90	--
F5635	☐	0.00	0.00	0.00	-0.99	1.25	--	-0.90	0.90	--
F5636	☐	0.00	0.00	0.00	0.97	1.25	--	-0.90	0.90	--
F5637	☐	0.00	0.00	0.00	0.97	1.25	--	-0.90	0.90	--
F5638	☐	0.00	0.00	0.00	0.97	1.25	--	-0.90	0.90	--

Figure 10. Feeder Input Display

The Feeder Input display allows the instructor to set the Feeder Amps and the Feeder Power Factor values for each phase.

Note

The Feeder Amp values are shown as 0.0 until a manual value is entered.

To change the feeder loading:

1. Select the Enable Manual Feeder Header Flow checkbox for each required feeder.
2. Enter the required Feeder Head Flow values for each phase.

The Feeder Head Flows are set to these values; any load curves that are modeled in the simulation will be ignored.

The new Feeder Head Flows are applied the next time the distribution power flow is run for the simulation power system (every 15 minutes by default).

To apply the load change immediately, the instructor must run the distribution power flow calculation process manually (see [Scaling Station Loads](#)).

2.3.5. Changing Device Parameters

In simulation mode, the instructor can set parameters for specific devices using toolbar, data entry pop-ups, and grid tabular displays.

For example, the instructor can change capacitor parameters on the following displays:

- Capacitor Input display (can be accessed from the Application Input Summary display).

RAVEN - Capacitor Input Display										
Name	ID	Locate	Nominal Feeder	Real Time Feeder	Control Capability	Remote Enabled	Local Control Type	Nominal On Level	Enable Manual On Level	Manual On
5699	77837500		F8865	F8865	Remote Only	☒	None	--	☒	
3931	10219064		F8865	F8865	Remote Only	☒	None	--	☒	
8271	10218700		F8864	F8864	Fixed	☒	None	--	☒	
5964	158266219		F8864	F8864	Remote Only	☒	None	--	☒	
3639	10218756		F8864	F8864	Remote Only	☒	None	--	☒	

RAVEN - Capacitor Input Display									
Name	ID	Locate	Has Local Voltage Override Capability	Operational Voltage Override High Level	Enable Manual Voltage Override High Level	Manual Voltage Override High Level	Operational Voltage Override Low Level	Enable Manual Voltage Override Low Level	Manual Voltage Override Low Level
5699	77837500		No	--	☒		--	☒	
3931	10219064		No	--	☒		--	☒	

RAVEN - Capacitor Input Display								
Name	ID	Nominal Hours Between Consecutive Switching	Manual Hours Between Consecutive Switching	Locate	Enable Manual Hours Between Consecutive Switching	Optimization Eligibility Status	Controller Operable	Steps
1656	10218686	0.25	1.00		☒	SC	☒	
3639	10218756	0.25	2.00		☒	SC	☒	
3931	10219064	0.25	2.00		☒	SC	☒	
3989	10217916	0.25	0.00		☒	SC	☒	
4017	10217706	0.25	0.00		☒	SC	☒	

- Capacitor Parameters display (can be accessed from the capacitor toolbar).

PEBBLE BEACH F6533 C4059 ABC REMOTE OFF --

Capacitor Parameters - 10217692

Parameter Name	Nominal Value	Override Enable	Override Value
Control Capability	Remote with Local		
Local Control Type	Voltage		
Control Settings			
Remote Enabled		☒	
Controller Operable		☒	
Upper Control Limit	1.05 per unit	☐	0.00
Lower Control Limit	0.98 per unit	☐	0.00
Voltage Override High Limit			
Voltage Override Low Limit			
Size			
Nominal Size per Phase (KVAR)	100.00	☐	300.00
Optimization Parameters			
Hours Between Consecutive Switching Operations	0.25	☐	0
Maximum Number of Daily Switching Operations	4	☐	0
Movement Cost	250.00	☐	0.00

Apply OK Cancel

- Capacitor toolbar (for example, Remote/Local status).

Device parameters are used for DNAF calculations in the simulated power system.

Note

Device parameters entered in the simulation system are not applied to devices in the real-time system. Exceptions are transformer tap positions, voltage targets, advanced regulation mode changes, and KVAR and PF setpoint measurements and controls for DER devices. The changes

are applied to the real-time mode after DPF is calculated in the simulation mode.

2.3.6. Running the Distribution Power Flow Function

System and device parameter changes are applied after the Distribution Power Flow (DPF) solution is run in SIM (either manually, or automatically by a periodic or event trigger).

You can also run periodic and event-triggered DPF solution with a specific evaluation time. This feature calculates DPF for the current network conditions for different dates with different loads schedule values.

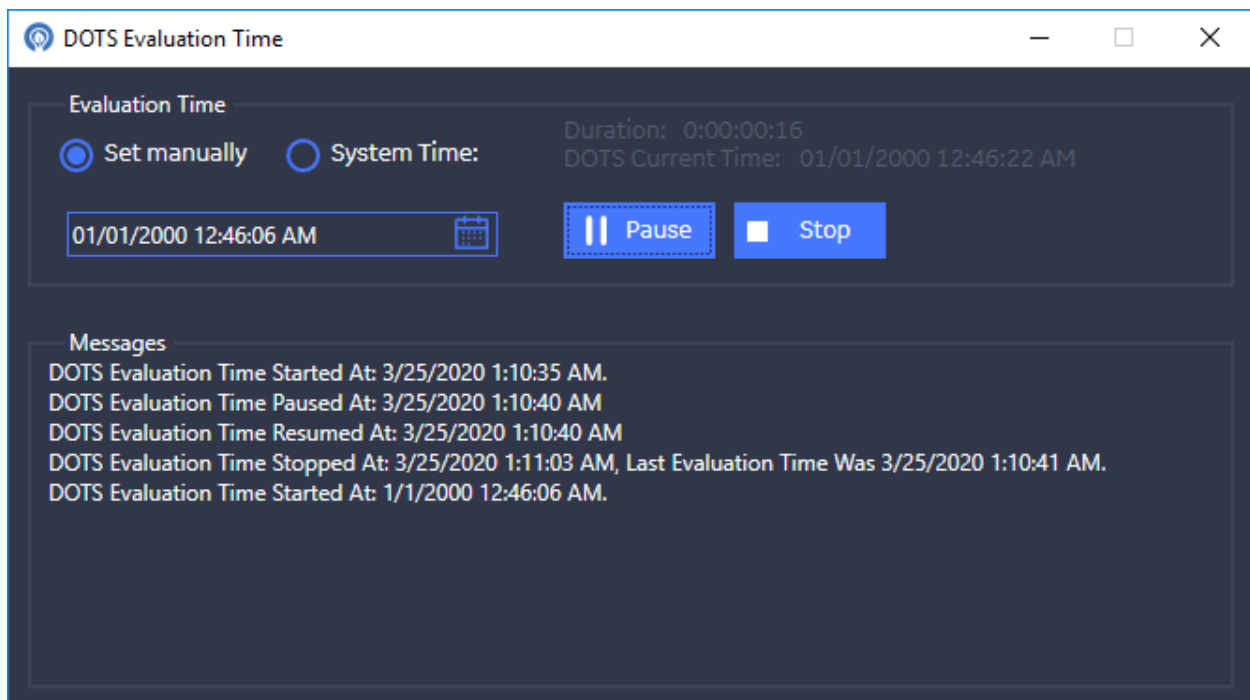


Figure 11. DOTS Evaluation Time

You can specify the system time for running DPF in SIM using the DOTS Evaluation Time display. This display can be opened from the SIM menu. For more details, refer to the *DNAF User Guide*.

2.3.7. Changing the Network Configuration

In the simulation mode, the instructor can change the configuration of the simulated network by placing cuts and jumpers in the same manner as in the real-time view (figure below).

The power flow for the simulated network accounts for the topology changes and updates the feeder loading and flows accordingly.

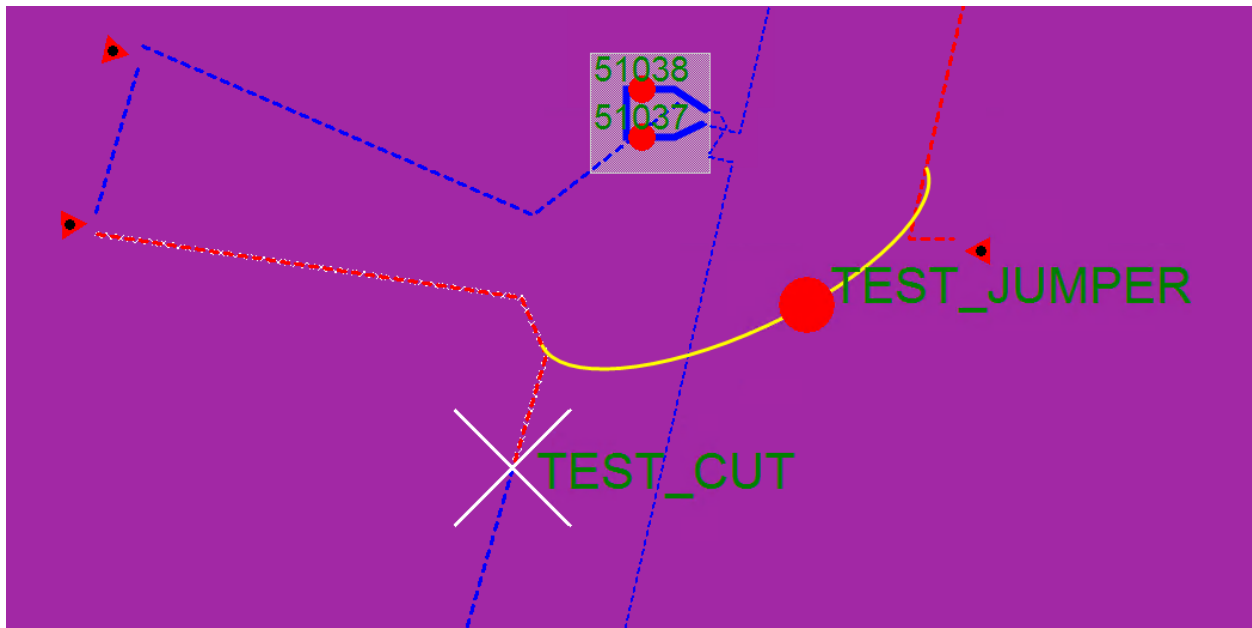


Figure 12. Cut and Jumper Applied to a Simulated Network

The cuts and jumpers created in the simulation mode are not visible on the real-time network view. The only changes visible on the real-time network view are the Scada values for feeder flows.

The instructor must apply temporary modifications to the simulated network with caution to ensure that the desired result is achieved and the trainee is able to recognize the nature of the problem.

2.3.8. Simulating Device Control Failure

In simulation mode, the instructor can simulate a Scada control failure for a switching device, capacitor, generator, energy resource, tap changing power transformer, or regulator. This helps train operators on how to handle device malfunctions in real-time operations.

Simulating a control failure is designed to provide the trainee with the same feedback as in the real operations when there is a failure of a device in the field. For example, in real-time operations the operator may send a control to Scada and realize that the switch is broken and does not open; or the operator may request the field crew to manually open a switch while the switch may be broken, welded, or rusted shut; or a tap changing transformer motor drive may not be operating.

Simulating a control failure can also be used to test DNAF applications, which send various controls, to see how DNAF functions react when a control does not work.

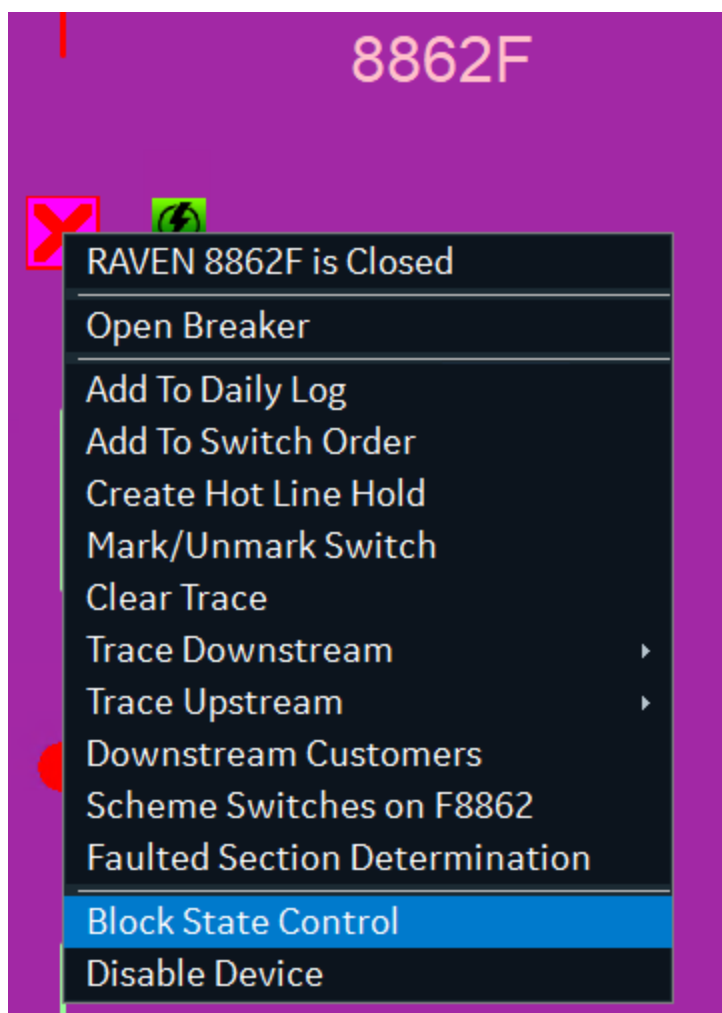
To simulate a Scada control failure from the network view:

1. Open the simulator view in the client or open the network view in the Event Scriptor.
2. Navigate to a Scada switching device, capacitor, generator, energy resource, tap changing power transformer, or regulator on the network or station internals view.
3. Select the Block State Control or Device Disabled option from the device's right-click menu.

- **Block State Control:** Sets the Block State Control status to Block. This specifies that device change state Scada controls are blocked. For example, Scada controls to open or close a switch, connect or disconnect a capacitor, generator, or energy resource, or raise or lower a transformer tap are blocked.
- **Disable Device:** Changes the Device Disabled status to Disable. This specifies that any Scada changes and controls to the device are disabled except Enable, but in the simulator server only. For example, the open/close, reclosing enable, tap position raise/lower, voltage setpoint, and power factor setpoint Scada controls are disabled.

Note

There is no blocking of non-SCADA changes in real time or simulator modes.



The Block State Control or Disable Device halo appears around the device in simulator mode only.

Block State Control

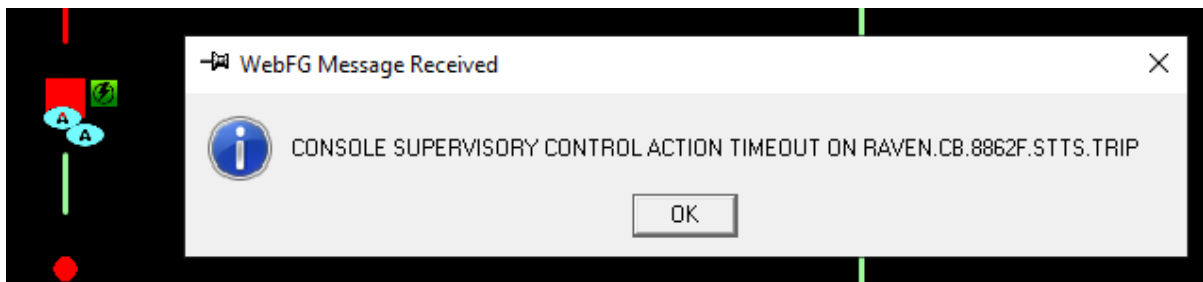


Disable Device



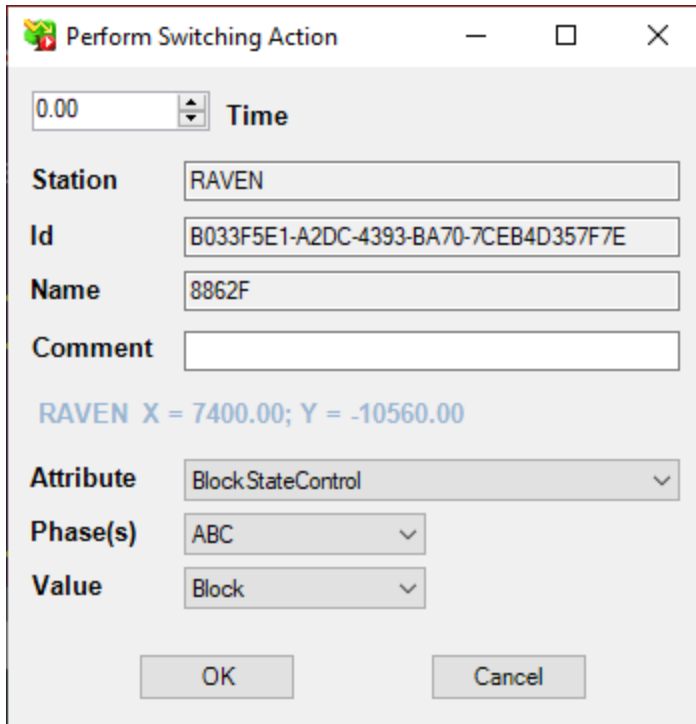
4. In real time mode, send a Scada control to the device.

The Scada control fails to be implemented within the timeout period.



To simulate a Scada control failure using an Event Scripter event.

1. Open the Event Scripter.
2. Navigate to a Scada device on the network or station internals view.
3. Create an event with the Block State Control or Device Disabled status change. These options are available for switching devices, capacitors, generators, energy resources, tap changing power transformers, and regulators. For more details, see [Creating a Switching Event](#).



The image shows a software dialog box titled "Perform Switching Action". It contains several input fields and dropdown menus. At the top, there is a "Time" field with a spinner set to "0.00". Below this are fields for "Station" (containing "RAVEN"), "Id" (containing "B033F5E1-A2DC-4393-BA70-7CEB4D357F7E"), "Name" (containing "8862F"), and "Comment" (empty). A blue text label displays "RAVEN X = 7400.00; Y = -10560.00". Further down are three dropdown menus: "Attribute" (set to "Block StateControl"), "Phase(s)" (set to "ABC"), and "Value" (set to "Block"). At the bottom are "OK" and "Cancel" buttons.

4. Execute the event to block the device state control or disable the device.

To restore the device functionality, do one of the following:

- Select the Allow State Control or Enable Device option from the device's right-click menu.
- Execute an event to allow the device state control or enable the device.

2.4. Running the Crew Simulator (RoboCrew)

The RoboCrew tool simulates behavior of field crews. It works together with the crew server and the operator simulator for automatic changes of crew status, crew position, damage assessment, and outage restoration.

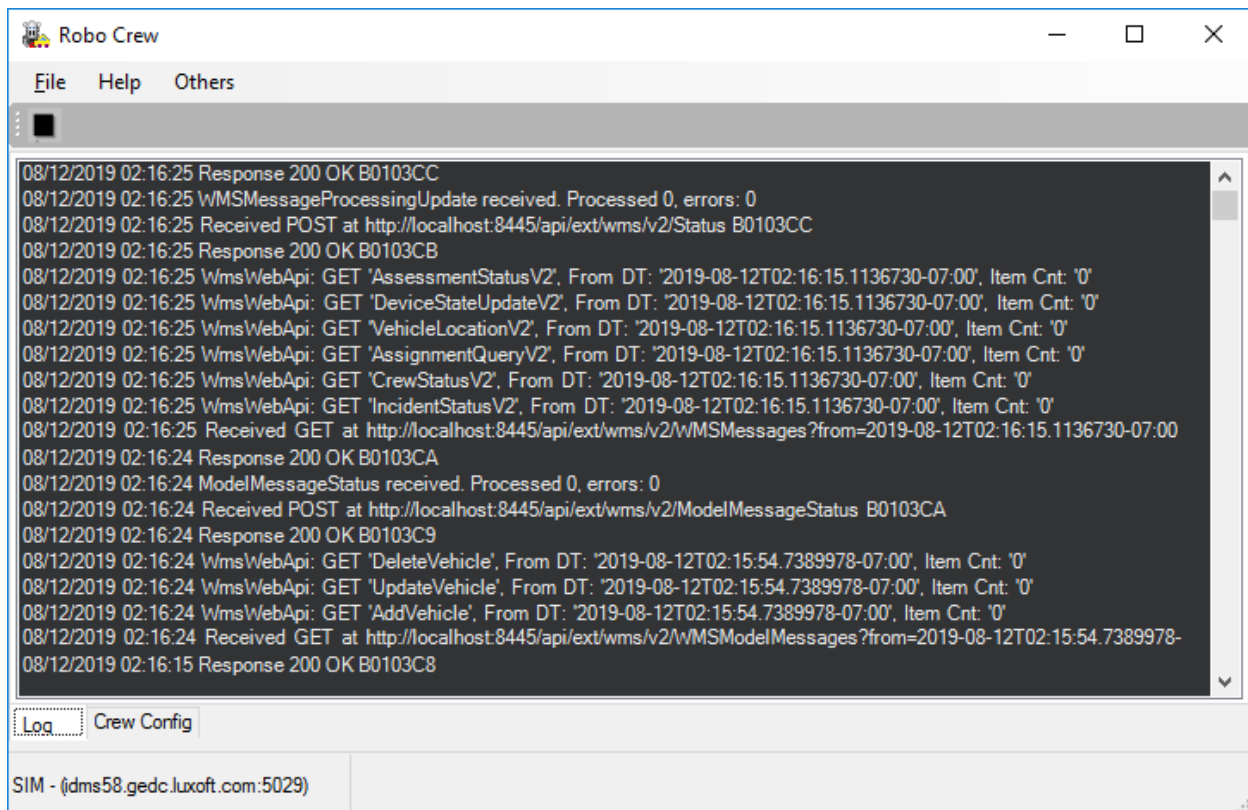


Figure 13. Crew Simulator (RoboCrew) Tool

In real life, when the control assigns a crew to an incident, the crew drives to the site of trouble. In the training environment, the RoboCrew moves the crew to the location of the trouble.

If the crew is equipped with an Automated Vehicle Locator (AVL) device, the crew simulator progressively moves the crew to the location of the fault. In cases when the crews assigned to the incident have no AVL devices, the crew simulator places the crews at the fault site without the moving simulation. You can view the details of the crews and change parameters using the crew displays of the Outage Management System, as described in the *Outage Management System User Guide*.

When the crew simulator moves the crew to the incident's location and the trainee wants the crew to open a switch, this operation must be performed by the instructor on the simulator screen.

Note

The Crew Simulator tool lists and operates only the crews equipped with the Mobile Data Terminal (MDT). The crew's MDT flag must be set to True in the OMS Crew Management System.

Note

A good practice is for the instructor or the instructor's assistant to play the role of a lineman to mimic operations from the field on the simulator. This provides effective training for

coordinating crews in the field. For example, they could train to open switches.

The RoboCrew tool includes the Log tab, described in section [Starting the Crew Simulator](#), and the Crew Config tab, described in section [Simulating Crew Behavior](#).

The RoboCrew tool controls and toolbar include the following options:

- **Stop/Start Sending Hello Messages Button:** Click to stop or start sending Hello messages to simulate the interaction with the work force management system.
- **File > Save Log to File:** Creates a new `RoboCrew_LOG_[current date and time].txt` file and saves the log records each time this option is selected.
- **File > Credentials:** Allows you to upload the `RoboCrewCredentialInfo.xml` file with valid credential data to simulate access to the Interface Design Description Schema (IDDS). The file contains an encryption key for the secured communication between the RoboCrew and the Crew server.
- **File > Automatic Crew Updates:** Enable this flag for the Crew Simulator to automatically simulate crew behavior. If disabled, MDT crews do not change position and status; new MDT crew statuses need to be manually sent by the operator.
- **File > Automatic Fault Clearance:** Set this flag to enable the automatic fault clearance by the crew simulator.
- **File > Don't Wait Onsite:** Set this flag to enable the fast mode, where the crews do not wait onsite for a pre-defined time period before clearing the next incident.
- **File > Clear All (Back to Nominal):** Select this option to return the simulator network view to the initial state. This functionality is similar to the Return Back to Normal function of the Event Scripter tool, described in section [Clearing Changes via the Event Scripter](#).
- **File > Send HelloV1 with SystemStatus 0:** Sends Hello message with System Status 0 or System Status 1. This function is needed to test the status of the Crew Management System.
- **Help > About RoboCrew:** Shows the RoboCrew application information.
- **Others > Vehicle Model:** Opens the RoboCrew Vehicle Modeler display to update vehicle data in the crew model.

2.4.1. Starting the Crew Simulator

Note

The RoboCrew application must be run by a user with administrator privileges.

To start the crew simulator tool:

- Click Start > Eterra > Distribution > Client > Crew Simulator.

The RoboCrew application starts and the RoboCrew icon  appears on the task bar.

The Crew Simulator starts on default with the Log tab open. The Log tab displays all the RoboCrew actions and processes records, including automatic Hello messages.

To enable automatic crew simulation:

- From the File menu, enable the Automatic Crew Updates option.

2.4.2. Simulating Crew Behavior

You can create a more-realistic simulation of the interaction with the work force management system by configuring the availability and responses for the selected crews using the Crew Config tab of the Crew Simulator tool.

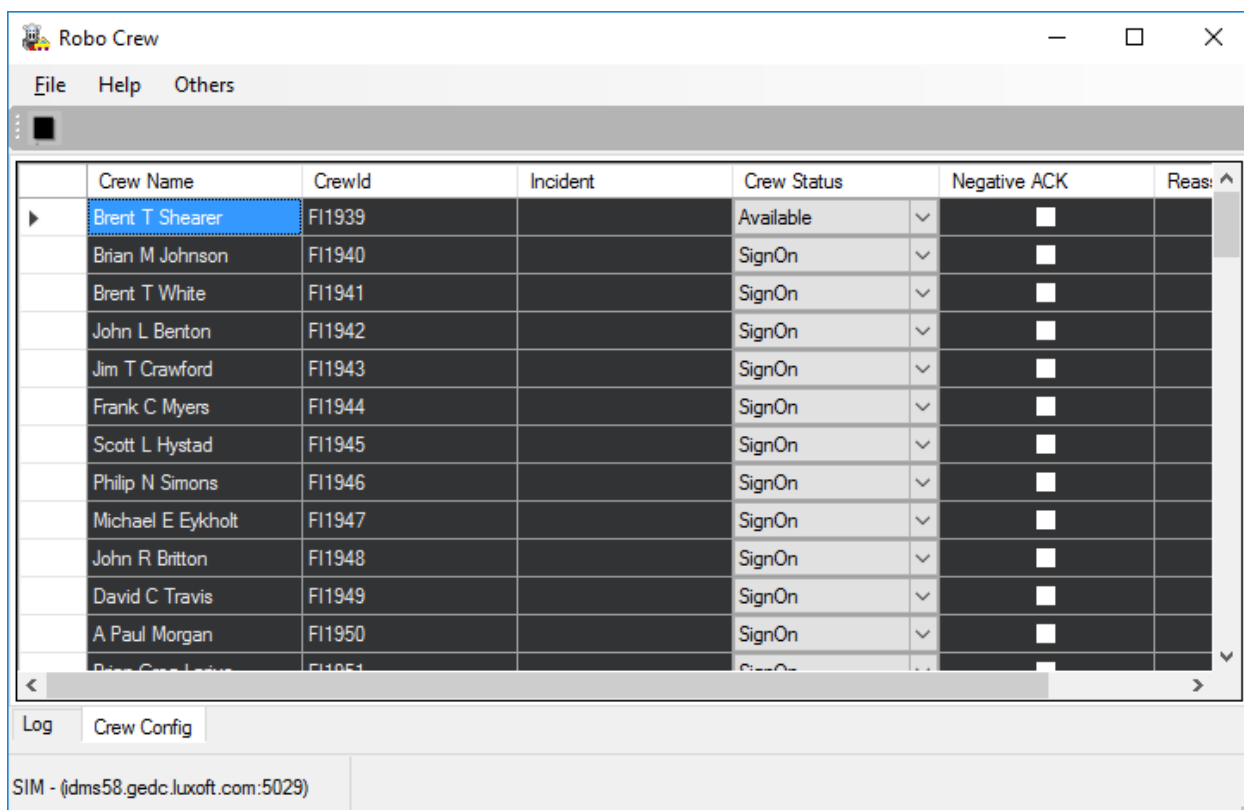


Figure 14. Crew Config Tab of the Crew Simulator Tool

The Crew Config tab of the Crew Simulator tool includes the following fields and controls:

- **Crew Name:** The first and last name of the crew lead (the primary employee).
- **CrewID:** The radio number of the crew.
- **Incident:** The incident number that the crew is assigned to.
- **Crew Status:** Current status of the crew (dispatched, en route, on-site, and so on). Legacy functionality.

Note

In ADMS 3.13, crew status is split into work status, shift status, communication status, and availability status. For backward compatibility, the legacy Status field is still supported. However, either Status or Work Status can be set. Use the Send Crew Status dialog to set Work Status.

- **Negative ACK:** When this option is set, a negative acknowledgment is sent from the Crew Management System, and the crew is not assigned to the incident in the OMS summary.
- **Reassign:** When this option is set, the crew rejects all the assignments and assigns another random available crew.

Right-click a crew row to open the context menu.

	Crew Name	CrewId	Incident	Crew Status	Negative
▶	Brent T Shearer	FI1939	0950000007	OnSite	
	Brian M Johnson	FI1940			
	Brent T White	FI1941			
	John L Benton	FI1942			
	Jim T Crawford	FI1943			
	Frank C Myers	FI1944			
	Scott L Hystad	FI1945			
	Philip N Simons	FI1946			
	Michael E Eykholt	FI1947			
	John R Britton	FI1948		SignOn	

- Incident Status
- Crew Status
- Damage Assessment
- Device State
- Confirm Device Open
- Restore Incident
- Complete Incident
- Clear Incident

Figure 15. Crew Config Tab Context Menu

The Crew Config context menu contains the following commands:

- **Incident Status:** Invokes the Update Incident Status pop-up window for the crew.

Update Incident Status

Incident Name: 2210000001 Outage Type: NotSelected

Crew Id: F11939 Failure Type:

Status: NotSelected Action Taken: None

SystemAffected: None # Poles Down: -1

Cause Group: No Reason Assigned # Transformers: -1

Cause: No Reason Assigned # Conductor Span: -1

Customer Count: 0 # Services Down: -1

Comment:

Work Order Id:

Cross Arms: 0 # Tree Trim Spans: -1

Actual Restoration Time:

ETR from WFM:

Tree Crew Needed: ☐

Line Crew Needed: ☐

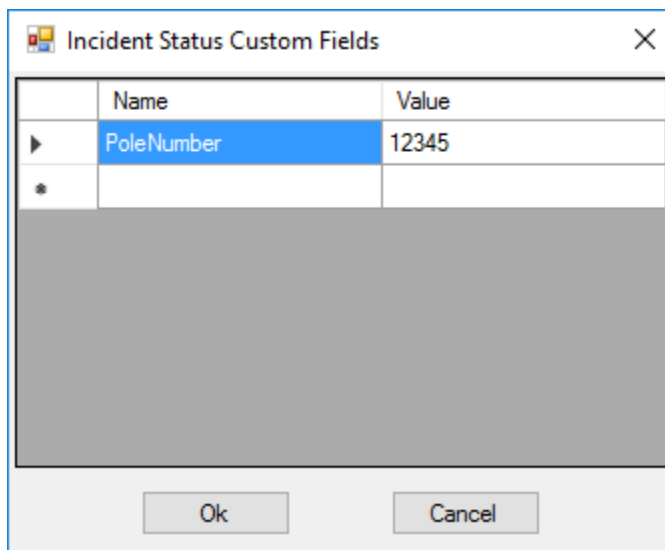
Evaluator Needed: ☐

Custom Fields Send Cancel

The Update Incident Status pop-up window contains a list of crew responses to an incident, including the following:

- **Incident Name:** Name of the incident, assigned to the crew. This field is populated automatically.
- **Crew ID:** ID of the crew. This field is populated automatically.
- **Status:** The incident status.
- **System Affected:** The system type of the network element, such as distribution lines.
- **Cause Group:** The incident's cause.
- **Cause:** The incident's sub-cause.
- **Customer Count:** Number of customers affected by the incident.
- **Comment:** Crew's comment on the incident.
- **Work Order ID:** The work order ID for the incident.
- **Outage Type:** Select the type of an outage.

- **Failure Type:** Select the failed device.
- **Action Taken:** The action that was taken for this incident.
- **# Poles Down:** Number of poles down.
- **# Transformers:** Number of transformers down.
- **# Conductor Span:** Number of spans of wire down.
- **# Services Down:** Number of services down.
- **# Tree Trim Spans:** Number of tree trim spans.
- **# Cross Arms:** Number of cross arms down.
- **Actual Restoration Time:** The actual restoration time.
- **ETR from WFM:** The estimated time of restoration.
- **Tree Crew Needed:** Select this option if the tree crew is needed for the incident.
- **Line Crew Needed:** Select this option if the line crew is needed for the incident.
- **Evaluator Needed:** Select this option if an evaluator is needed for the incident.
- **Custom Fields:** Opens the Incident Status Custom Fields dialog to update incident custom fields. In the Name field, use custom fields Field Name values that are defined in the OMS Server Configuration Editor.



The dialog box titled "Incident Status Custom Fields" contains a table with two columns: "Name" and "Value". The first row has "PoleNumber" in the Name column and "12345" in the Value column. Below the table is a large grey rectangular area. At the bottom of the dialog are "Ok" and "Cancel" buttons.

	Name	Value
▶	PoleNumber	12345
*		

- **Send:** Click to send the Update Incident Status message. The Update Incident pop-up of the OMS display will be updated accordingly.
- **Cancel:** Click to cancel the Update Incident Status message.
- **Crew Status:** Invokes the Send Crew Status pop-up window for the crew.

The Send Crew Status pop-up window contains a list of crew status updates for an incident, including the following:

- **Crew ID:** ID of the crew. This field is populated automatically.
- **Incident ID:** Name of the incident assigned to the crew. This field is populated automatically.
- **ETA:** The estimated time for arrival (ETA).
- **Work Order ID:** The work order ID of the incident.
- **Vehicle Info:** Information about crew vehicle.
- **Area:** The current crew area name.
- **Comment:** Crew comments about the incident.
- **Legacy Status:** Current status of the crew (dispatched, en route, on-site, and so on). Legacy functionality.
- **Work Status:** Current work status of the crew (dispatched, en route, on-site, and so on).

Note

In ADMS 3.13, crew status was split into work status, shift status, communication status, and availability status. For backward compatibility, the legacy Status field is still supported. However, either Status or Work Status can be set.

- **Shift Status:** Current shift status of the crew (for example, on duty).
- **Communication Status:** Current communication status of the crew (for example, in coverage).
- **Availability Status:** Current availability status of the crew (for example, on break).
- **Damage Assessment:** Invokes the Damage Assessment Status pop-up window where a crew

can create or update a damage assessment status report.

Damage Assessment Status

Incident: 1060000003

Location Guid: [dropdown]

XCoordinate: 951394

YCoordinate: 790487

Radius: [text box]

Status: Not Repaired

Repair Hours: [text box]

TreeTrimming: Manual

Truck Accessible: No

Poles Down: [text box]

Transformers: [text box]

Services Down: [text box]

Devices Down: [text box]

Conductor Span: [text box]

Comment: [text area]

Attachments: [dropdown]

Buttons: CustomFields, Send, Cancel, New/Delete

The Damage Assessment Status pop-up window contains a list of damage assessment responses, including the following:

- **Incident:** Name of the incident that is assigned to the crew. This field is populated automatically.

A crew can send a damage assessment status when assigned to an incident and the incident status is not completed.

- **Location Guid:** Location Guid is generated automatically when sending the first damage assessment status.

When updating an existing damage assessment status, Location Guid can be selected from the list box.

- **XCoordinate:** X coordinate of the damage assessment location. By default, the X and Y coordinates of a crew are used and the damage assessment symbol is placed on the network view immediately under the crew symbol.
- **YCoordinate:** Y coordinate of the damage assessment location.
- **Radius:** Incident radius.
- **Status:** Incident status (Repaired or Not Repaired).
- **Repair Hours:** Estimated incident restoration period in hours. If the status is set to Repaired, the Repair Hours number sent via WFM is 0.
- **Tree Trimming:** Tree trimming status (Need Truck or Manual).
- **Truck Accessible:** Truck accessibility status (Yes or No).
- **Poles Down:** Count of poles down.
- **Transformers:** Count of damaged transformers.
- **Services Down:** Count of services down.
- **Devices Down:** Count of devices down.
- **Conductor Span:** Count of spans of wire down.
- **Image:** Image.
- **Comment:** Crew comment.
- **Attachment:** Damage assessment attachment reference, including link, file name and type, and coordinates.

The Damage Assessment Status pop-up window contains the following controls:

- **New:** Opens the Attach File window to add a new damage assessment attachment reference.
- **Delete:** Removes the selected damage assessment attachment information.
- **Custom Fields:** Opens the Damage Assessment's Custom Fields dialog to update custom fields. In the Name field, use custom fields Field Name values that are defined in the OMS Server Configuration Editor.

	Name	Value
▶	FieldName	Value
*		

Ok Cancel

- **Send:** Sends the damage assessment status message with the specified information.
- **Cancel:** Cancels sending the damage assessment status message.
- **Device State:** Opens a display to change the state of the device of record.

Device ID:

Device State:

Phases Out:

Station:

User Name:

Feeder:

Send Cancel

- **Confirm Device Open:** Opens the device of record. This option is applicable only to predicted incidents on devices that do not have Scada status measurements. Available only if field clearing is enabled for this type of device of record.
- **Restore Incident:** Closes the device of record and changes the incident status to Incident is Waiting for Clearance by Dispatcher. This option is applicable only to devices that do not have Scada status measurements. Available only if field clearing is enabled for this type of device of

record.

- **Complete Incident:** Changes the incident status to Incident Completed by WMS. For crews modeled as evaluators (for example, engineers), changes the incident status to Incident has been Evaluated.
- **Clear Incident:** Automatically clears the pending incident. This option is applicable only to incidents on devices that do not have Scada status measurements and is available only if field clearing is enabled for this type of device of record.

2.4.3. Updating Vehicle Data in the Crew Model

You can update vehicle data in the crew model using the RoboCrew Vehicle Modeler display. To access this display, from the Others menu, select Vehicle Model.

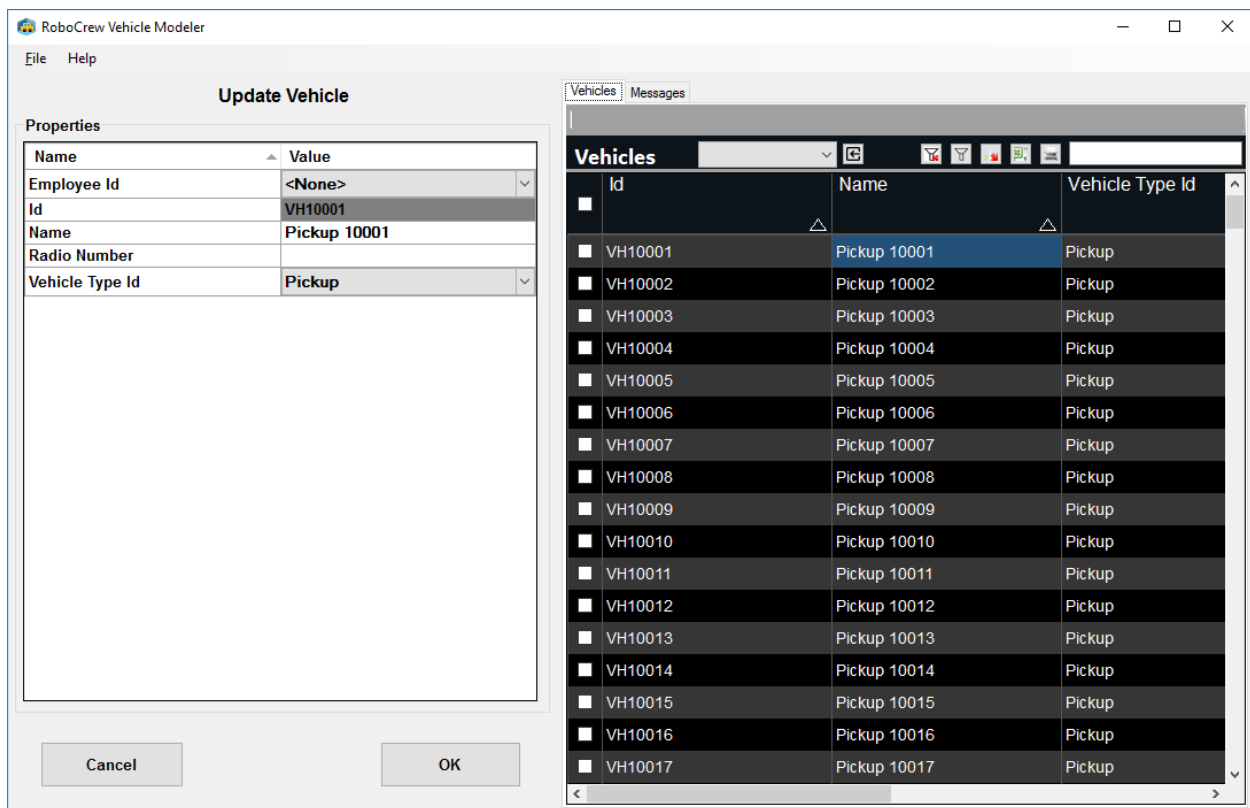


Figure 16. Updating Vehicle Data in the Crew Model

The RoboCrew Vehicle Modeler display has the same vehicle modeling functionality as the Crew Modeling tool. For more information, refer to the *Crew Modeling Tool User Guide*.

To update the crew model, modify the data and click OK. The updates will be sent to the Crew Server using the web interface (if enabled).

2.5. Running the Operator Simulator (RoboDispatcher)

The operator simulator (RoboDispatcher) simulates the control room operators' tasks by automatically assigning crews to the incident, keeping track of the crew work, assessing damage, and clearing incidents. The operator simulators build up the activity of the whole control room for a more realistic simulation of real-life operations.

To start the operator simulator tool: from the Windows Start menu, click Start > Programs > Eterra > Distribution > Operator Simulator.

Note

The RoboDispatcher application must be run by a user with administrator privileges.

The RoboDispatcher icon  appears on the task bar when the operator simulator tool is running. When the application is launched, it begins working with the incidents according to the applied configuration without further operator intervention.

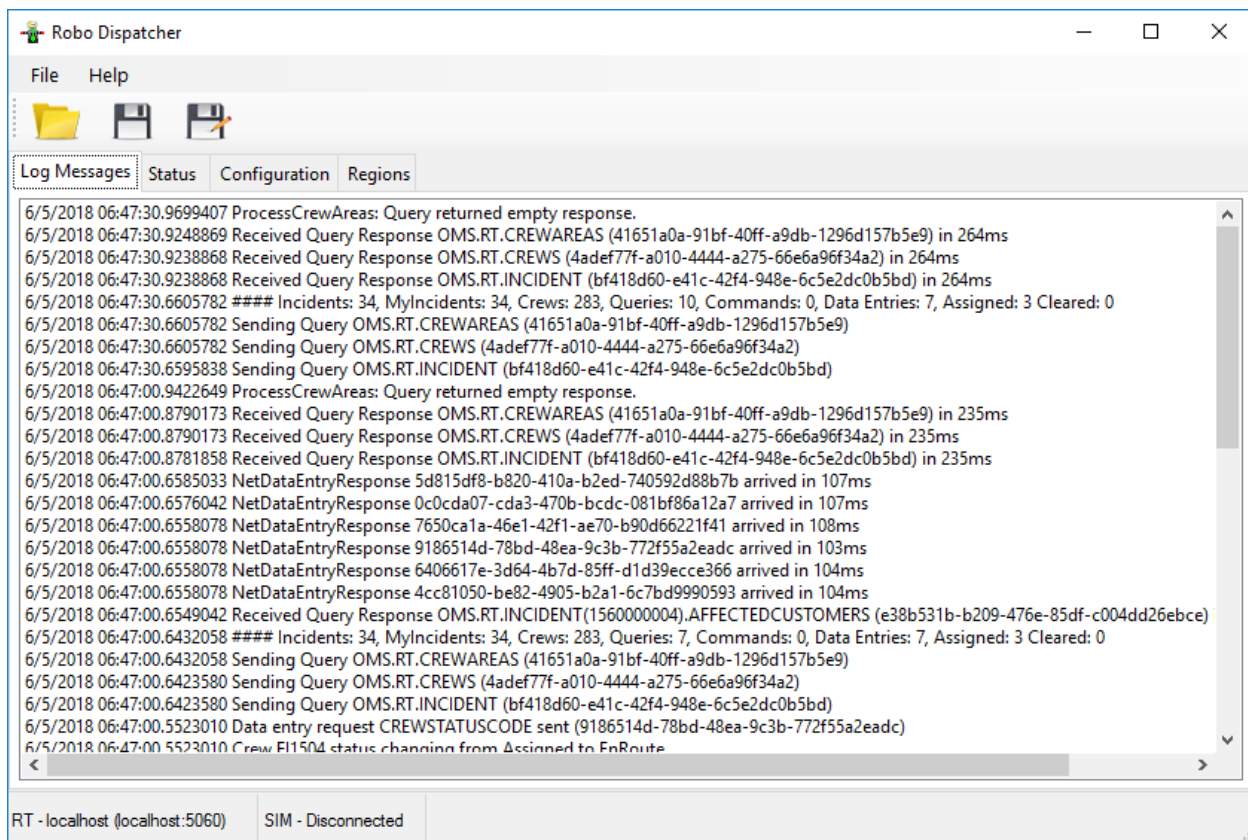


Figure 17. Operator Simulator (RoboDispatcher) Tool

By default, the Operator Simulator uses the RoboDispatchConfig.xml configuration file.

The tool starts with the Log tab open. The Log Messages tab displays all the actions and processes records.

The Operator Simulator's controls and the toolbar include the following options:

- **File > Automatic Incident Clearance:** Enables the automatic incident clearance by the Operator Simulator.
- **File > Save Configuration:** Saves the currently open configuration.
- **File > Save Configuration As:** Saves the currently open configuration with the different name.
- **Open:** Opens a configuration.

2.5.1. Viewing the Operator Simulator Status

On the Status tab, you can review the status of the Operator Simulator and change the settings of the run-time tool. The editable fields are shown in bold.

Note

Run-time configuration changes apply only to the current session. They are not saved in the configuration file.

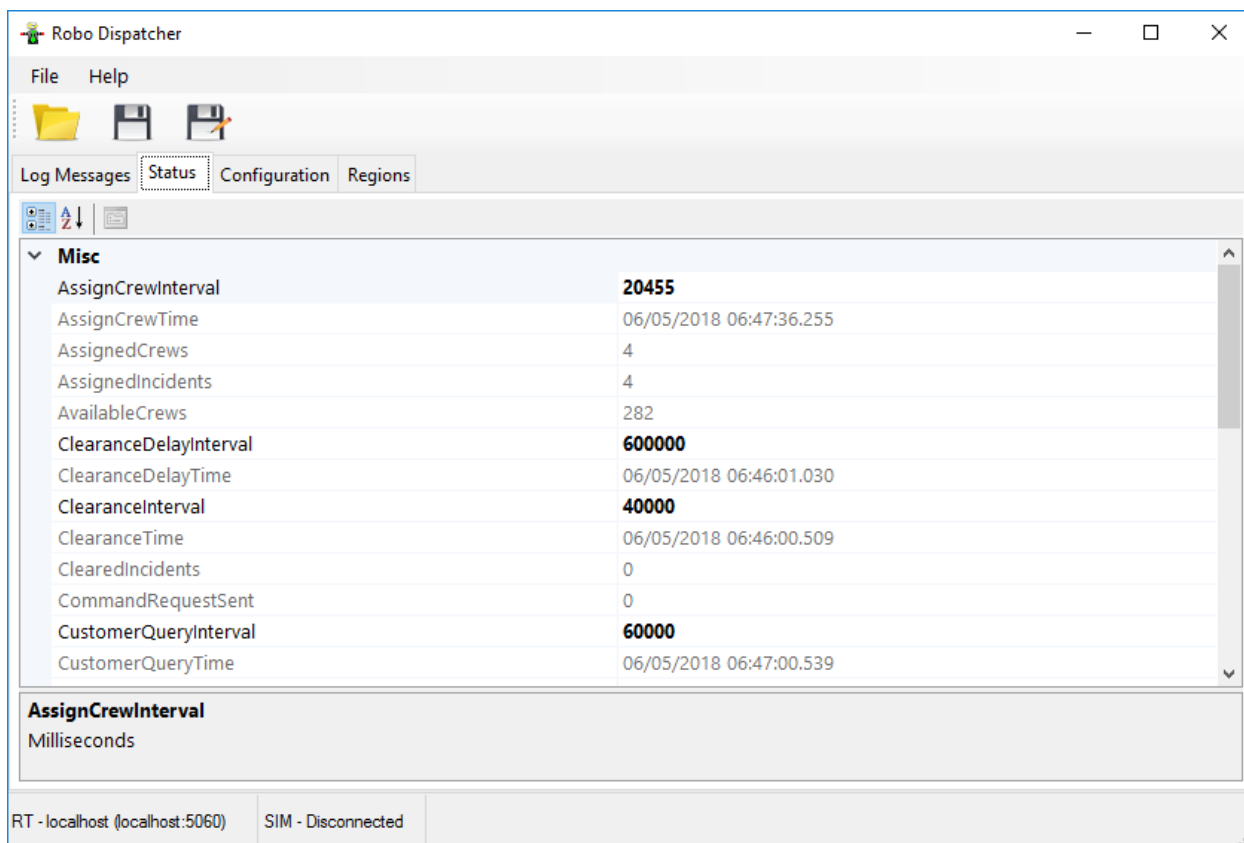


Figure 18. Viewing Operator Statistics in the Operator Simulator

Note

The time values are provided in milliseconds (0.001 second).

2.5.2. Configuring the Operator Simulator

On the Configuration tab, you can configure the Operator Simulator.

Note

Saved RoboDispatchConfig.xml configuration changes also apply to the Operator Simulator at run-time.

The screenshot shows the Robo Dispatcher application window with the Configuration tab selected. The interface includes a menu bar (File, Help) and a toolbar with icons for Log Messages, Status, Configuration, and Regions. The Configuration tab contains several sections:

- Queries:** Three spinners for Refresh Rate (30), Customer Query Rate (60), and Query Timeout (600), all in seconds.
- Incident Clear Interval:** Radio buttons for Auto Start (checked), Random, and Fixed. The Random option has sub-controls for Delay (10 minutes), Minimum (100 incidents per hour), Maximum (200 incidents per hour), and Rate (90 incidents per hour).
- Sort Order for Incident Cleaning:** A dropdown menu set to IncidentName.
- Cause Code:** Radio buttons for Random (checked) and Fixed. The Fixed option has a Cause Code dropdown set to Equipment Failure - No Reason Assigned.

The status bar at the bottom shows "RT - localhost (localhost:5060)" and "SIM - Disconnected".

Figure 19. Configuring the Operator Simulator

The fields and controls on the Configuration tab include the following:

- **Queries:**
 - **Refresh Rate:** The time period when the Operator Simulator queries the Net Server for data on incidents and crews. This affects the refreshing rate of the Status tab.
 - **Customer Query Rate:** The time period to query for affected customers. The time when a query is sent and the time when a query response is received are logged on the Log

Messages tab.

- **Query Timeout:** The time period to query for incident, crew, and crew area information from the OMS server.
- **Incident Clear Interval:**
 - **Auto Start:** If selected, the Operator Simulator begins clearing incidents after it starts.
 - **Delay:** Time delay before the Operator Simulator begins clearing incidents after it starts.
 - **Random:** If selected, the Operator Simulator clears a random number of incidents (between the minimum and maximum numbers) per hour.
 - **Minimum:** The minimum number of incidents to be cleared per hour.
 - **Maximum:** The maximum number of incidents to be cleared per hour.
 - **Fixed:** If selected, the Operator Simulator clears the fixed number of incidents per hour, as specified in the Rate field.
 - **Rate:** The fixed number of incidents to be cleared per hour.
 - **Sort Order for Incident Clearing:** Specifies the incident clearing priority. Can be Incident Name, Customers Out, Start Time, or Random.

 Note

For a predicted incident, the Operator Simulator first confirms the device out, and then confirms the device in. The Operator Simulator clears the incident after the incident status changes to Pending Clear.

- **Cause Code:**
 - **Random:** If selected, a random cause code is used for each cleared incident.
 - **Fixed:** If selected, a fixed cause code is used for each cleared incident.
 - **Cause Code:** Allows you to select the fixed cause code.
- **Type Code:**
 - **Random:** If selected, a random type code is used for each cleared incident.
 - **Fixed:** If selected, a fixed type code is used for each cleared incident.
 - **Type Code:** Allows you to select the fixed type code.
- **Damage Assessment Mode:** When running in the damage assessment mode, the Operator Simulator can enter a random number of poles and wires down for each incident. The random number for poles is between 0 and 10. The random number for wire is between 1 and 1,000.
 - **Enable:** If selected, the Operator Simulator performs damage assessment for incidents.
 - **Random:** If selected, the Operator Simulator performs damage assessment for a random number of incidents (between the minimum and maximum numbers) per hour.
 - **Minimum:** The minimum number of incidents to perform damage assessment per hour.

- **Maximum:** The maximum number of incidents to perform damage assessment per hour.
 - **Fixed:** If selected, the Operator Simulator performs damage assessment for a fixed number of incidents per hour, as specified in the Rate field.
 - **Rate:** The fixed number of incidents to perform damage assessment per hour.
- **Crew Assignment Interval:**
 - **Assign All Available Crews:** If selected, the Operator Simulator assigns all available crews to incidents immediately.
 - **Random:** If selected, the Operator Simulator assigns a random number of crews to a random number of incidents (between the minimum and maximum numbers per hour). A single available crew is assigned within a random interval.
 - **Minimum:** The minimum number of incidents to be assigned per hour.
 - **Maximum:** The maximum number of incidents to be assigned per hour.
 - **Fixed:** If selected, the Operator Simulator assigns a fixed number of incidents per hour, as specified in the Rate field. A single available crew is assigned at the fixed rate.
 - **Rate:** The fixed number of incidents to be assigned per hour.
 - **Assign Crew to Multiple Incidents:** If selected, a crew can be assigned to multiple incidents.
 - **Assign Multiple Crews to Incidents:** If selected, multiple crews can be assigned to an incident.
 - **Assign Non-MDT Crews:** If selected, a non-MDT crew can be assigned to an incident.
 - **Update Every Minutes:** Specifies the timeout in minutes for refreshing the incident status when a non-MDT crew is assigned to an incident.
 - **Assign MDT Crews:** If selected, an MDT crew can be assigned to an incident.
 - **Use Crew Areas:** If selected, the Operator Simulator can assign only crews, which are in the incident adjacent crew areas. If not selected, the Operator Simulator can assign crews from all areas.
- **Operator Notes:**
 - **Cleared by RoboDispatch:** If selected, the "Cleared by RoboDispatch" comment will be applied in the Dispatcher Comments field to each cleared incident.
 - **Custom Comment:** If selected, the comment entered in the text field will be applied in the Dispatcher Comments field to each cleared incident.
 - **Operator Name:** Specifies the operator name.
- **Region File Path:** Path to the Operator Simulator's region file (by default, %IDMS_ROOT%\config\server\RoboDispatchRegionsConfig.xml).
- **Log File Path:** The path and name base for the Operator Simulator's log file. A log file name follows the pattern <Name base + timestamp> (for example, OperatorSimulator_2016_11_21_05_39_50_079.txt), and is created in the specified folder each

time the tool starts.

2.5.3. Specifying the Operator Simulator Area of Responsibility

On the Regions tab of the Operator Simulator, you can specify the area of the Operator Simulator operation.

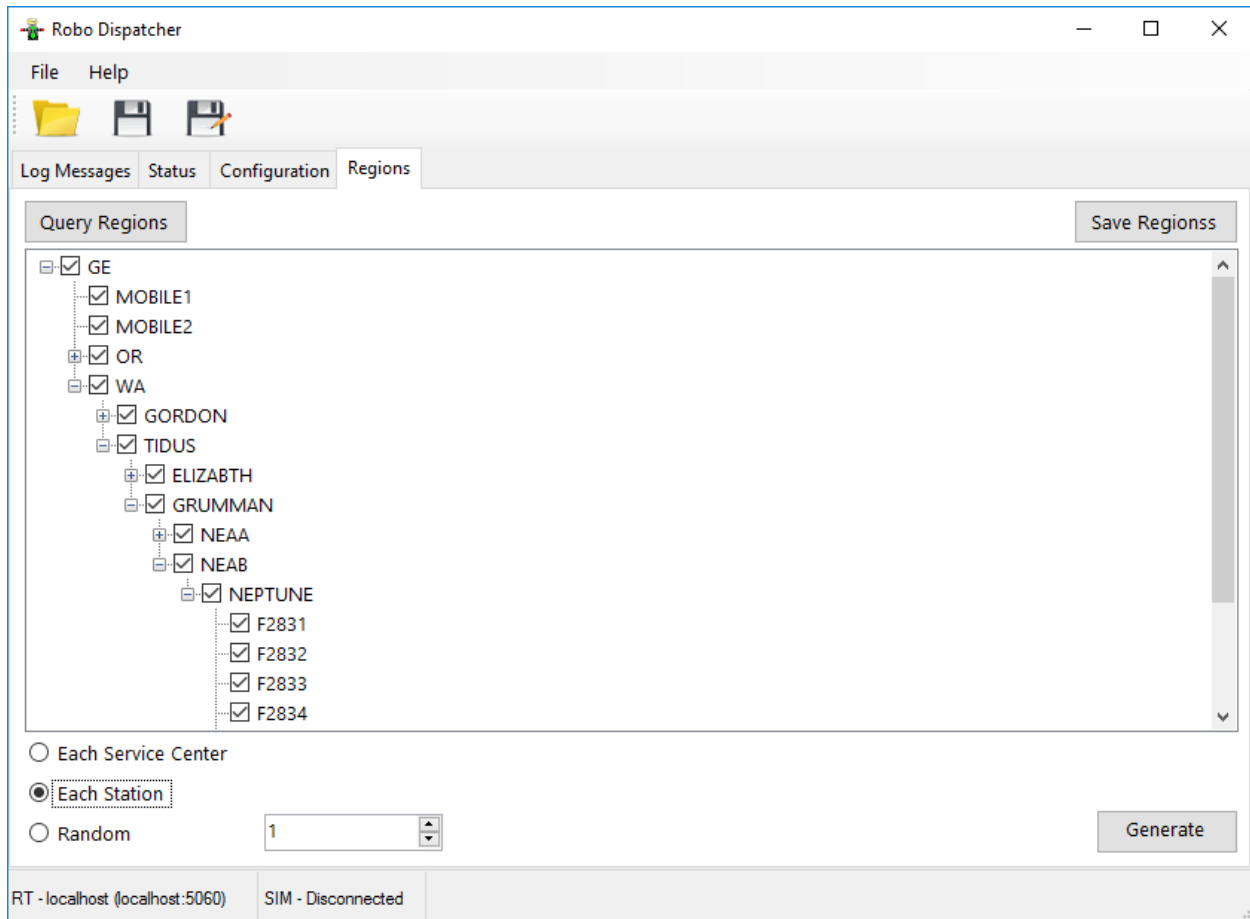


Figure 20. Specifying the Operator Simulator Area of Responsibility

You can specify the Operator Simulator working area by selecting or clearing relevant checkboxes on the tree view structure (Company > State > County > City > Service Center > Substation > Feeder) on the Regions tab. You can also generate separate configuration files for each service center, for each substation, or for a random number of stations.

The areas are loaded from the region file (located at %IDMS_ROOT%\config\server\RoboDispatchRegionsConfig.xml by default). The path to this file is specified in the RoboDispatchConfig.xml file.

The fields and controls on the Regions tab include the following:

- **Query Areas:** Loads the areas list from the region file.
-

Save Areas: Saves the areas specified on the Operator Simulator's Regions tab to the areas file, `RoboDispatchRegionsConfig.xml`.

Note

To save Operator Simulator's working areas, click Save on the toolbar. This updates the working areas information in the `RoboDispatchConfig.xml` file.

- **Generate:** Generates one or more Operator Simulator configuration files for the specified areas and saves them in the folder that contains the Operator Simulator executable file (`RoboDispatch.exe`).

The Generate Number field specifies the number of Operator Simulator configuration files to be generated (applicable only to the Each Substation and Random options). For several configuration files, the specified operation area is evenly distributed between the Operator Simulator configuration files.

- **Each Service Center:** Generates separate Operator Simulator configuration files for each service center (for example, `RoboDispatchConfigCTLA.xml` and `RoboDispatchConfigWSTC.xml`).
- **Each Substation:** Generates Operator Simulator configuration files to cover all substations (for example, `RoboDispatchConfig01.xml` and `RoboDispatchConfig02.xml`).
- **Random:** Generates Operator Simulator configuration files to cover a random area (for example, `RoboDispatchConfig0001.xml` and `RoboDispatchConfig0002.xml`).

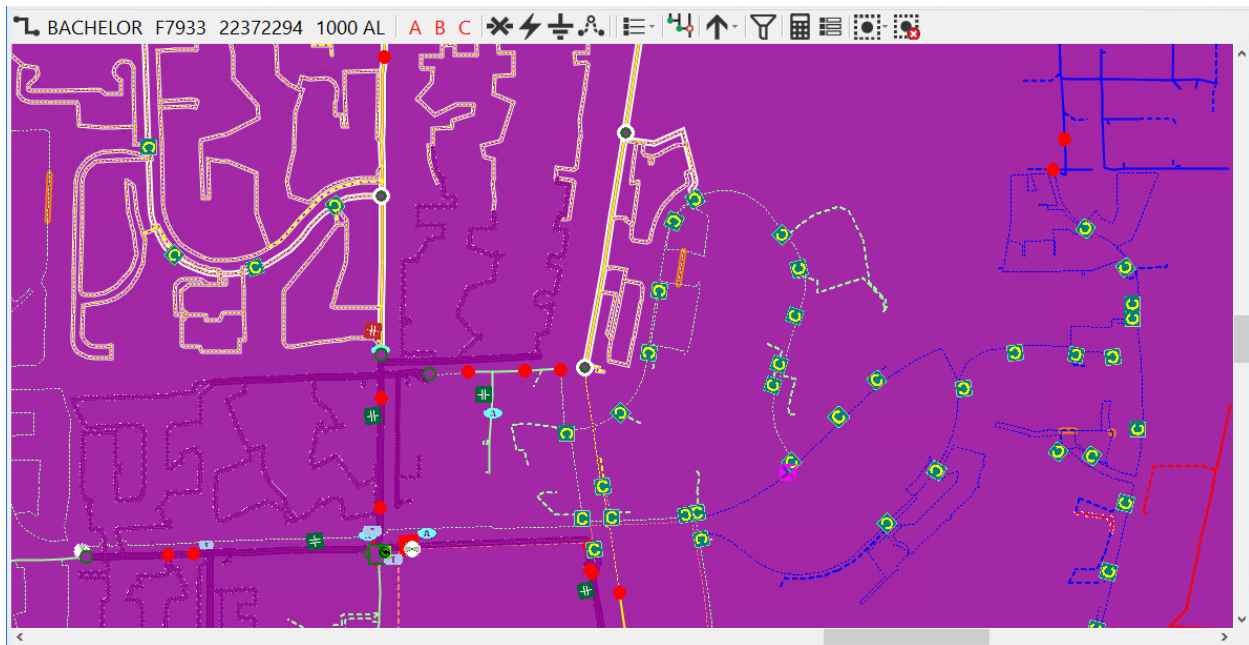
2.6. Running Training Scenarios

Running training scenarios means allowing the instructor to make changes to the physical network using the simulator view and allowing the operator in the simulated control room to react to those changes using the real-time view.

The instructor can create and run various training scenarios from the simulator view, including:

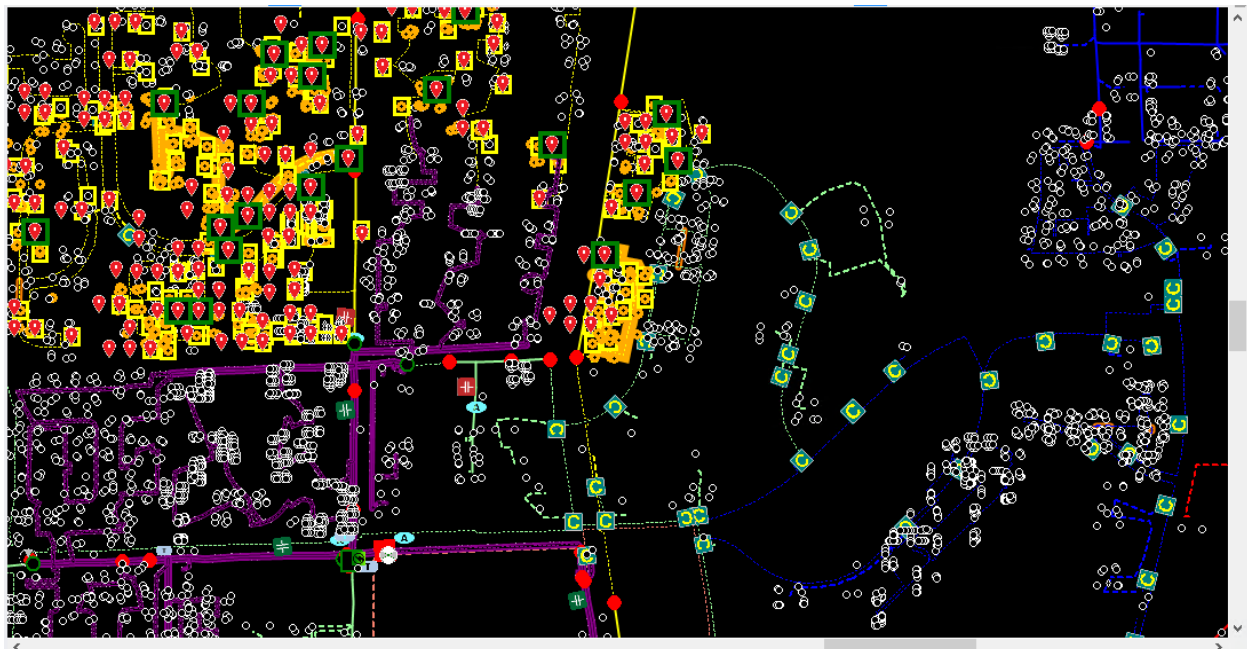
- Creating incidents
- Placing faults
- Changing switch status

For example, if the instructor creates a fault in the simulator view, the system responds to the changes by tracing upstream to the first protective device and operating it. If the first protective device is a fuse, the system opens the fuse and de-energizes it in the same way as it happens in the real-life system, as shown in the image below.



Note that all of this activity happens on the simulator view on the instructor's console. The trainees in the simulated control room do not see these operations on their screens.

The system generates customer calls in accordance with the default parameters. The trainee operator receives these calls on the real-time screen (see the image below) and reacts to these changes.



i Note

Detailed instructor and trainee actions include all standard ADMS operations described in the

Distribution Management System User Guide. They are not detailed in this document.

2.7. Running Event Scripts

The Event Scriptor tool can be used to create and replay various events (including major storms) for training and demo purposes. An event script is a separate process that can apply to the simulator and real-time models.

[Event Scriptor](#) describes how to use the Event Scriptor.

3. Event Scripter

Use the Event Scripter to create and play time-based scripts of power system events that are injected into the simulated power system model. Most of these events are common disturbances that occur on the distribution power system. Event scripts allow the instructor to save and repeat complex and/or lengthy training scenarios as required. You can also use scripts for demo purposes.

The scripts are not able to accommodate interactive operations such as responding to an operator's actions; the instructor must perform these responses. This tool does, however, provide an essential aid to the training process.

Note

You can execute Event Scripter's scripts for the real-time system model.

3.1. Enable EventScripter to connect to realtime DMS

To enable the execution of event scripts for a real-time system model, it is necessary to enable EventScripter to connect to the DMS server. By default, this feature is disabled for security reasons. To enable a connection with the real-time DMS, the following additional step is required in addition to adding the NetDynCtrl element for RT to the configuration file.

Add the following registry key with string value of the GUID of the NetDynCtrl RT section from the EventScripter's configuration. The key is:

Path: Computer\HKEY_LOCAL_MACHINE\SOFTWARE\GE

Key: DmsEventScripterIds = {2CC2A4E1-973C-4068-BDE4-89CE8D547F76}

To start the Event Scripter, click Start > Programs > Eterra > Distribution > Client > DOTS > Event Scripter.

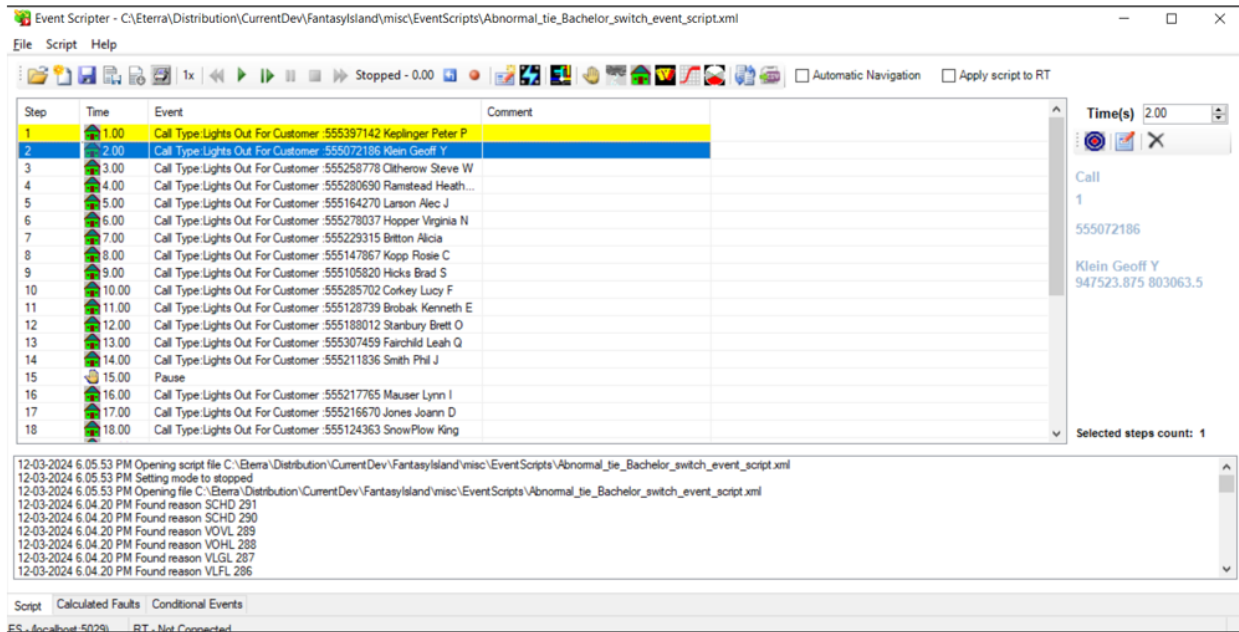


Figure 21. Event Scripter Tool

3.2. Event Scripter Displays and Controls

This section describes the Event Scripter's tabs, toolbars, and menu options.

3.2.1. Script Tab

By default, the Event Scripter tool starts with the Script tab open (figure below).

The Script tab allows the instructor to create and play a script with various types of events, and to upload and modify existing scripts.

The top panel of the Script tab displays steps; the bottom panel lists action records.

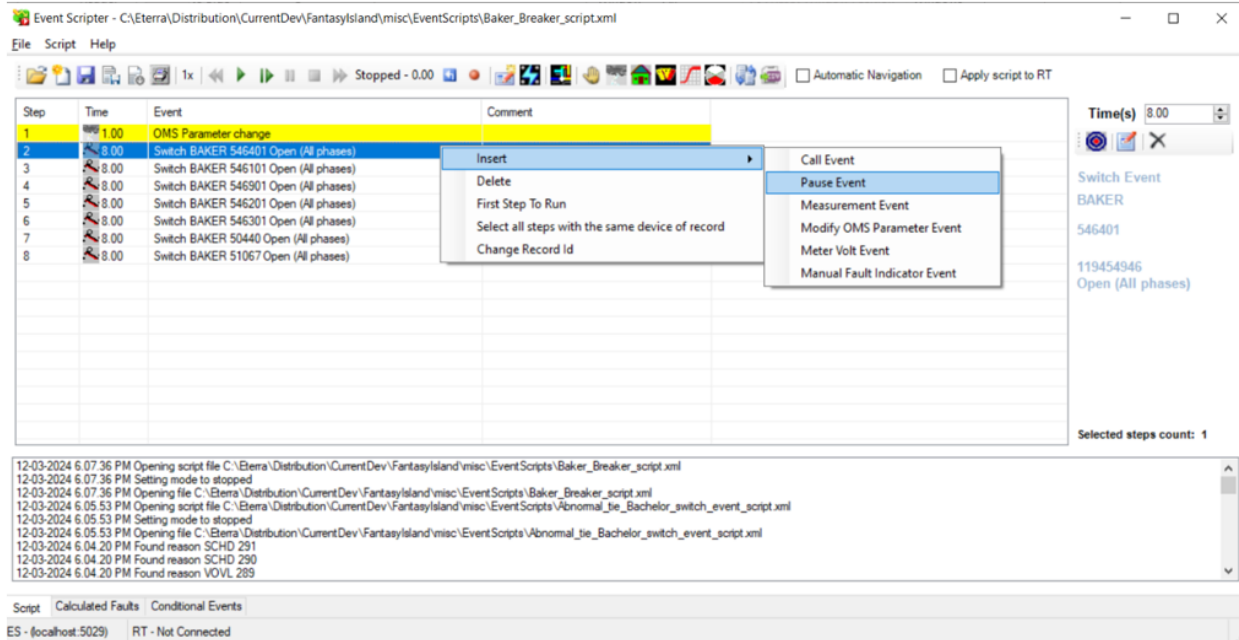


Figure 22. Script Tab of the Event Scripter Tool

3.2.1.1. Script Tab Main Toolbar

On the Script tab, the toolbar includes the following options:



Open: Opens an existing script from the script library.

Create a New Script: Creates a new script file.

Save: Saves the current configuration file.

Save Selected Steps as Template: Saves script steps as a template script.

Insert Template: Inserts a regular script or a template script.

Print: Prints the current script file.

1x Replay Speed: Shows the current replay speed of the script.

Speed Down: Decreases the replay speed by 1.

Play: Executes the script.

Execute the Next Step: Executes the next step in the list. The next step is highlighted in yellow.

Speed Up: Increases the replay speed by 1.

Pause the Script: Pauses the script's execution.

Stop: Stops execution of the script.

 **Status:** Shows the status of the script execution for each step (running/paused/stopped).

 **Rewind:** Rewinds the script back to the beginning, and sets the current script time to zero.

 **Record:** Starts the Record mode, where the Event Scripiter writes the trainer's actions on the Network View as steps in a script.

 **Go Back to Normal:** Clears the changes after running the scripts.

 **Clear Faults Only:** Removes all faults from the simulator network view.

 **Geographic View:** Calls up the network display for the Event Scripiter. You can navigate to the needed station and manually create events (create a fault, open switches, add customer calls, add a switching step, and so on). The steps appear in the currently open script file.

 **Pause:** Adds a pause event to the script. This event causes the simulator to automatically pause until the instructor continues the simulation by clicking the Play button.

 **OMS Parameters:** Adds an event to the script to modify the OMS call generating parameters for the current script.

 **Customer Call:** Opens the Add Customer Call pop-up display, to add a specific customer call event in addition to any calls that are automatically generated.

Note

A better practice is to create the customer call event from the Event Scripiter geographic network view.

 **Measurement:** Opens the Measurement Event form, which allows injecting a Scada measurement event.

 **Meter Measurement Event:** Opens the Meter Measurement form, which allows adding a customer meter measurement event.

 **Fault Indicator Event:** Opens the Manual Fault Indicator Event form, which allows adding a fault indicator event.

 **Rescale:** Opens the Rescale Current Script pop-up display, which allows compressing or expanding the time range of the script.

 **Export to CSV:** Exports the script details to the CSV file.

☐ **Automatic Navigation** : Enables automatic navigation to event steps on the geographic view. This feature is applicable when executing steps using the "Execute the Next Step" and "Execute This Step" options.

☐ **Apply script to RT** : Enables executing scripts for both simulator and real-time network models.

3.2.1.2. Script Tab Menu

On the Script tab, the menu includes the following options:

- **File > New Script:** Creates a new script file.
- **File > Open Script:** Opens an existing script from the script library.
- **File > Save Script:** Saves the current configuration file.
- **File > Save Script As:** Saves the current configuration file under the new name.
- **File > Open and Merge Script:** Allows you to open an existing script and add it to the currently open script file.
- **File > Open and Merge Dynamics:** Allows you to add steps from a dynamics file to the currently open script file.
- **File > Insert Template:** Inserts a regular script or a template script.
- **File > Print Preview:** Displays the print preview of the current script file.
- **File > Print:** Prints the current script file.
- **File > Exit:** Closes the Event Scriptor application.
- **Script > Play:** Executes the script.
- **Script > Pause:** Pauses the script's execution.
- **Script > Stop:** Stops execution of the script.
- **Script > Rewind:** Rewinds the script back to the beginning.
- **Script > Go Back to Normal:** Clears the changes after running the scripts to the simulator's nominal state.
- **Script > Clear Faults Only:** Removes all faults from the simulator network view.

3.2.1.3. Script Tab Context Menu

On the Script tab, the context menu includes the following options:

- **Insert > Call Event:** Adds a call event.
- **Insert > Pause Event:** Adds a pause event.
- **Insert > Measurement Event:** Adds a customer meter measurement event.
- **Insert > Modify OMS Parameter Event:** Adds an OMS parameters event.
- **Insert > Meter Volt Event:** Adds a customer meter event.
- **Insert > Manual Fault Indicator Event:** Adds a manual fault indicator event.

For more information about events, see [Types of Events](#).

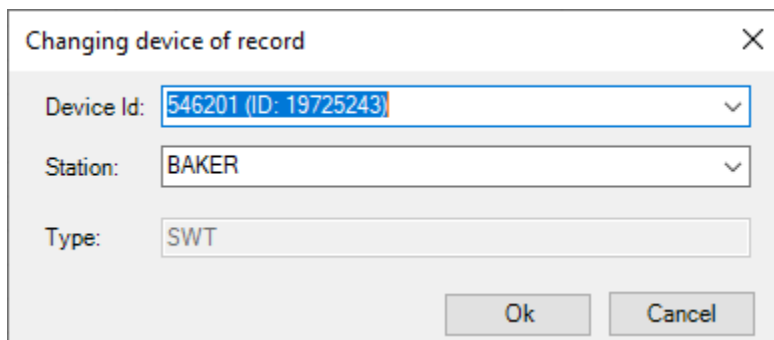
- **Delete:** Deletes selected events.
- **First Step To Run:** Sets the script start time to the selected step's time. Use this option to

resume a stopped script from the last executed event or replay only the required part of a script.

- **Select All Steps with the Same Device of Record (Ctrl + Shift + A):** Selects all script steps that are associated with the same object as the selected step.
- **Change Record ID:** Opens the "Changing device of record" dialog to associate the selected script steps with another object (for example, device name, ID, or customer account number). You can change the associated object using the drop-down menus, or by dragging an object from the map into the dialog.

Note

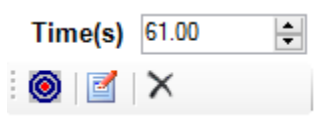
When changing an object for multiple steps, the validation function checks that all selected script steps are associated with the same object types (for example, switching devices).



The dialog box titled "Changing device of record" has a close button (X) in the top right corner. It contains three input fields: "Device Id:" with a dropdown menu showing "546201 (ID: 19725243)", "Station:" with a dropdown menu showing "BAKER", and "Type:" with a text field showing "SWT". At the bottom right are "Ok" and "Cancel" buttons.

3.2.1.4. Script Tab Editing Toolbar

The editing toolbar for the Script tab and the Calculated Faults tab allows editing of the currently selected step/fault properties:



The toolbar includes a "Time(s)" field with a value of "61.00" and a spin button. Below it are three icons: a target icon, a pencil icon, and a delete icon (X).

- **Time(s):** Number of seconds after the start of the script when the step executes. Using this field, you can set and modify the time for each step.

The time is based on the number of seconds from the start, rather than on a fixed time format, so that the script works regardless of the start time.

Note

Script steps are executed with a time step of one second. For example, a Time of 8.5 means that the step executes (that is, a fault is placed) 8 seconds after the start of the script. Multiple steps can have the same time. In this case, they are executed together in the order of the steps in the list.

- **Execute This Step:** Allows you to execute a single selected step from the list.
- **Edit:** Allows you to edit the event parameters for the selected step.
- **Delete:** Deletes the selected step.

3.2.2. Calculated Faults Tab

The fault calculator allows you to generate a sequence of network faults for the various scenarios that can be used in the event scripts. For example, if you are creating a major storm scenario, this tool can generate a set of faults that simulate the effect of the storm. For more information on creating high activity event scripts using the fault calculator, refer to [Using the Fault Calculator for Creating Scripts](#).

The Calculated Faults tab consists of three sections:

- **Reasons:** Allows you to enter the probability of a fault (faults per hour) occurring on different equipment types for a range of possible causes. For example, an ice/snow storm (reason) has high probabilities of faults on overhead equipment, but low probabilities on underground equipment.
- **Calculations:** Allows you to select the station to place the faults.
- **Fault List:** Shows the list of faults generated by the calculator and allows you to edit the faults using the editing toolbar.

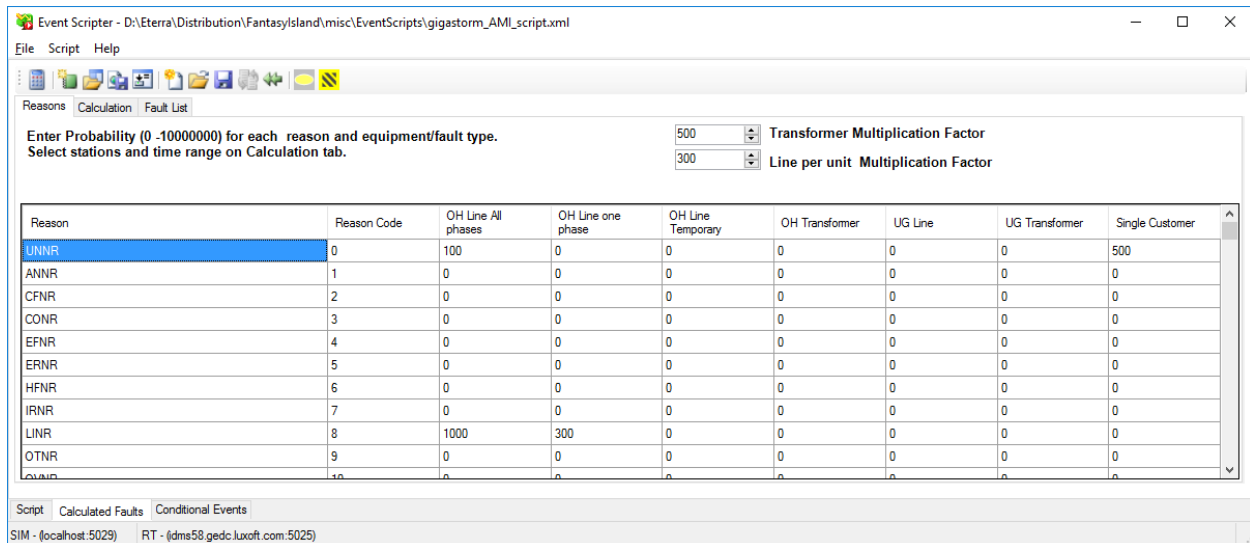


Figure 23. Event Scripter - Calculated Faults on the Reasons Tab

You can switch among these sections by using the tabs below the Event Scripter toolbar.

Both the scenario (table of probabilities) and the generated fault list can be saved separately, or the fault list can be merged with the main script opened on the Script tab.

3.2.2.1. Calculated Faults Tab Main Toolbar

On the Calculated Faults tab, the toolbar includes the following options:



- **Calculate Faults:** Generates a list of faults based on the probabilistic scenario for each equipment and fault type for the selected stations over the specified time range.
- **Create New Scenario:** Creates a new blank scenario file.
- **Save:** Saves the current scenario.
- **Save As:** Saves the current scenario under the new name.
- **Clear:** Clears the current fault list.
- **Open:** Opens an existing fault list.
- **Save:** Saves the current fault list.
- **Rescale:** Opens the Rescale Current Script pop-up display, which allows compressing or expanding the time range of the fault list.
- **Merge:** Merges the current fault list into the main script.
- **Create a script for one customer for every transformer:** Creates a script with call events coming from one customer for every transformer.
- **Create a script with all customers:** Creates a script with call events coming from all existing customers.

3.2.2.2. Calculated Faults Menu

On the Calculated Faults tab, the menu includes the following options:

- **File > Open and Merge Script:** Allows you to open an existing script and add it to the currently open script file.
- **File > New Scenario:** Creates a new blank scenario file.
- **File > Open Scenario:** Opens an existing scenario.
- **File > Save Scenario:** Saves the current scenario.
- **File > Save Scenario As:** Saves the current scenario under the new name.
- **File > New Fault List:** Creates a new blank fault list.
- **File > Open Fault List:** Opens an existing fault list.
- **File > Save Fault List:** Saves the current fault list.
- **File > Save Fault List As:** Saves the current fault list under the new name.
- **Script > Play:** Executes the fault list.
- **Script > Pause the Script:** Pauses the fault list's execution.

- **Script > Stop:** Stops execution of the fault list.
- **Script > Rewind:** Rewinds the fault list back to the beginning.
- **Script > Go Back to Normal:** Clears the changes after running the scripts to the simulator's nominal state.
- **Script > Clear Faults Only:** Removes all faults from the simulator network view.

3.2.3. Conditional Events Tab

The Conditional Events tab allows you to create scripts with event steps that are triggered depending on certain network conditions. For example, you can create an event with a customer call that triggers if the customer's transformer is disconnected.

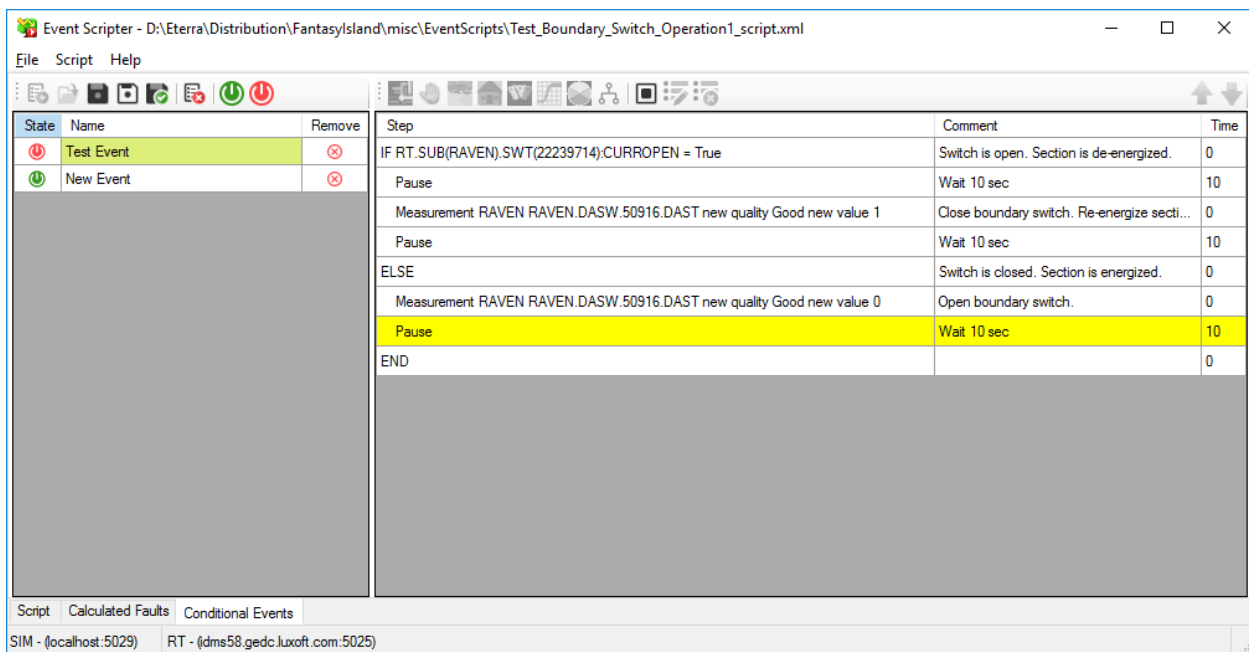
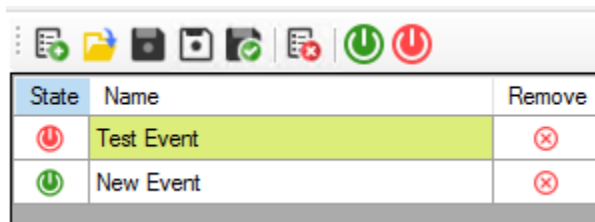


Figure 24. Event Scripter - Conditional Events Tab

3.2.3.1. Conditional Events Tab Script Events Section

The Script Events section shows events that are included in the conditional event script. Each script can include multiple events, which include individual steps and behave as separate scripts.



The Script Events toolbar includes the following options:

-  **Add New Event:** Adds a new event to the script.
-  **Open:** Opens an existing conditional event script from the library.
-  **Save:** Saves the current conditional event script.
-  **Save As:** Saves the current conditional event script with the new name.
-  **Save Selected Events:** Saves the selected conditional events to a new script.
-  **Clear:** Removes all events from the script.
-  **Execute All:** Executes all events.
-  **Stop All:** Stops execution for all events.

The Script Events grid includes the following fields and controls:

- **State:** Shows the event execution state. The state bitmap is red for executing events, and green for stopped events. Click this bitmap to toggle the execution state.
- **Name:** The event name.
- **Remove:** Click to remove the selected event.

3.2.3.2. Conditional Events Tab Event Steps Section

The Event Steps section shows event steps that are included in the selected script event.

  		
Event	Comment	Time
IF RT.SUB(RAVEN).SWT(22239714).CURROPEN = True	Switch is open. Section is de-energized.	0
Pause	Wait 10 sec	10
Measurement RAVEN RAVEN.DASW.50916.DAST new quality Good new value 1	Close boundary switch. Re-energize section.	0
Pause	Wait 10 sec	10
ELSE	Switch is closed. Section is energized.	0
Measurement RAVEN RAVEN.DASW.50916.DAST new quality Good new value 0	Open boundary switch	0
Pause	Wait 10 sec	10
END		0

Use the Event Steps toolbar to manage event steps. The toolbar includes the following options:

 **Geographic View:** Calls up the network display for the Event Scriptor. You can navigate to the needed station and manually create events steps (for example, create a fault, open switches, add customer calls, add a switching step, and so on).

 **Pause:** Adds a pause step to the event. This step causes the simulator to pause the event execution for the specified number of seconds.

 **OMS Parameters Event:** Adds an OMS call generating parameters event step.

 **Customer Call Event:** Adds customer call event step.

 **Measurement Event:** Adds a measurement event step.

 **Meter Event:** Adds a customer meter measurement event step.

 **Fault Indicator Event:** Adds a fault indicator event step.

 **Conditional Event:** Adds a condition step.

 **Stop:** Stops execution of the current event.

 **Edit Step:** Opens the selected event step for editing.

 **Remove Step:** Removes the selected event step.

 **Move Up:** Moves the selected step up in the list.

 **Move Down:** Moves the selected step down in the list.

3.3. Managing Scripts

You can manually create scripts, calculate faults, and merge with the existing faults and training scenarios. The sections below provide detailed instructions.

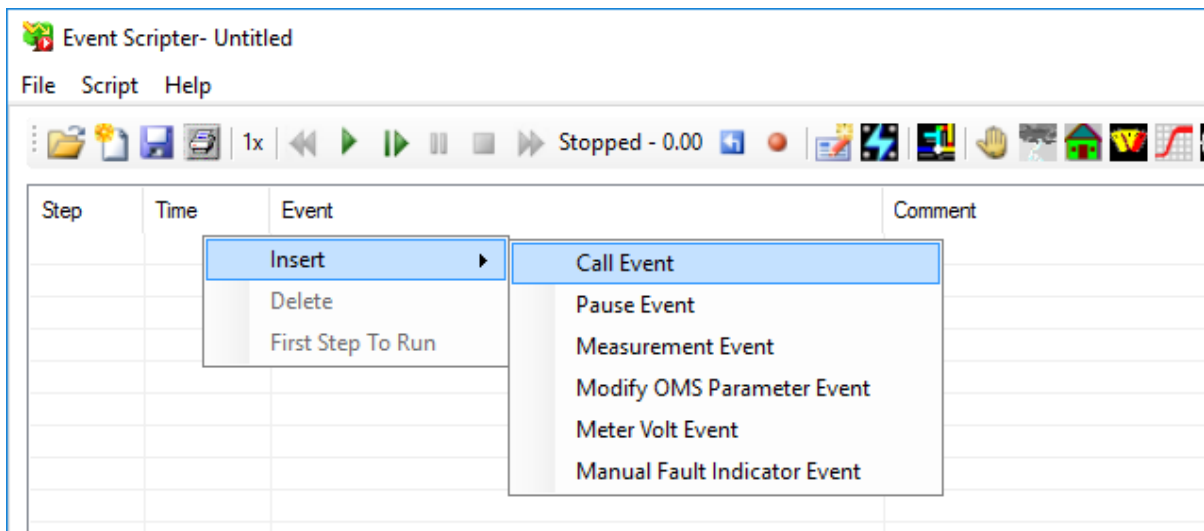
A time-based script is a series of events with a time offset representing the number of seconds from the start of the simulation. The script is stored in an XML file. It is possible to re-order the script by changing the time offset of various events. If the time offsets are changed while the simulation is running, and a re-sort is requested, it may cause some events to be skipped or duplicated. Any step that gets its time changed to the past before it executes, or any step that is skipped because the simulation counter is advanced past its time, is not executed.

For details about events, see [Creating Simulation Events and Scenarios](#).

3.3.1. Creating a Simple Script

To create a new script:

1. On the Script tab toolbar, click the New Script button .
2. Select the event type from the Script tab main toolbar or right-click menu.

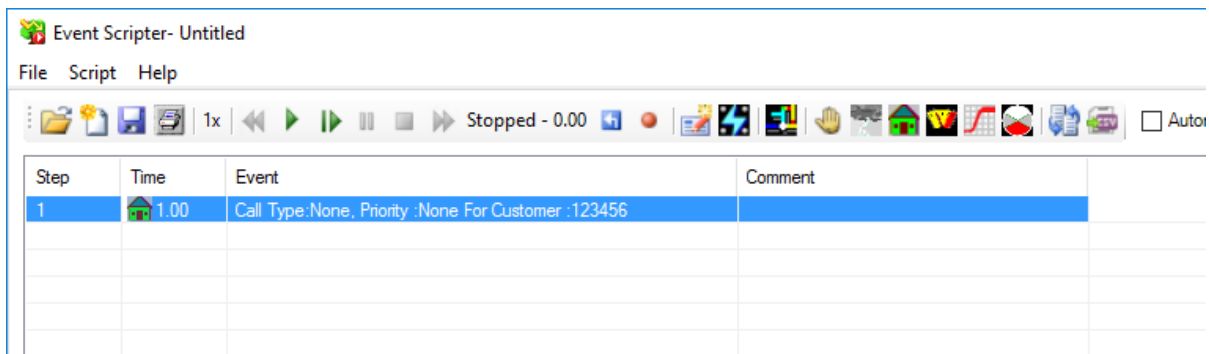


3. Configure the event and click OK.

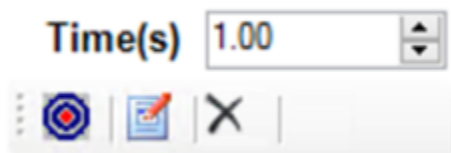
The screenshot shows the 'CustomerEventForm' dialog box. It contains the following fields and options:

- Time:** A numeric field with the value '1.00' and a spinner control.
- Customer Event:** A group box containing three radio buttons:
 - ☒ Call
 - ☐ AMI Power Out Report
 - ☐ AMI Power On Report
- Customer Name:** A text field.
- Account Number:** A text field containing the value '123456'.
- Customer Key:** A text field.

The step appears in the top panel of the Script tab.



- Adjust where the event should be inserted in the current script by adjusting the time on the Editing toolbar via the Time field.



In the example above, time 1 means that the step is executed one second after the start of the script.

- Add more event steps as needed.
- Click File > Save Script to save the new script in the [script unique name]_script.xml format.

3.3.2. Creating a Template Script

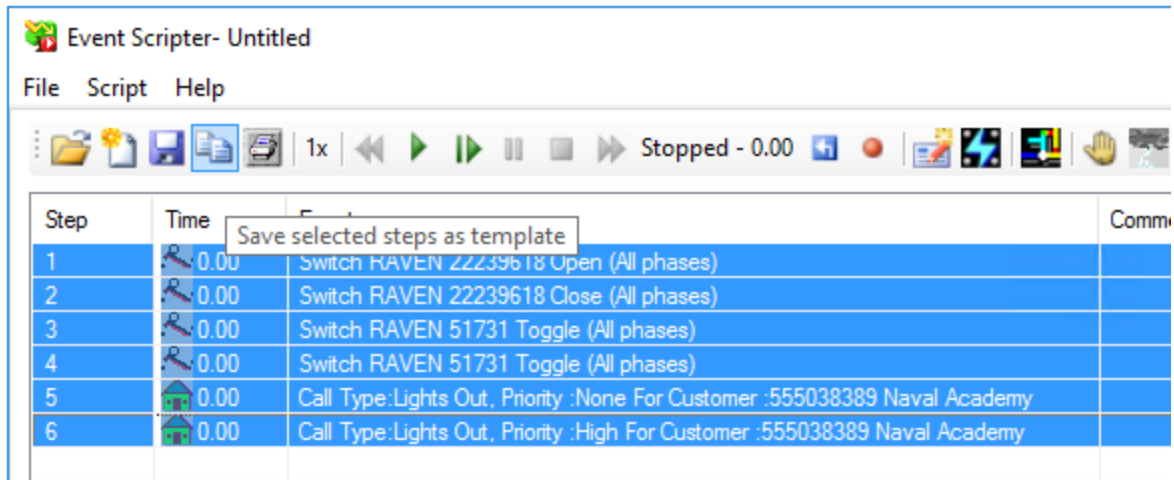
To create a new template script:

- On the Script tab, open a script or create a script. For more information, refer to the [Creating a Simple Script](#).
- Select one or more script steps. To select all steps that are associated with the same object, select a step and push Ctrl + Shift + A or click "Select All Steps with the Same Device of Record" from the context menu.

Note

When no script steps are selected, all script steps are saved in the template script.

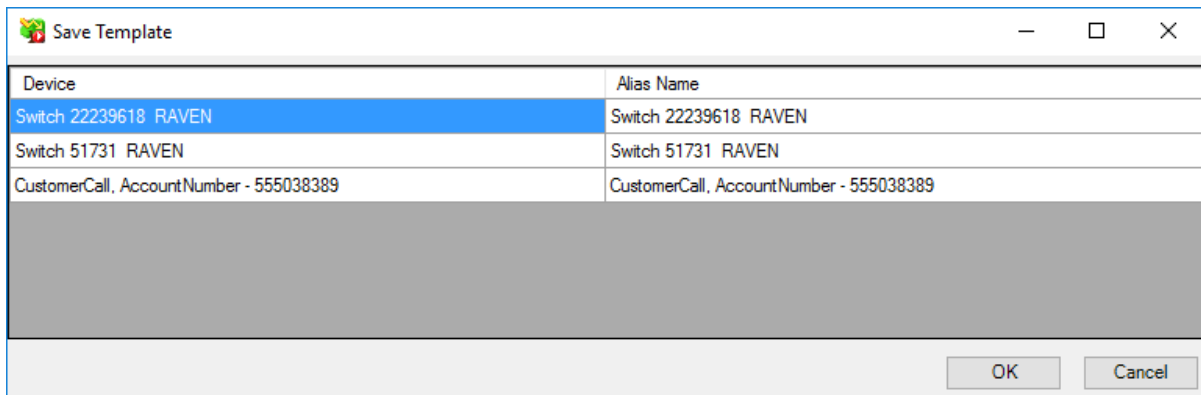
- Click Save Selected Steps as Template on the Script tab main toolbar.



The Save Template dialog appears.

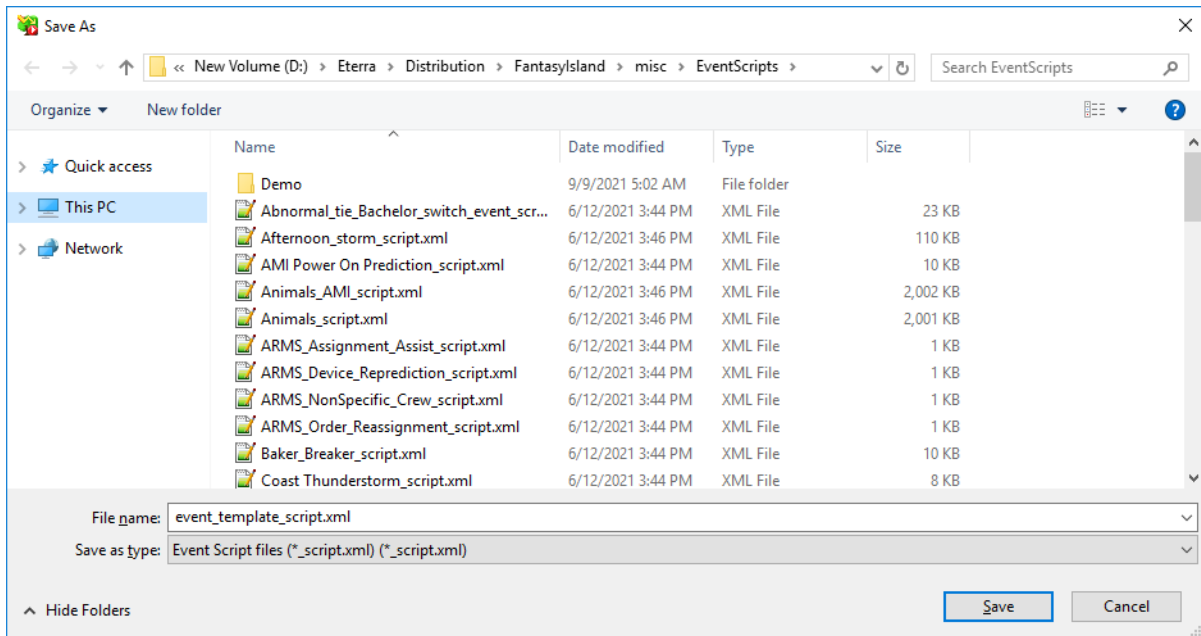
In the Save Template dialog, the script steps are grouped by the associated objects, such as customers or devices. Each step group is assigned an alias name, which by default includes key object properties. Alias names are shown when inserting a regular script or a template script into a script.

4. Modify alias names, as appropriate, and click OK.



Save As dialog appears.

5. Enter script file name and click Save.



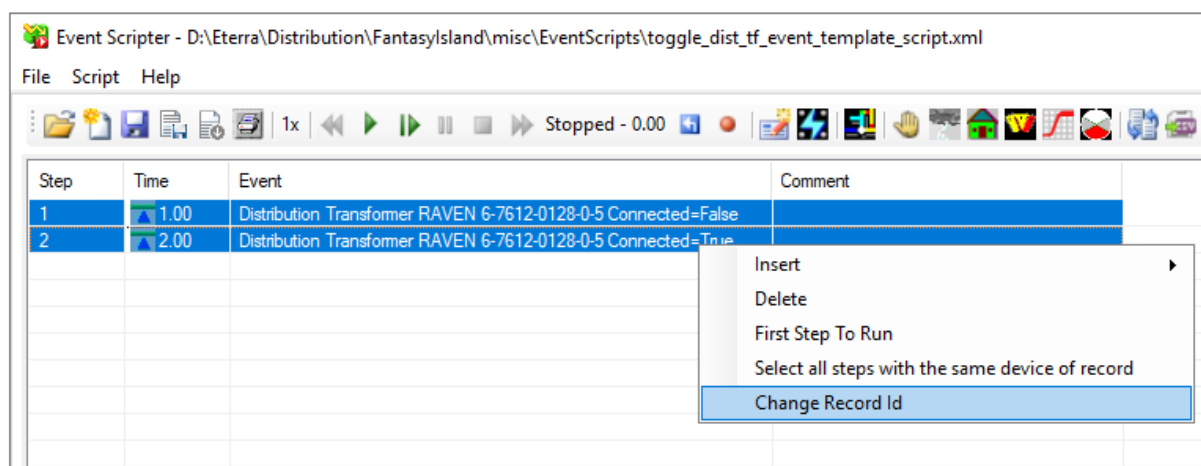
The template script is saved.

3.3.3. Creating a Script using a Template

To create a new script using a template script:

1. Open a regular script or template script.
2. Edit the script if necessary.

For example, you may need to associate script steps with other objects using the "Select All Steps with the Same Device of Record" and "Change Record ID" right-click menu options.



Changing device of record

Device Id: 6-7612-0128-0-5 (ID: 13298244)

Station: RAVEN

Type: DISTTF

Ok Cancel

Optionally, you can insert another script or template using the "Insert Template" button on the Script tab toolbar. When inserting a script or template, you can associate script steps with other objects using the drop-down menus or by dragging an object from the map into the associated entry.

Event Scripter - D:\Eterra\Distribution\FantasyIsland\misc\EventScripts\toggle_dist_tf_event_template_script.xml

File Script Help

1x Stopped - 0.00

Step	Time	Event	Comment
1	1.00	Distribution Transformer RAVEN 6-7612-0128-0-5 Connected=False	
2	2.00	Distribution Transformer RAVEN 6-7612-0128-0-5 Connected=True	

Load Template "toggle_switch_event_template_script"

Alias Name	Device	Station
Switch 52721 RAVEN	52721 (ID: 22233938)	RAVEN

OK Cancel

Event Scripter - D:\Eterra\Distribution\FantasyIsland\misc\EventScripts\toggle_dist_tf_event_template_script.xml

File Script Help

1x Stopped - 0.00

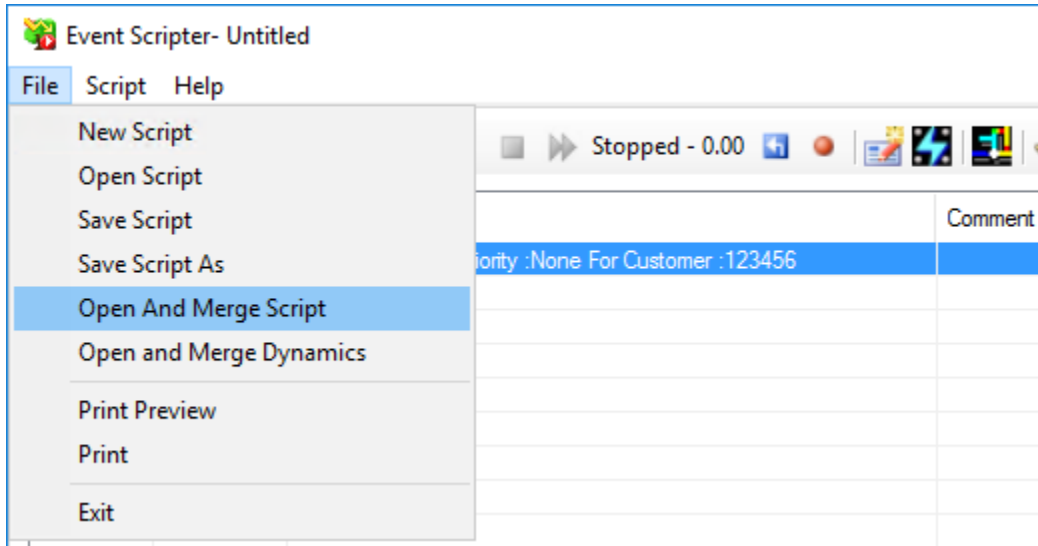
Step	Time	Event	Comment
1	1.00	Distribution Transformer RAVEN 6-7612-0128-0-5 Connected=False	
2	2.00	Distribution Transformer RAVEN 6-7612-0128-0-5 Connected=True	
3	0.00	Switch RAVEN 52721 Toggle (phase ")	
4	0.00	Switch RAVEN 52721 Toggle (phase ")	

3. Click File > Save Script As, and save the script with the new name.

3.3.4. Merging Scripts

To merge two scripts:

1. Open the first script, and edit it if necessary.
2. Click File > Open and Merge Script, and select the second script to merge with.



This opens a window where you can browse to the script to merge with the currently open script.

3. Select a script to merge, and click Open.

The original script contains the steps from the second script. The steps' precedence is adjusted in accordance with the original time from both scripts.

4. Click File > Save Script to save the changes.

3.3.5. Merging Dynamics

To merge a script with a dynamics file:

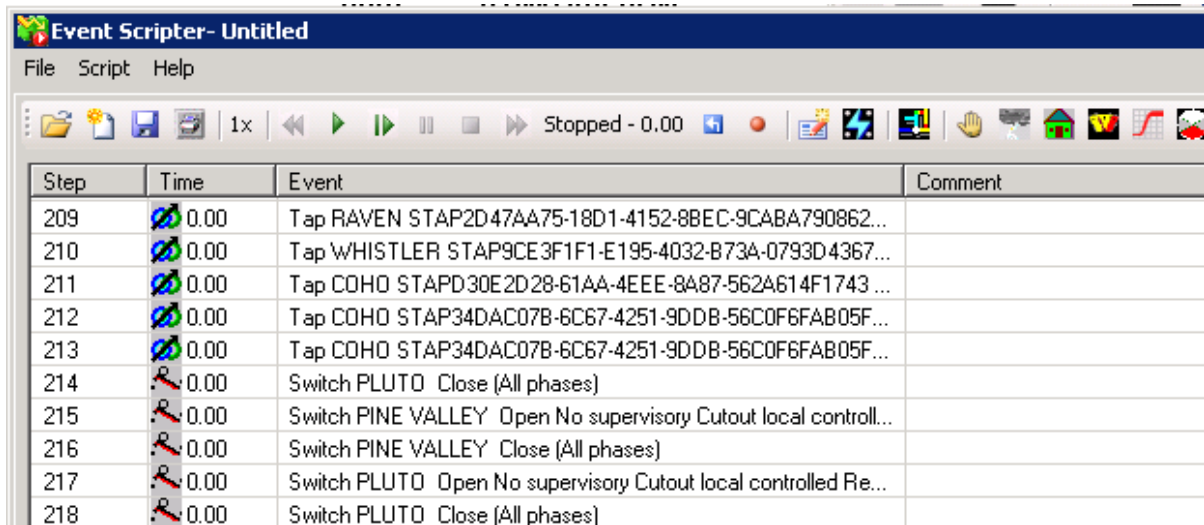
1. Open an event script, and edit it if necessary.
2. Click File > Open and Merge Dynamics, and select the second script to merge with.

This opens a window where you can browse to the dynamics file to merge with the currently open script. By default, the dynamics files are located at %IDMS_ROOT%/dynamics.

3. Select the dynamics file to merge, and click Open.

The steps from the dynamics are added to the original script, the execution time is set to 0, and

the steps are grouped by device type.



4. Click File > Save Script to save the changes.

✓ Tip

The Open and Merge Dynamics option does not reflect the actual dynamics step sequence. Therefore, use this option to only simulate a certain system condition before executing a script. For retrieving network events with their actual time, see [Exporting Events from Dynamics](#).

3.3.6. Modifying and Executing a Single Step

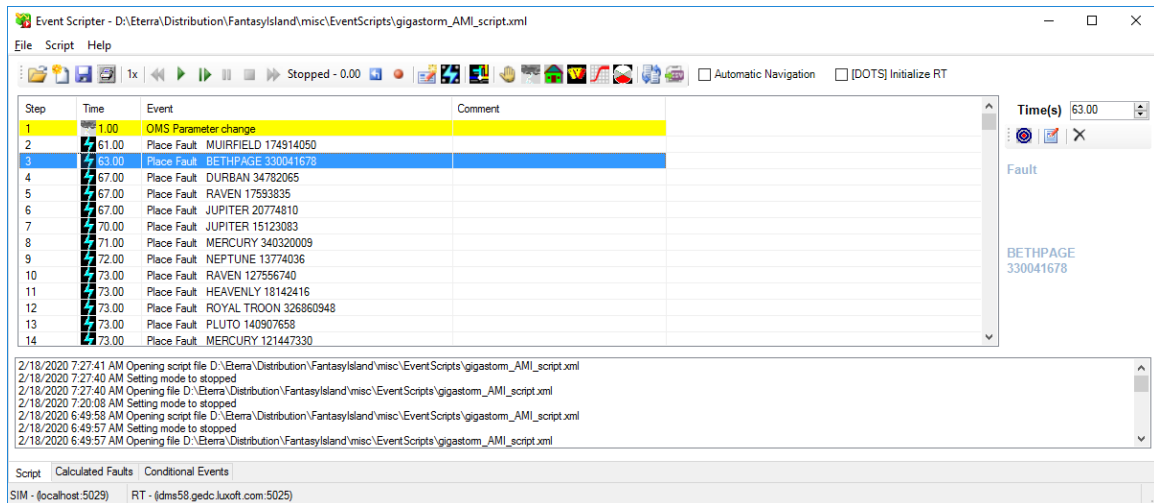
You can edit, delete, or execute a single step from the script using options on the editing toolbar next to the steps panel.

To modify or execute a single step:

1. Open a script using the File > Open Script menu options.
2. Select a step to modify.

Detailed event information appears under the editing toolbar.

- To adjust the precedence of the step, modify the time for the selected step in the Time (s) field of the editing toolbar.
- To edit the step's event parameters, click the Edit button .
- To delete the selected step, click the Delete button .
- To execute a single step, select the step and click the Execute This Step button  on the editing toolbar.



3. Click File > Save Script to save the changes.

3.3.7. Rescaling the Time Range

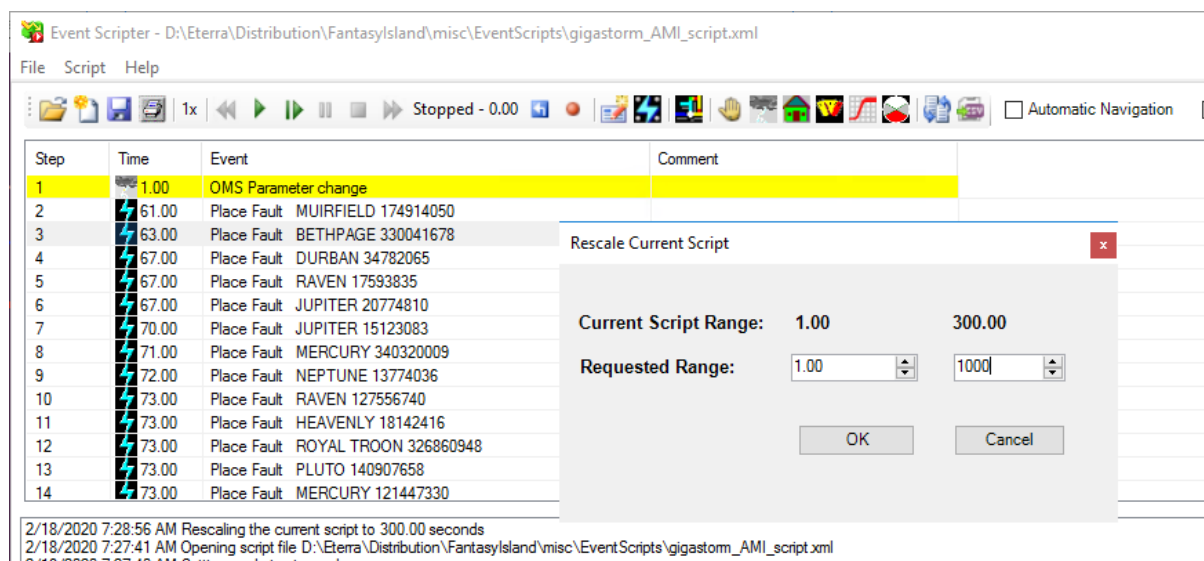
You can change the start and end times of a script or scenario. The Rescale option proportionately rescales the timing of all steps in the script. If the timescale is reduced, some steps may have the same time value.

To rescale the time range:

1. Open the script to rescale, and click the Rescale Script to a New Time Range  button.

The Rescale pop-up window appears.

2. The Current Script Range field displays the current start and stop time of the script steps. Adjust the time range in the Requested Range fields as needed.



3. Click OK.

The steps' time is adjusted proportionally to fit the new time range.

Step	Time	Event
1	1.00	OMS Parameter change
2	201.47	Place Fault MUIRFIELD 174914050
3	208.15	Place Fault BETHPAGE 330041678
4	221.52	Place Fault DURBAN 34782065
5	221.52	Place Fault RAVEN 17593835
6	221.52	Place Fault JUPITER 20774810

Note

The time is based on a number of seconds from the start, rather than on a fixed time format, so that the script works regardless of the start time. Multiple steps can have the same time.

3.3.8. Running Scripts

A script normally executes from its first step until the end; however, you can choose another step as the starting point. In addition, you can execute a script step by step, or execute a single step from the script. You can also pause a script and stop the simulation counter, and increase the execution speed.

To run scripts, use these Event Scriptor toolbar buttons:

- Click the Play button  to execute the script.
- Click the Speed Up button  to increase the replay speed by 1.
- Click the Speed Down button  to decrease the replay speed by 1.

The supported replay speed range is 1-50.

- Click the Pause button  to pause the script's execution.

Pausing only affects the script execution. While the script is paused, the simulator is fully functional, but the script counter does not advance.

- Click the Execute the Next Step button  to execute the next step in the script.

Use this option to manually execute a script step-by-step, regardless of the event time settings. You can use this option together with the "Execute the First Step to Run" context menu option to execute a certain part of a script step by step.

- Click the Stop button  to stop the execution of the script.
- Click the Rewind button  to rewind the script back to the beginning, and to set the current script time to zero.

The Status field **Stopped - 4.00** shows the status of the script's execution for each step (running/paused/stopped) and the last executed step's time.

The next step to be executed is shown in yellow.

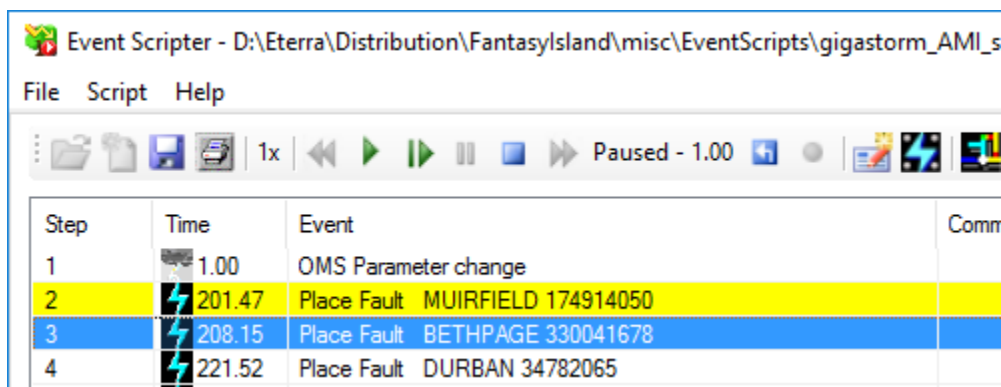


Figure 25. Event Scripter - Running Scripts

3.3.9. Using the Event Scripter's Network View

Event Scripter's network view allows you to easily navigate to network elements, create events, and perform other operations. On this network view you can perform the same operations as on the simulator view of the ADMS client. In addition, you can enable automatic navigation to steps when running a script step by step or when executing a single step in the script.

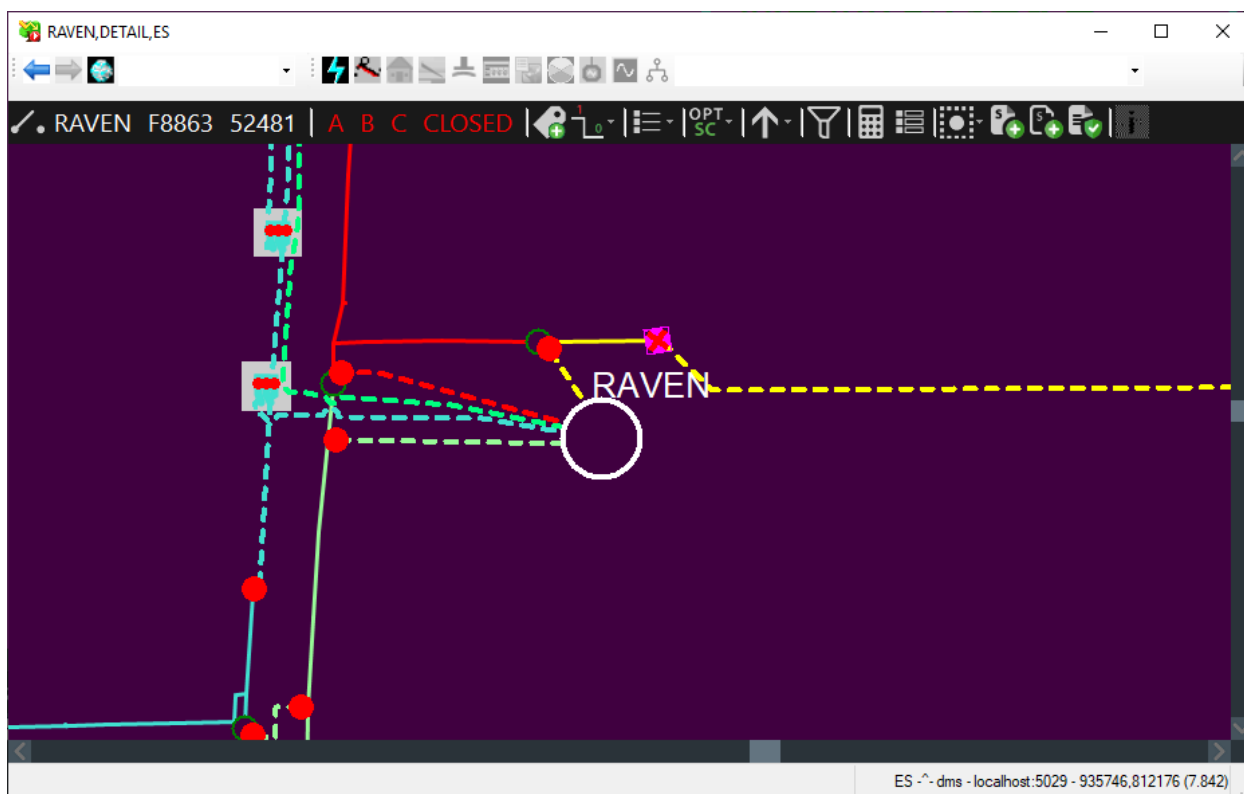


Figure 26. Event Scripter - Network View

On the Event Scripter network view, the toolbar includes the following options:



 Navigates to the previous display.

 Navigates to the next display.

 Navigates to the substation entered in the text box.

— Text box to enter a substation name.



Text box to enter Netview commands.

Events created from the network view are added to the script list. You can also record manual instructor actions as described in [Recording Actions on the Event Scriptor's Network View](#).

3.3.10. Clearing the System

For instructions on resetting the system to the initial state and clearing dynamics after running the training scenarios, refer to [Returning the Simulator to the Initial State](#).

4. Creating Simulation Events and Scenarios

This chapter describes event types and provides details about how to add event steps to scripts and create simulation scenarios.

4.1. Types of Events

The Event Scriptor allows for the following event types:

- **Pause:** Causes the simulator to automatically pause the running script until the instructor resumes the simulation by executing a play function.
- **Create/Remove Fault:** Creates a fault that causes the network to trip the upstream protective device. Removing a fault simulates that a crew has repaired the problem or that the fault was transient and fixed itself. An instructor can also add or remove faults manually using the simulator's geographic network view, as described in chapter [Using the Training Simulator](#).
- **Change Switch Status:** Sets a switch status. The options include Closed/Open, with or without a momentary operation, and with or without a device failure.

For example, setting a switch to open plus momentary, when the switch is currently closed, causes a TRIP-CLOSE-TRIP transition. A failed switching device cannot be operated until the failed status is removed. An instructor can also change the switch status manually using the simulator's geographic network view, as described in chapter [Using the Training Simulator](#).

- **Set OMS Parameters:** Changes the OMS call generating parameters from the default.

For example, if customers know that a hurricane caused their outage, they are less likely to call to report an outage. Another example is to set the AMR parameters.

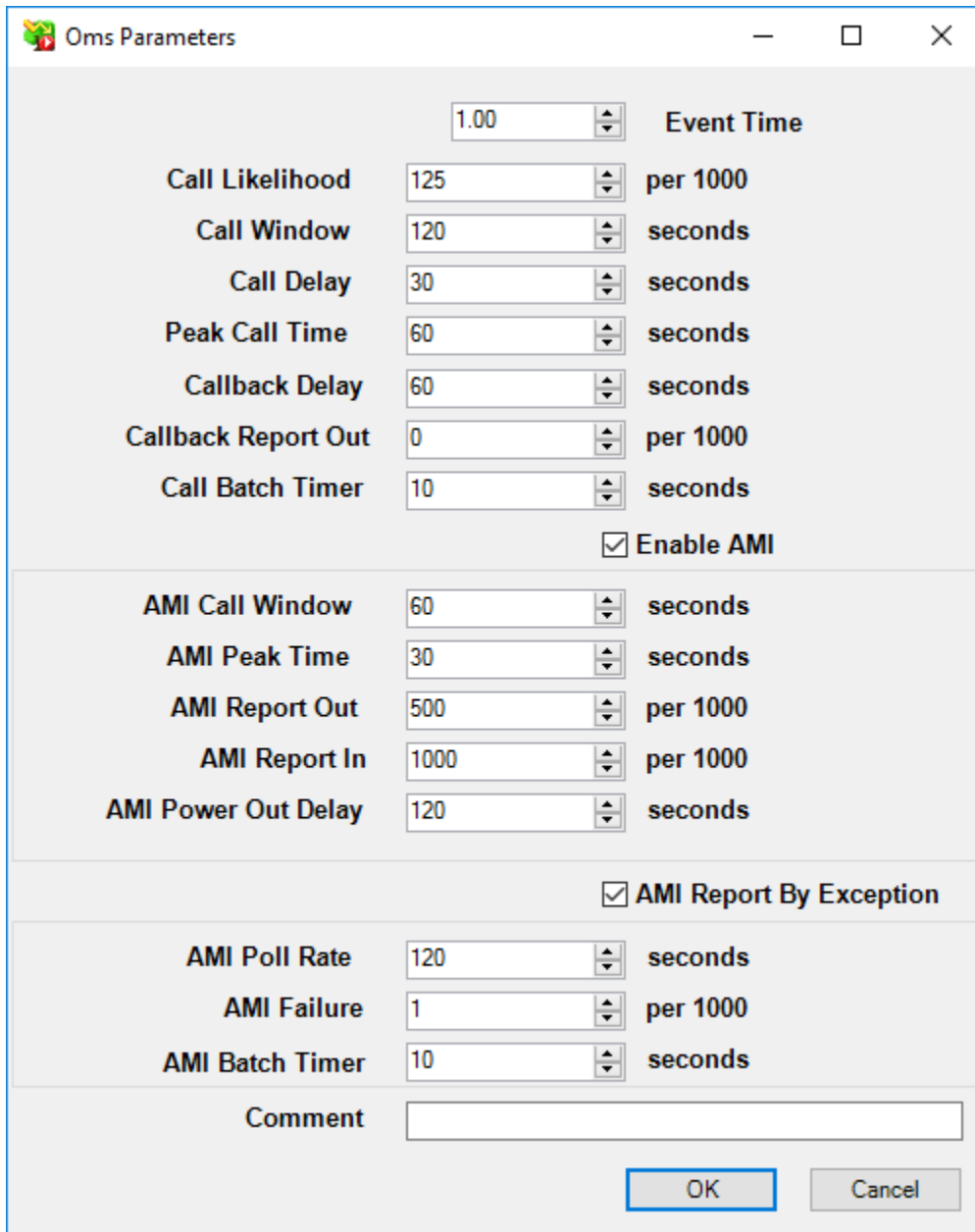
- **Generate Customer Calls:** Generates one or more specific customer calls in addition to any calls that are automatically generated.
- **Inject Measurement Event:** Allows the instructor to enter noise on measurements, generate suspect data quality, and inject bias.
- **Inject Meter Measurement Event:** Allows the instructor to enter AMI meter measurements.
- **Inject Fault Indicator Event:** Allows the instructor to enter a fault indicator status change event.
- **Inject Device Input Event:** Allows the instructor to set parameters for substations, transformers, feeders, and capacitors.

4.2. Setting OMS Parameters for the Script

In most cases, event scripts include a step to specify OMS call generating parameters. This option allows simulating the expected customer call environment.

To add an event that specifies OMS call generating parameters:

1. From the Event Scripter, click the Add an Event to Modify OMS Parameters  toolbar button. The OMS Parameters dialog box opens.



Parameter	Value	Unit
Event Time	1.00	
Call Likelihood	125	per 1000
Call Window	120	seconds
Call Delay	30	seconds
Peak Call Time	60	seconds
Callback Delay	60	seconds
Callback Report Out	0	per 1000
Call Batch Timer	10	seconds
Enable AMI	☒	
AMI Call Window	60	seconds
AMI Peak Time	30	seconds
AMI Report Out	500	per 1000
AMI Report In	1000	per 1000
AMI Power Out Delay	120	seconds
AMI Report By Exception	☒	
AMI Poll Rate	120	seconds
AMI Failure	1	per 1000
AMI Batch Timer	10	seconds
Comment		

2. In the Event Time field, select the time of the step in seconds from the start of the script.
3. Configure the call and the AMI report settings.
4. Add a comment in the Comment field.
5. Click OK.

4.3. Injecting a Scada Measurement Event

The Event Scripter supports the creation of a "Scada measurement event", which allows users to modify the value, quality, noise, and deadband of one or more Scada measurements. These measurement events are injected into the Scada system and affect the values presented to the real-time model and the displays visible to the trainee. They do not affect the simulated power system model.

When the Measurement Event pop-up appears, the parameter fields for each event type are hidden until the relevant checkbox is selected. The event types are displayed regardless of whether they are applicable to the selected measurement.

The screenshot shows a window titled "MeasurementEventForm" with standard Windows window controls. Inside the window, there is a "Time" field with a spinner set to "1.00". Below this are three dropdown menus: "Station" set to "RAVEN", "Measurement type" set to "Analog", and "Measurement Id" set to "RAVEN.FDR.8861.MVAR". There is an unchecked checkbox for "Simulate Only". Below that is a checked checkbox for "Apply all in station". Following this are four checked checkboxes, each with an associated input field: "Quality change" (set to "Good"), "Value change" (set to "0"), "Noise change" (set to "0" with a "%" suffix), and "Deadband change" (set to "0" with a "%" suffix). At the bottom is a "Comment" text box. Finally, there are "OK" and "Cancel" buttons at the bottom right.

Figure 27. Event Scripter - Measurement Event Form

The Measurement Event pop-up allows you to specify some details related to the event, in particular:

- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Station:** The station name of the measurement as used in ADMS.
-

Measurement Type: Select the Analog or Status type. Other measurement types are not supported.

- **Measurement ID:** The ID of the measurement in DNOM in the format [SUBSTATION NAME].[EQUIPMENT GROUP].[DEVICE NAME].[MEASUREMENT ID] (for example, BAKER.BUS.8812_HI.KV).

You can also search for available IDs on the Scada Station Tabular display for the relevant substation.

Note

You can simulate events only for measurements that are listed in the Measurement ID drop-down list. Typically, all Analog and Status measurements for the selected substation are supported. Other measurement types, such as Set Point and Control Point may be supported as simulate-only measurements.

- **Simulate Only:** Use this option to filter the Measurement ID drop-down list by simulate-only measurement points (all measurement points modeled in Scada are shown, even if they are not modeled in DNOM). For more details, refer to section 4.3.4 Creating a Measurement Event.
- **Apply All in Station:** Selecting this option applies the measurement event to all measurements of the same type in the station. For example, if the measurement ID is xxx.xxx.xxx.AMPA, all measurements of AMPA will be affected. This does not apply to the Noise Event, where all analog measurements in the station are similarly affected.

For simulate-only measurement points, selecting this option applies the measurement event to all simulate-only measurement points in the station.

- **Command Operation:** Selecting this option sets the IsCommanded bit for the status measurement, which means that the status change is caused by a known (commanded) operation. This option only appears when the Status measurement type is selected.
- **Quality Change:** Selecting this option changes the Scada Quality Flags for the measurement point. Select the quality type: Good, Replaced, Bad, or Suspect.
- **Value Change:** Selecting this option changes the value of the measurement as part of this event. This option can be used to set values for both analog measurements and status points. The new value remains fixed until another value is applied, or the system is re-initialized.
- **Multi Phases:** This option can be used to set the status values for a multi-phase point. This option will be displayed only when the "Status" measurement type is selected. When the "Multi Phases" option is selected, the "Value Change" option will be disabled and vice-versa.

MeasurementEventForm

Time: 1.00

Station: ROYAL TROON

Measurement type: Status

Measurement Id: RYLTROON.RECL.22229749.MPP1

☐ Simulate Only

☐ Apply all in station ☐ Commanded Operation

☐ Quality change

☐ Value change

☒ Multi Phases

Comment:

0 - Adis=0, Bdis=0, Cdis=0

0 - Adis=0, Bdis=0, Cdis=0

1 - Adis=1, Bdis=0, Cdis=0

2 - Adis=0, Bdis=1, Cdis=0

3 - Adis=0, Bdis=0, Cdis=1

4 - Adis=1, Bdis=1, Cdis=0

5 - Adis=1, Bdis=0, Cdis=1

6 - Adis=0, Bdis=1, Cdis=1

7 - Adis=1, Bdis=1, Cdis=1

OK Cancel

- **Noise Change:** This option adds a random variation (in percentage of current value) to the value in analog measurement (applicable for the analog measurement type only).
- **Deadband Change:** This option applies a deadband (in percentage) to the noise of the analog measurement (applicable for the analog measurement type only). For example, if the Noise Change is set to 10% and the Deadband Change is set to 5%, only the measurement variations between 5% and 10% will be seen by the Scada system.
- **Comment:** An optional instructor's comment for this event. This comment appears in the script only.

4.3.1. Creating an Analog Type Measurement Event

The example below demonstrates how to create a noise change measurement event for a station power transformer.

1. On the real-time network view display, select a substation and navigate to the Scada One Line display.
2. Click the rectangle with the substation name in the upper-left corner of the display.

The Station Tabular display appears.

3. Open the "Analog w limits" tab.
4. Navigate to the T1 transformer and take note of its MVA value.

Eqp Group	Name	Device	ID	Value	Ignore	Data Quality	Inhibit Alarm	Options	Not In Service	TEST Mode	Meas Tag	Hier Tag
TF	T1	MW		-29.15		Good	☐	☐	☐			
TF	T1	MVAR		-11.19		Good	☐	☐	☐			
TF	T1	MVA		31.81		Good	☒	☐	☐			
TF	T2	MW		-19.90		Good	☐	☐	☐			
TF	T2	MVAR		-7.23		Good	☐	☐	☐			
TF	T2	MVA		21.17		Good	☐	☐	☐			

5. From the Event Scripter, click the Add a Measurement Event  toolbar button.
6. In the Time field, select the time of the step (in seconds) from the start of the script.
7. Select the substation from the Station drop-down list.
8. Select the Analog type from the Measurement Type drop-down list.
9. Select the RAVEN.TF.T1.MVA measurement from the Measurement ID drop-down list.
10. Set the Noise Change check box, and enter the value in percent in the field next to it.
11. Click OK.

MeasurementEventForm

1.00 Time

Station: RAVEN

Measurement type: Analog

Measurement Id: RAVEN.TF.T1.MVA

☐ Simulate Only

☐ Apply all in station

☐ Quality change

☐ Value change

☒ Noise change: 30 %

☐ Deadband change

Comment:

OK Cancel

12. Run the event or implement this measurement step.

13. Check that the T1 MVA value is constantly changing within the specified range.

SCADA RAVEN

Test Mode: OFF Inhibit: OFF NIS: OFF

Equipment Group Filter: none

Eqp Group	Name	ID	Value	Ignore	Data Quality	Inhibit Alarm	Options	Not In Service	TEST Mode	Meas Tag	Hier Tag
TF	T1	MW	-29.15		Good						
TF	T1	MVAR	-11.19		Good						
TF	T1	MVA	27.02		Good						
TF	T2	MW	-19.90		Good						
TF	T2	MVAR	-7.23		Good						
TF	T2	MVA	21.17		Good						

The Analog Measurement Event changes for a given measurement ID will remain in effect until a subsequent measurement event is received that resets the changes.

For example, to remove a noise or deadband percentage, a new measurement event with the noise or deadband set to a value of 0 must be used.

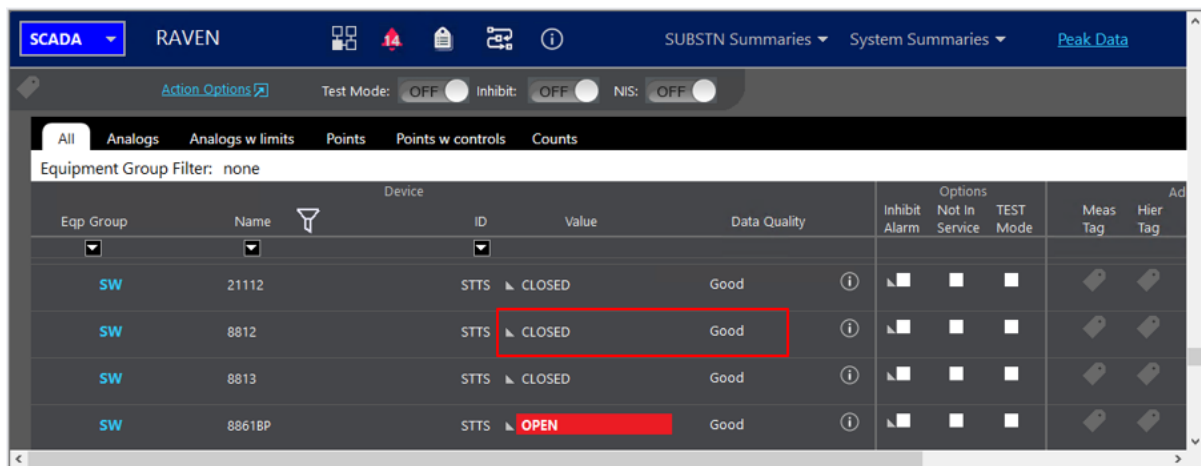
4.3.2. Creating a Status Type Measurement Event

The example below demonstrates how to create a measurement status and data quality change event for a status point (for example, a switch).

1. On the real-time network view display, select a substation and navigate to the Scada One Line display.
2. Click the rectangle with the substation name in the upper-left corner of the display.

The Station Tabular display appears.

3. Select a status point (for example, RAVEN.SW.8812.STTS) and note its current state.



Eqp Group	Name	ID	Value	Data Quality	Inhibit Alarm	Options Not In Service	TEST Mode	Meas Tag	Hier Tag	Ad
SW	21112	STTS	CLOSED	Good	☐	☐	☐			
SW	8812	STTS	CLOSED	Good	☐	☐	☐			
SW	8813	STTS	CLOSED	Good	☐	☐	☐			
SW	8861BP	STTS	OPEN	Good	☐	☐	☐			

4. From the Event Scriptor, click the Add a Measurement Event  toolbar button.
5. In the Time field, select the time of the step (in seconds) from the start of the script.
6. Select the substation from the Station drop-down list.
7. Select the Status type from the Measurement Type drop-down list.
8. Select the RAVEN.SW.8812.STTS measurement from the drop-down list.
9. Select the Quality Change check box, and select the new type from the drop-down list.
10. Select the Value Change check box and set the value to "0 - SDIS=0, XDIS=0", which is the "Open" command. Or, if the measurement is a multi-phase point measurement type, select "Multi Phases" and select "0 - Adis=0, Bdis=0, Cdis=0" from the dropdown (example images are shown below.)

Non multi-phase measurement event:

MeasurementEventForm

1.00 Time

Station RAVEN

Measurement type Status

Measurement Id RAVEN.SW.8812.STTS

☐ Simulate Only

☐ Apply all in station ☐ Commanded Operation

☐ Quality change

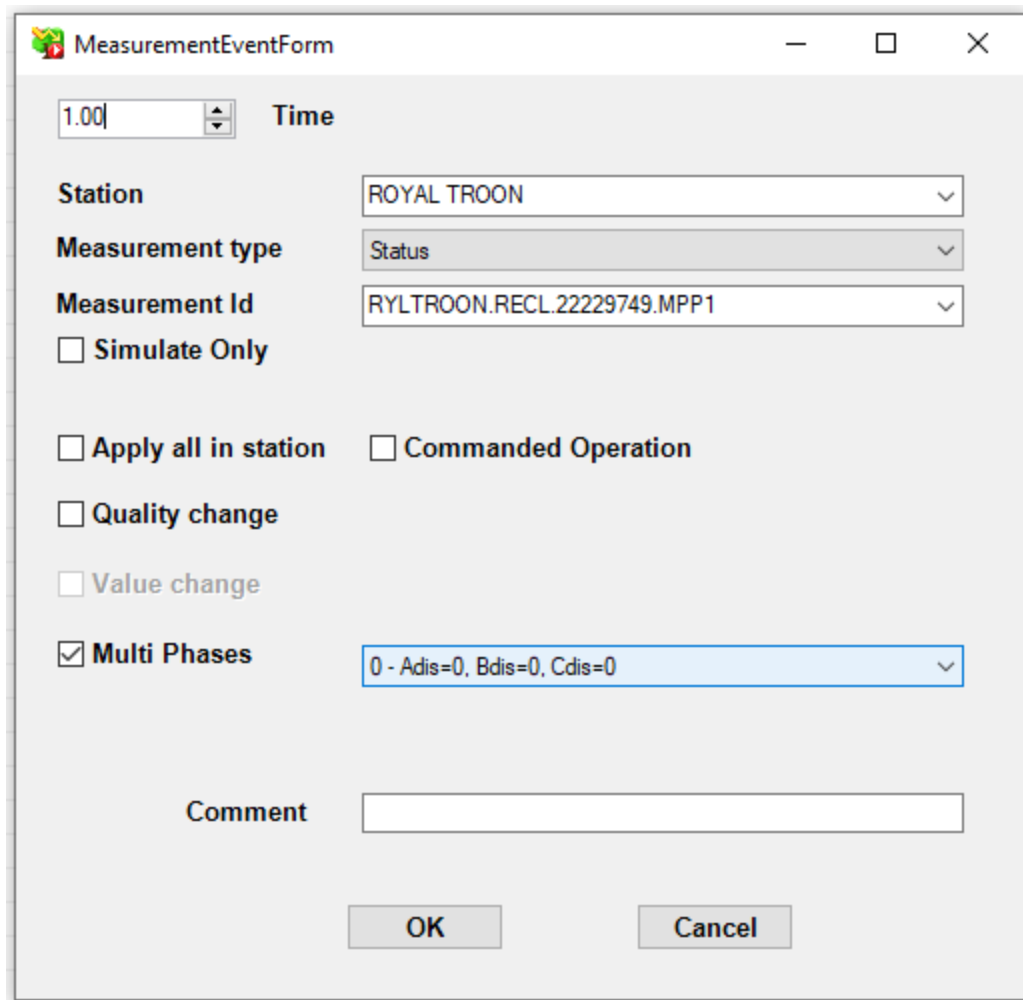
☒ Value change 0 - SDIS=0, XDIS=0

☐ Multi Phases

Comment

OK Cancel

Multi-phase measurement event:



MeasurementEventForm

1.00 Time

Station: ROYAL TROON

Measurement type: Status

Measurement Id: RYLTROON.RECL.22229749.MPP1

☐ Simulate Only

☐ Apply all in station ☐ Commanded Operation

☐ Quality change

☐ Value change

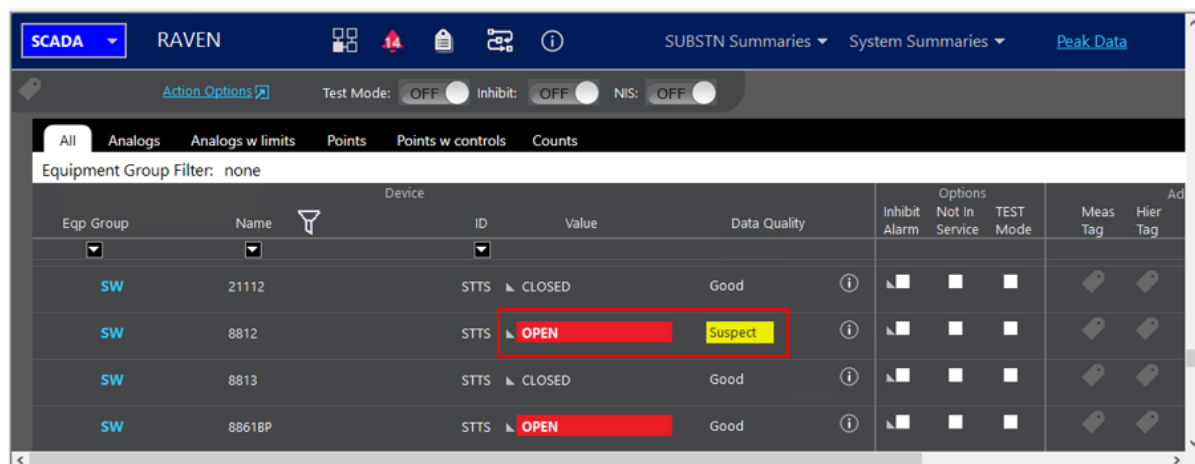
☒ Multi Phases: 0 - Adis=0, Bdis=0, Cdis=0

Comment:

OK Cancel

11. Click OK.

12. Run the event. Note that the device's Value and Data Quality values have changed.



SCADA RAVEN

Test Mode: OFF Inhibit: OFF NIS: OFF

Equipment Group Filter: none

Eqp Group	Name	Device	ID	Value	Data Quality	Inhibit Alarm	Not In Service	TEST Mode	Meas Tag	Hier Tag
SW	21112	STTS	CLOSED	Good						
SW	8812	STTS	OPEN	Suspect						
SW	8813	STTS	CLOSED	Good						
SW	8861BP	STTS	OPEN	Good						

4.3.3. Creating a Momentary Status Change Event

This section describes how to create a multiple change detection (MCD) event.

To include a "momentary" event signal for a Scada measurement on a Scada breaker, create a step with the momentary MDIS bit set:

1. On the Real-Time Network View display, select a substation and navigate to the Scada One Line display.
2. Click the rectangle with the substation name in the upper-left corner of the display.

The Station Tabular display appears.

3. Find a closed Scada breaker status point (for example, RAVEN.CB.8861F.STTS).
4. From the Event Scriptor, click the Add a Measurement Event  toolbar button.
5. In the Time field, select the time of the step (in seconds) from the start of the script.
6. Select the RAVEN substation from the Station drop-down list.
7. Select Status from the Measurement Type drop-down list.
8. Select the RAVEN.CB.8861F.STTS measurement from the Measurement ID drop-down list.
9. Select the Value Change check box, and set the value to "5 - SDIS=1, XDIS=0, MDIS=1", which is the "Close" command with the MDIS bit.

MeasurementEventForm

1.00 Time

Station RAVEN

Measurement type Status

Measurement Id RAVEN.CB.8861F.STTS

☐ Simulate Only

☐ Apply all in station ☐ Commanded Operation

☐ Quality change

☒ Value change 5 - SDIS=1, XDIS=0, MDIS=1

☐ Multi Phases

Comment

OK Cancel

10. Click OK.

11. Run the script.

Check that a momentary incident for the breaker 8861F is created, with the Current Customers count equal to zero.

The MDIS bit is the "trigger" for the OMS system to process an outage that may no longer exist. Consider the case where a breaker opens and then closes very quickly. By the time the OMS system runs, the breaker is closed and no customers are de-energized. However, with the MDIS bit set on a breaker measurement, the system knows that the switch was recently open and customers were de-energized. The system calculates the outage and marks the incident as Momentary with no currently affected customers.

Another way to simulate a momentary status change is to simulate a breaker reclosing cycle (Open, Close) within 3 seconds. However, the MDIS bit may not be sent on the measurement from the device. In addition, the OMS system may see the open switch and solve the outage normally (not like the momentary open/close case).

Note

The simulator does not allow different measurement events to be placed at the same time or within 1 second. In this case, one measurement is overwritten by another measurement before they are sent from the SIM server to the RT server.

4.3.4. Creating a Measurement Event for a Simulate-Only Point

Scada measurement points that do not correspond to an electrical network device attribute (for example, the open/close switch status) are called simulate-only, non-simulated, or generic measurements points. These Scada points may be not defined in the simulated power system model but are driven directly from the Event Scriptor (for example, a low battery alarm on a feeder automation switch).

Event Scriptor supports creating measurement events for the following simulate-only Scada points:

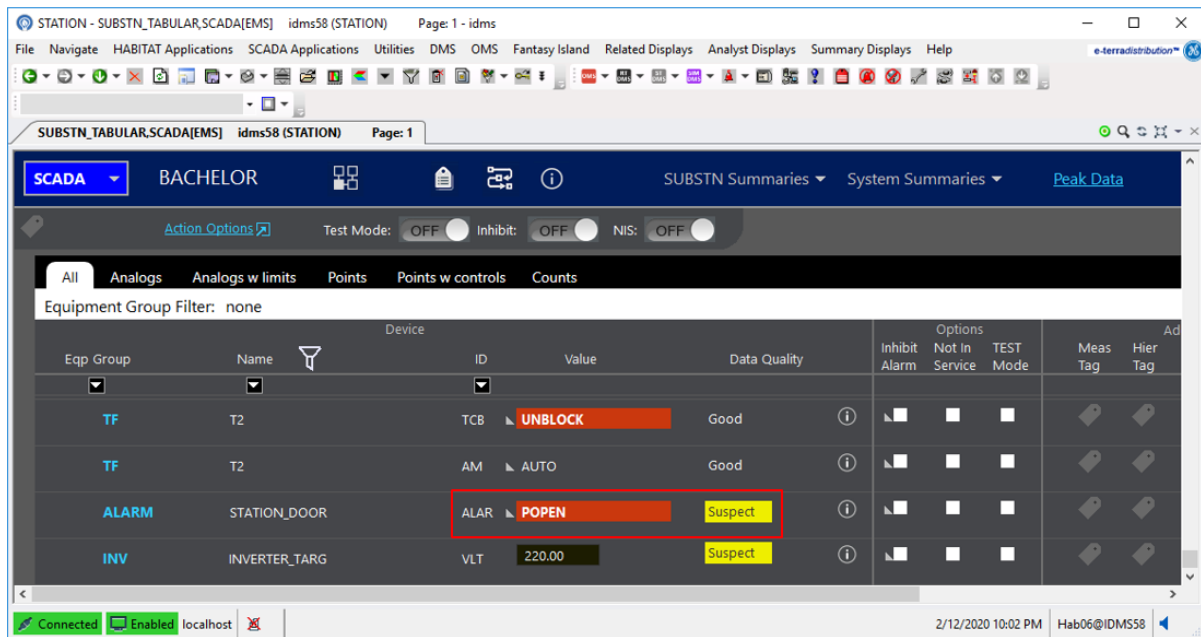
- Defined in the Substation Editor by setting the "Measurement: Simulate Only" property to True for the relevant Scada measurement. Such measurements have DMS mapping.
- Defined in the SCADAMOM database (shown on the Scada Station Tabular display). Such measurements may not have DMS mapping.

The example below demonstrates how to create a measurement event for a simulate-only status point (for example, a station door):

1. On the real-time network view display, select a substation and navigate to the Scada One Line display.
2. Click the rectangle with the substation name in the upper-left corner of the display.

The Station Tabular display appears.

3. Select a simulate-only status point (for example, BACHELOR.ALARM.STATION_DOOR.ALAR) and note its current state.

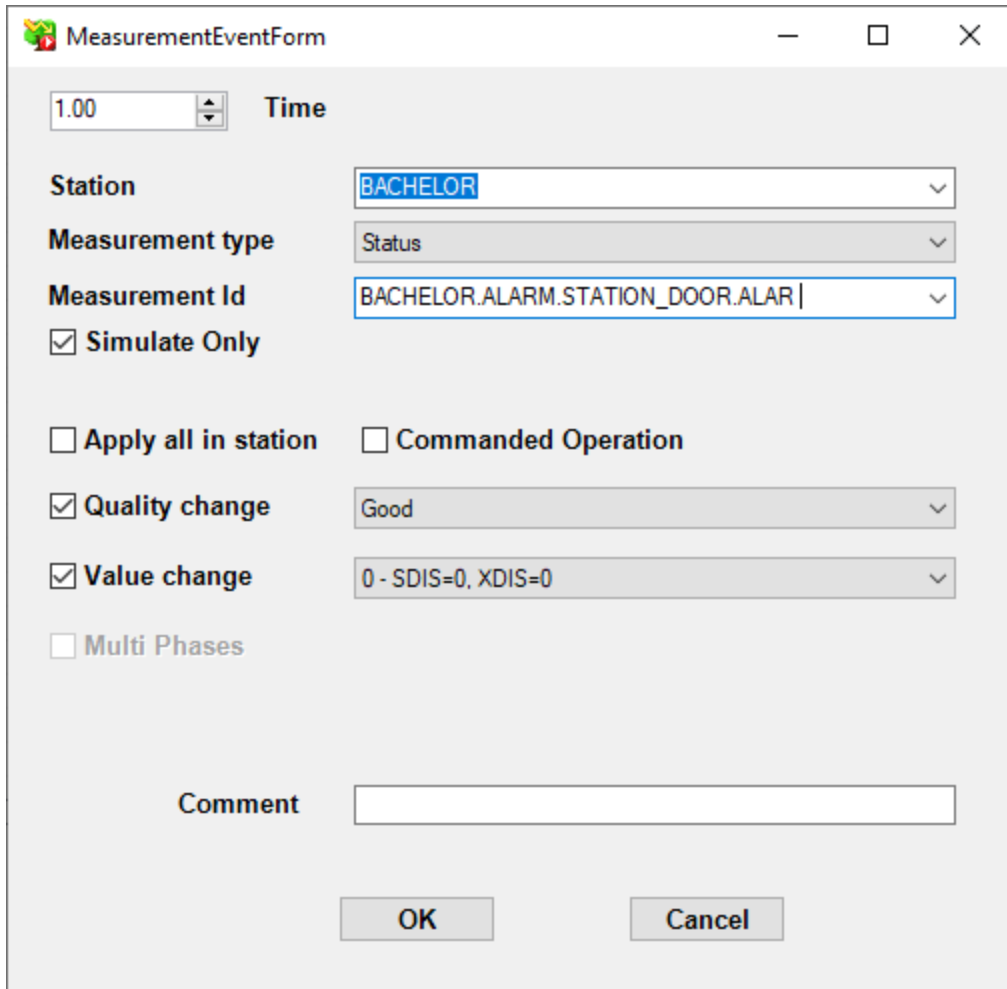


4. From the Event Scriptor, click the Add a Measurement Event toolbar button.
5. In the Time field, select the time of the step (in seconds) from the start of the script.
6. Select the substation from the Station drop-down list.
7. Select the Status type from the Measurement Type drop-down list.
8. Select the Simulate Only option.
9. Select the BACHELOR.ALARM.STATION_DOOR.ALAR measurement from the drop-down list.

Note

Selecting the Apply All in Station option applies the measurement event to all simulate-only measurement points in the station.

10. Select the Quality Change check box, and select the new type from the drop-down list.
11. Select the Value Change check box, and set the value to "1 - SDIS=1, XDIS=0", which is the "Close" command



The image shows a Windows-style dialog box titled "MeasurementEventForm". It contains several input fields and checkboxes. At the top left, there is a numeric spinner set to "1.00" with a "Time" label. Below this are three dropdown menus: "Station" (set to "BACHELOR"), "Measurement type" (set to "Status"), and "Measurement Id" (set to "BACHELOR.ALARM.STATION_DOOR.ALAR"). There are three checkboxes: "Simulate Only" (checked), "Apply all in station" (unchecked), and "Commanded Operation" (unchecked). Below these are two more dropdown menus: "Quality change" (set to "Good") and "Value change" (set to "0 - SDIS=0, XDIS=0"). A "Multi Phases" checkbox is also present and unchecked. At the bottom, there is a "Comment" text box and two buttons: "OK" and "Cancel".

1.00 Time

Station BACHELOR

Measurement type Status

Measurement Id BACHELOR.ALARM.STATION_DOOR.ALAR

☒ Simulate Only

☐ Apply all in station ☐ Commanded Operation

☒ Quality change Good

☒ Value change 0 - SDIS=0, XDIS=0

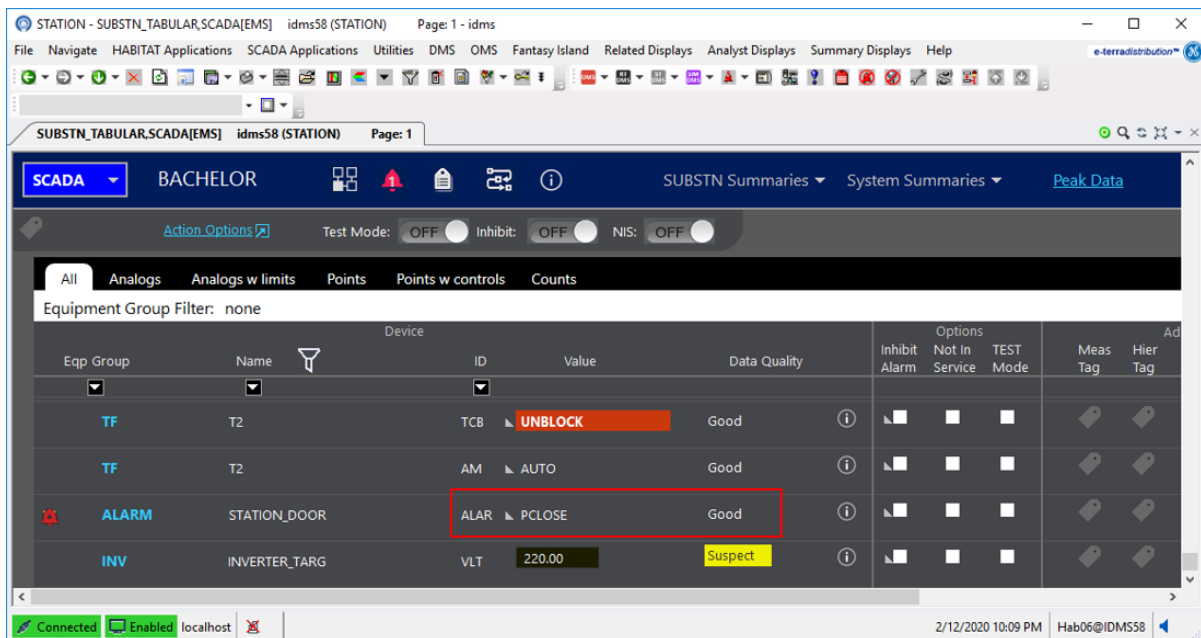
☐ Multi Phases

Comment

OK Cancel

12. Click OK.

13. Run the event. Note that the point's Value and Data Quality values changed.



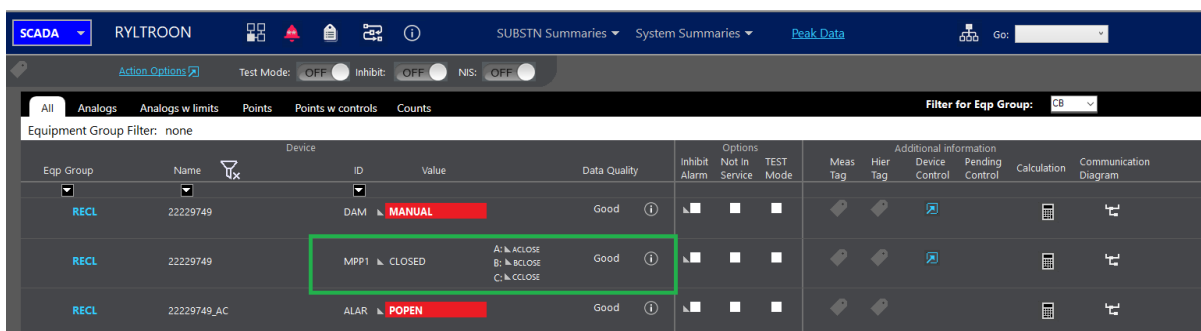
4.3.5. Creating a Multi-Phase Status Type Measurement Event

The example below demonstrates how to create a multi-phase measurement status and data quality change event for a switching device (for example, a recloser).

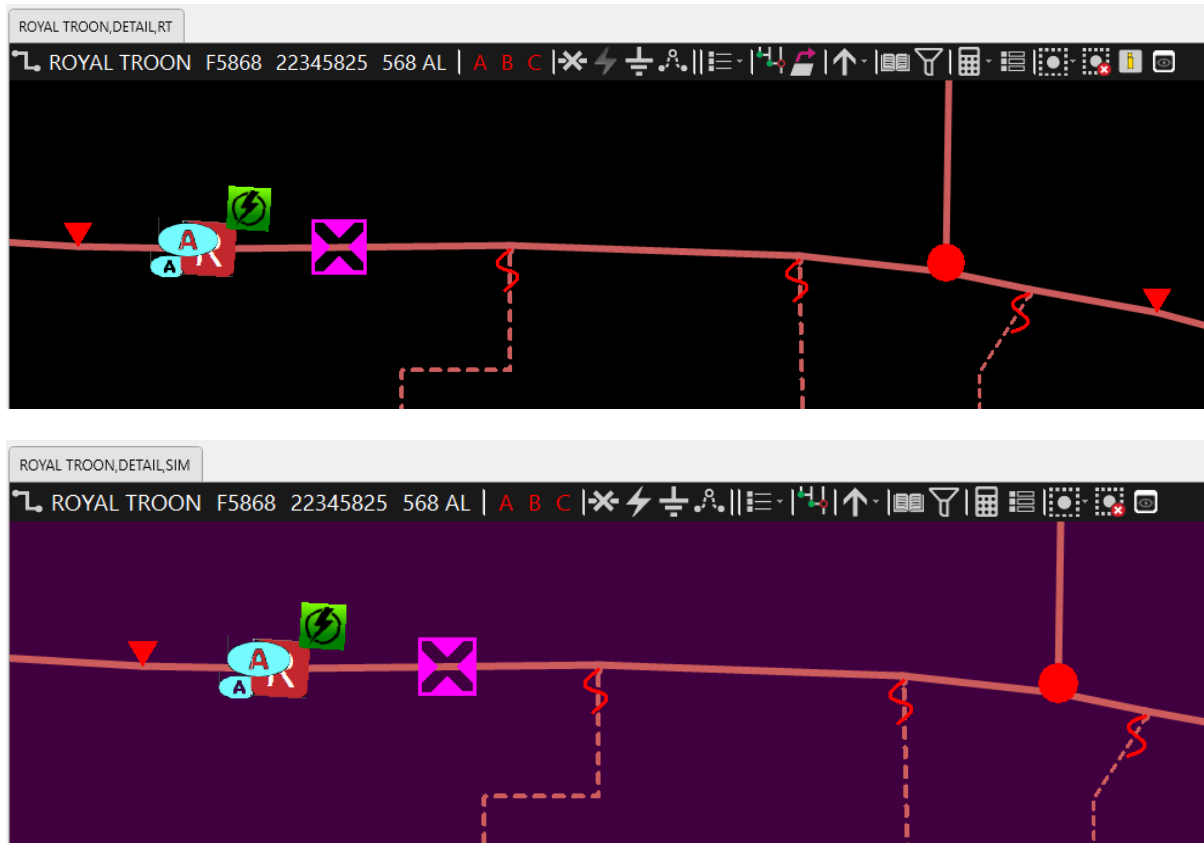
1. On the real-time network view display, select a substation (forexample, ROYAL TROON) and navigate to the Scada One Line display.
2. Click the rectangle with the substation name in the upper-left corner of the display.

The Station Tabular display appears.

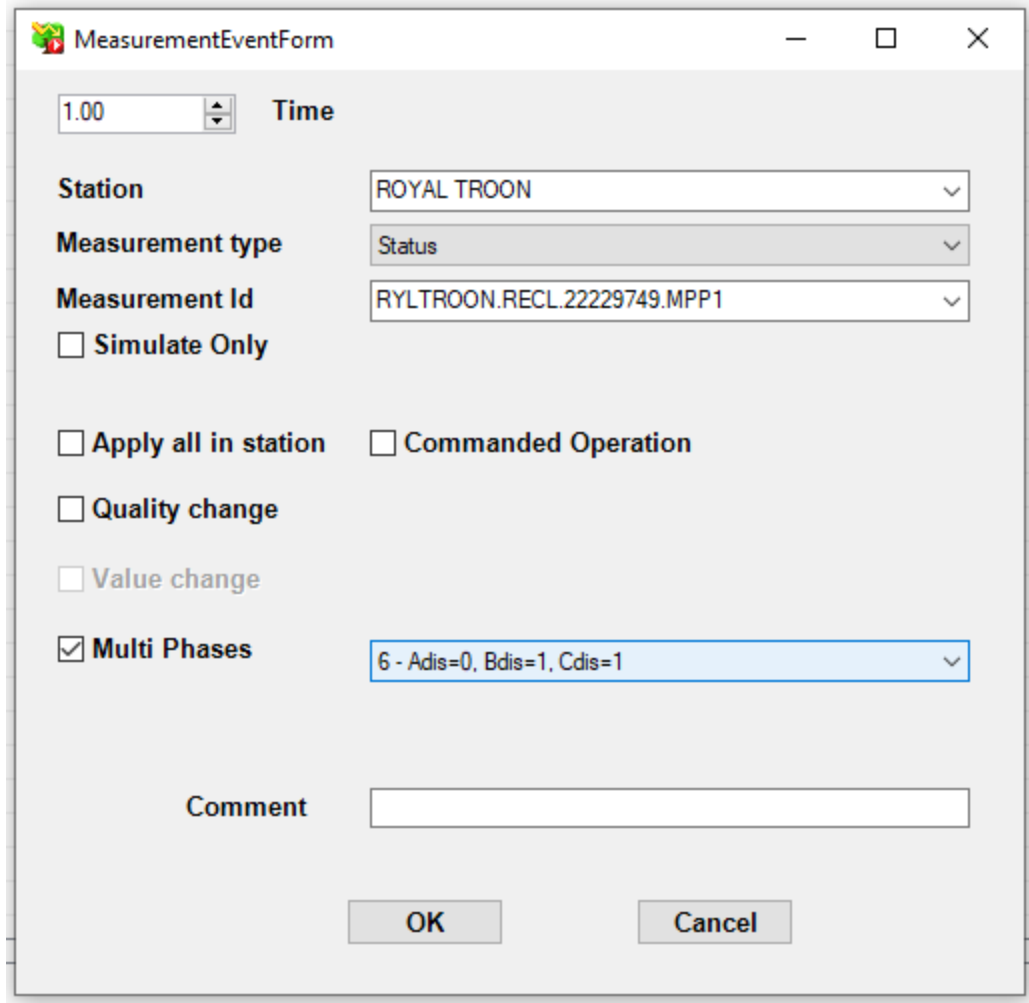
3. Select a multi-phase status point (for example, RYLTROON.RECL.22229749.MPP1) and note its current state.



Also make a note of surrounding real-time and simulator geographic views for the identified device (such as below: 22229749). The selected line (22345825) next to the recloser (22229749) has all phases (A, B, C) closed.



4. From the Event Scripter, click the Add a Measurement Event  toolbar button.
5. In the Time field, select the time of the step (in seconds) from the start of the script.
6. Select the ROYAL TROON substation from the Station drop-down list.
7. Select the Status type from the Measurement Type drop-down list.
8. Select the RYLTROON.RECL.22229749.MPP1 measurement from the drop-down list.
9. Select the "Multi Phases" check box. From the enabled drop-down select, option 6 as shown in the image below. (The A phase is opened whereas B and C phases are closed.)



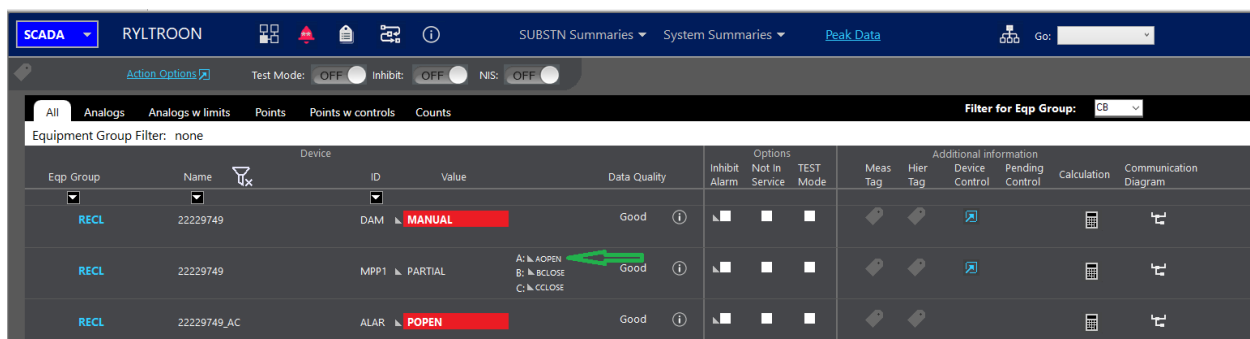
The MeasurementEventForm dialog box is shown with the following settings:

- Time:** 1.00
- Station:** ROYAL TROON
- Measurement type:** Status
- Measurement Id:** RYLTROON.RECL.22229749.MPP1
- ☐ Simulate Only
- ☐ Apply all in station
- ☐ Commanded Operation
- ☐ Quality change
- ☐ Value change
- ☒ Multi Phases: 6 - Adis=0, Bdis=1, Cdis=1
- Comment:** (empty text box)
- Buttons:** OK, Cancel

10. Click OK.

11. Run the script.

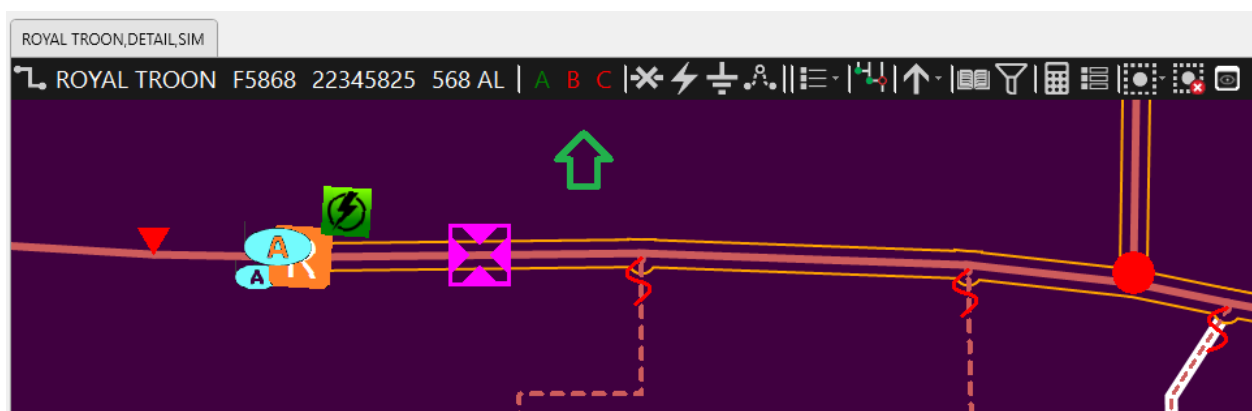
Note that the point status on SCADA UI is updated.



The SCADA UI screenshot shows the following table of equipment data:

Eqp Group	Name	Device	ID	Value	Data Quality	Inhibit Alarm	Options Not In Service	TEST Mode	Meas Tag	Hier Tag	Additional information Device Control	Pending Control	Calculation	Communication Diagram
RECL	22229749	DAM		MANUAL	Good									
RECL	22229749	MPP1	PARTIAL	A: AOPEN B: BCLOSE C: CCLOSE	Good									
RECL	22229749_AC	ALAR		POPEN	Good									

Also note the surrounding real-time and simulator geographic views of recloser 22229749. The selected downstream line, 22345825, has A phase deenergized.



☐ **Apply all in station** ☐ **Commanded Operation**
☒ **Quality change** Testmode

RECL	22229749	DAM	MANUAL	Good									
RECL	22229749	MPP1	PARTIAL	Good									
				Remote Testmode									
RECL	22229749_AC	ALAR	POPEN	Good									

ROYAL TROON - Measurement Data											
ID	Device Name	Device ID	Locate	Nominal Feeder	Energizin g Feeder	Type	Value	SCADA Quality	Measureme nt Validation (DNAP)	Phasing	Multi Phase
RYLTROON.RECL.22229749.MPP1	413141	22229749		F5868	F5868	Switch Status	Closed	Test Mode	Suspect	BC	Yes

ROYAL TROON - Measurement Data										
ID	Device Name	Device ID	Locate	Nominal Feeder	Energizing Feeder	Type	Value	SCADA Quality	Measurement Validation (DNAP)	Phasing
RVLTROON.RECL.22229749.MPP1	413141	22229749		F5868	F5868	Switch Status	Closed	Test Mode	Valid	BC
										Yes

4.4. Creating a Meter Measurement Event

The Event Scripter supports the creation of AMI measurement events, which allows you to simulate customer meter statuses, measurements, and signals, such as low voltage, return to normal, and high voltage.

You can set up a meter measurement event using the Meter Measurement form. To access the form:

1. From the Event Scripter, open the geographic view by clicking the Call Up a Geographic Display  button.
2. Select a customer, and then click the Add Meter Event button  on the upper toolbar.

The example below demonstrates how to create a measurement event for a customer AMI meter.

Figure 28. Event Scripter - Meter Measurement Form

The Meter Measurement form contains the following fields:

- **Time:** Select the time of the step in seconds from the start of the script.
- **Meter ID:** Customer meter ID. This field is populated automatically when adding a meter event from the geographic view.
- **Meter Event:** AMI meter voltage event.

- **Low Voltage:** The AMI meter reports low voltage. By default, this event is indicated with a brown house around the customer symbol.
- **Return to Normal:** The AMI meter reports nominal voltage. By default, this event is indicated with a cyan house around the customer symbol.
- **High Voltage:** The AMI meter reports high voltage. By default, this event is indicated with a yellow house around the customer symbol.
- **Manual Set:** Allows you to enter real power (Watt) and reactive power (VAR) measurements for phases A, B, and C. This event is not indicated on the network view.
- **Meter Status:** AMI meter status event.
 - **On:** The meter ping succeeded. By default, this event is indicated with a green circle around the customer symbol.
 - **Out:** The meter ping failed. By default, this event is indicated with a red circle around the customer symbol.
 - **Not Tried:** The meter ping is pending. By default, this event is indicated with a gray circle around the customer symbol.
 - **Undetermined:** The meter ping is undetermined. By default, this event is indicated with a gray circle around the customer symbol.
 - **Communication Error:** The meter had a communication error. By default, this event is indicated with a yellow circle around the customer symbol.
 - **Invalid Meter:** The meter is invalid. By default, this event is indicated with a yellow circle around the customer symbol.
 - **Expired:** The AMI meter ping request expired. By default, this event is indicated with a yellow circle around the customer symbol.
- **Volt (LN):** Linear voltage in phases A, B, and C in volts.
- **Watt:** Real power in phases A, B, and C in watts.
- **VAR:** Reactive power in phases A, B, and C in volt-ampere reactive.

In the figure below, meter voltage and meter status events are shown. The event shapes and colors are configurable by the Viewer Configuration Editor. For more information, refer to the *Viewer Configuration Editor User Guide*.

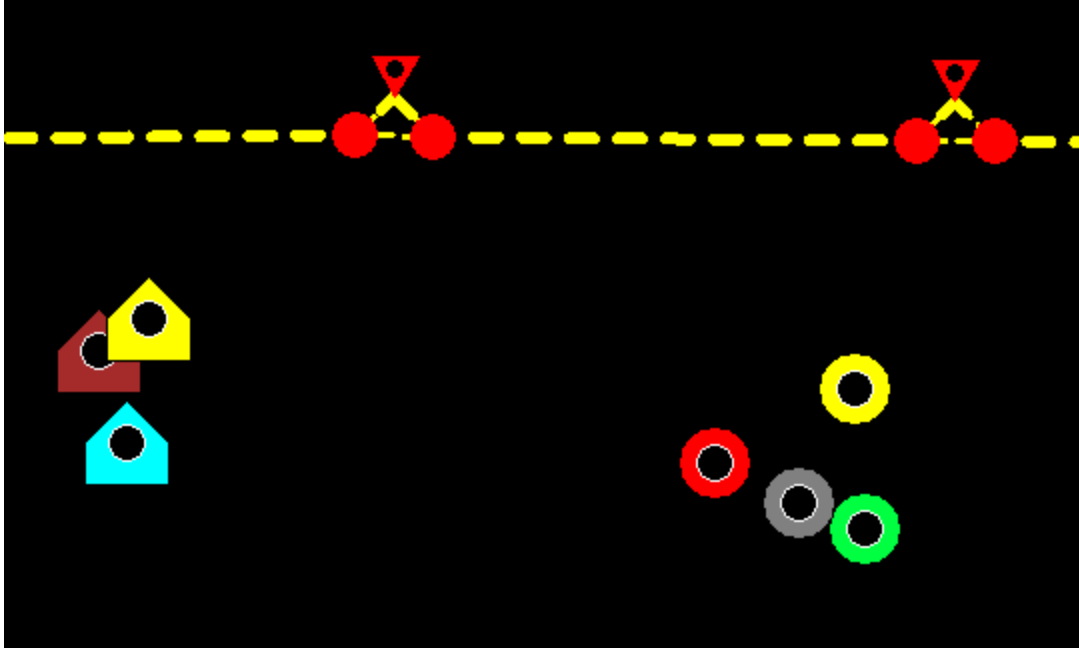


Figure 29. Meter Measurement Events

4.5. Creating a Switching Event

The Event Scripter supports the creation of a switching event. For example, you can set the switch status to Closed/Open, with or without a device failure.

You can set up a switching event using the Perform Switching Action form. To access the form:

1. From the Event Scripter, open the geographic view by clicking the Call Up a Geographic Display  button.
2. Select a switch, and then click the Add a Switching Step button  on the upper toolbar.

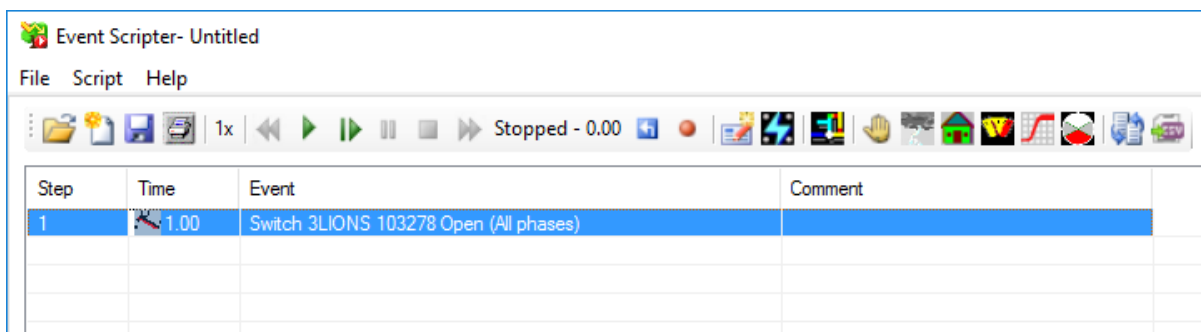
Figure 30. Event Scripter - Perform Switching Action Form

The Perform Switching Action form contains the following fields:

- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Station:** The name of the station.
- **ID:** Identification number of the selected switch.
- **Name:** Name of the selected switch.
- **Comment:** A comment, in free form.
- **Attribute:** Device attributes that can be modified.
 - **Switch State:** Changes the switch status to the selected value (Open/Close).
 - **Toggle Switch State:** Changes the switch status to the opposite status.
 - **Supervisory Cutout:** Changes the Supervisory Cutout state to the selected value (On/Off). This only applies to Scada controllable switches. If the Supervisory Cutout state is On, the switch is not controllable.
 - **Block Close:** Changes the Block Close state to the selected value (Block/Unblock). If the Block Close state is Block, the switch is blocked from closing action.
 - **Set Device Failure:** Changes the Device Failure state to the selected value (True/False). If the Device Failure state is True, the switch is not controllable.
 - **Switch Remote Controllable:** Changes the Remote Controllable state to the selected value (On/Off). If the Remote Controllable state is On, you can control the switch remotely. This only applies to controllable switches.
 - **Enable Ground Trip:** Changes the Ground Trip Enable state to the selected value

(Enable/Disable). The function is used to simulate ground trip protection on major switches, such as feeder breakers.

- **Enable Reclose:** Changes the Reclose Enable state to the selected value (Enable/Disable). This only applies to switches with reclosing capability such as breakers or reclosers. If reclosing is enabled and the simulator server is configured to simulate reclosing, the reclosing events follow the reclosing schedule defined in the NetServerSimConfig.txt file. For more information, see section [Fault Indicator, Recloser, Breaker, and Meter Parameters](#).
- **Enabled Scheme Auto:** Changes the Scheme Auto state to the selected value (Enable/Disable). If enabled, the switch can participate in an auto transfer scheme.
- **Switch Auto/Manual:** Changes the Auto/Manual status to the selected value (Auto/Manual). This defines whether the device operates in automatic or manual mode.
- **Toggle Pending/Permanent Status:** Changes the Pending/Permanent status to the opposite value. The Pending Permanent state indicates that a switching device currently has the "pending permanent" status. This means that the normal state of the switch has been changed, but the change has not yet been confirmed by an update of the GIS/AMS data. Network topology processing continues to use the existing normal state of the switch until the change is confirmed.
- **Breaker Lockout:** Changes the breaker or recloser lockout status to the selected value (Locked Out/Not Locked Out). The lockout status is considered by the Fault Isolation and Service Restoration application.
- **Block State Control:** Changes the Block State Control status to the selected value (Allow/Block). This defines whether device state controls are blocked.
- **Device Disabled:** Changes the Device Disabled status to the selected value (Enable/Disable). This defines whether the device is disabled for any controls except Enable. For example, the open/close and reclosing enable controls are disabled.
- **Phase(s):** Select phases to be switched (for example, A, AB, or ABC).
- **Value:** Select the value to be set for the selected attribute.
- **OK:** Click to add the event to the script.
- **Cancel:** Click to cancel adding the event.



4.6. Creating a Fault Event

You can create a fault event using the Event Scripter's Place Fault form. To access the form:

1. From the Event Scripter, open the geographic view by clicking the Call Up a Geographic Display  button.
2. Select a line or a transformer, and then click the Add a Fault Event button  on the upper toolbar.

Place Fault

0.00 Time

☒ Permanent
☐ Temporary 7 Time to Live
☐ Momentary

Fault Phases

☒ A
☒ B
☒ C
☐ G

☐ Trip Upstream Breaker/Recloser
☐ Skip Protective Devices 0
☐ Simulate Breaker Differential Relay
☐ Evaluate with Arc Suppression Coil (ASC)

0 - UNNR Reason

CURLEW Station

20535339 Component Name

344030854 Component ID

1 Per Unit Length

1 line DNOM Type

929125.6 740588.9 Placement X, Y

DETAIL Geometry Name

Comment

OK Cancel

Figure 31. Event Scripter - Place Fault Form

The Place Fault form contains the following fields:

- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Permanent:** Select to create a permanent fault event.
- **Temporary:** Select to create a temporary fault event. The temporary fault disappears after the

pre-configured period of time in the Time to Live field.

- **Momentary:** Select to create a momentary fault event. The momentary fault disappears immediately after the step is executed.
- **Time to Live:** Sets the time in seconds after which the temporary fault is removed. This only applies to the temporary fault event.
- **Phases:** Check phases from A, B, C, and G (ground) to place the fault on.
- **Trip Upstream Breaker/Recloser:** If checked, in addition to tripping the nearest protective devices, the upstream recloser or breaker is tripped when the fault event step is executed.
- **Skip Protective Devices:** This option allows skipping a specified number of protective devices. For example, if the fault is placed on a fused lateral, the fuse would normally operate. If Skip Protective Device "1" is selected, the fuse does not operate and the next upstream device (recloser or feeder breaker) operates. This option allows simulating more severe faults or faulty protective devices.
- **Simulate Breaker Differential Relay:** This option simulates a differential relay scheme between breakers.

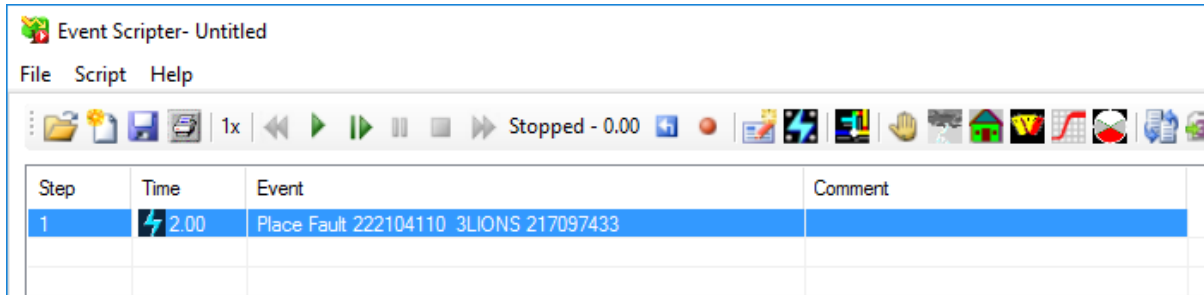
In a real world scenario, the input and exiting currents of a primary power transformer would be monitored for differences. If a discrepancy were detected, the upstream and downstream breakers would trip in order to protect the TF from a faulted or back flow condition.

When this option is chosen in the simulator and a fault is placed on a primary transformer in the station internals, the immediate upstream breaker and all immediate, normally connected downstream breakers will open.

- **Evaluate with Arc Suppression Coil (ASC):** This option simulates a station transformer Arc Suppression Coil (ASC) fault indicator event. This option is only available for lines that are associated with a power transformer with an arc suppression coil fault indicator connected to the secondary side.
- **Reason:** Select the reason of the fault. This reason is only shown on the Place Fault dialog and not passed to the OMS.
- **Station:** The station where the fault is located. This field is populated automatically.
- **Component Name:** The identification name of the network component where the fault is placed. This field is populated automatically.
- **Component ID:** The identification number of the network component where the fault is placed. This field is populated automatically.
- **Per Unit Length:** Length from the node of the branch. This field is populated automatically.
- **Placement X,Y:** The geographic coordinates of the fault location. This field is populated automatically.
- **Geometry Name:** The name of the geographic geometry. This field is populated automatically.
- **Comment:** Enter a free-form comment about the fault. This comment appears on the script list and is visible to the instructor only.
-

OK: Click to add the event to the script.

- **Cancel:** Click to cancel adding the event.



Note

You can also place faults on buses and power transformers from the Station Internals view via the device toolbar. These fault events will be added to the script if the Record mode is on.

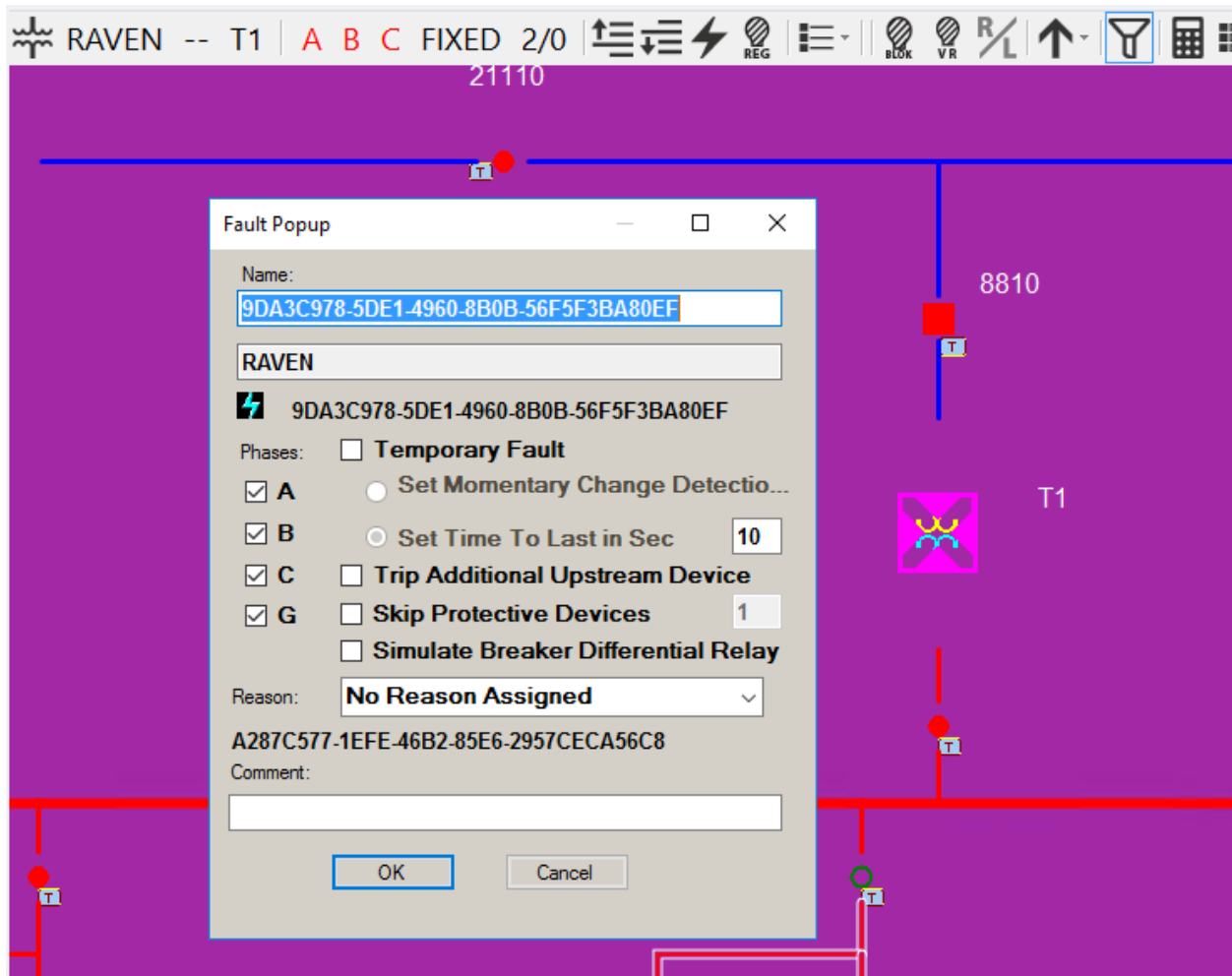


Figure 32. Placing a Fault from the Station Internals View

For the training simulator, ADMS supports the system-wide configuration of the fault indicator reset time. For more information, refer to section [Fault Indicator, Recloser, Breaker, and Meter Parameters](#).

4.7. Creating a Fault Indicator Event

Event Scriptor supports the creation of events for manual (non-telemetered) fault indicators. This allows you to change the switch indicator's status to Set (fault detected), Unset (no fault detected), or Unknown for the selected phases. When a manual fault indicator is set, a pink fault halo is shown around the line downstream from this fault indicator to the end point of the network or to the next Unset fault indicator.

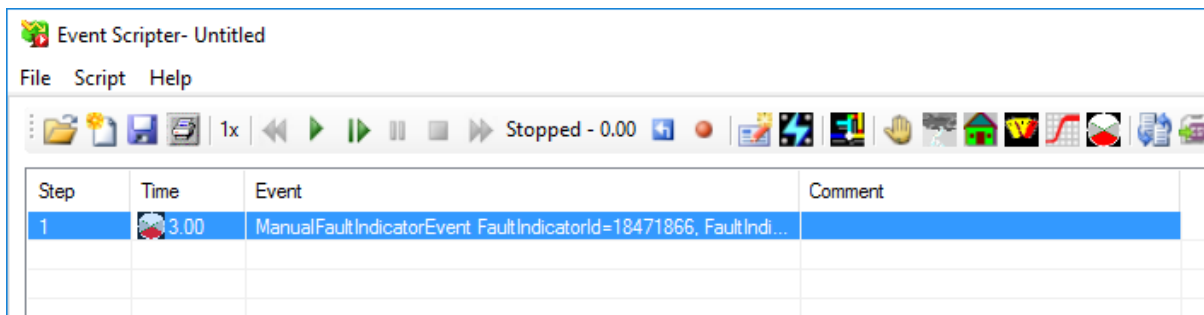
You can create a fault indicator event using the Event Scriptor's Manual Fault Indicator Event form. To access the form:

1. From the Event Scriptor, open the geographic view by clicking the Call Up a Geographic Display  button.
2. Select a fault indicator, and then click the Add a Manual Fault Indicator Event button  on the upper toolbar.

Figure 33. Event Scripter - Manual Fault Indicator Event Form

The Manual Fault Indicator Event contains the following fields:

- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Geometry Name:** The name of the geographic geometry that is used in the display call-up URLs.
- **Station Name:** The name of the station.
- **Fault Indicator ID:** Identification number of the selected fault indicator.
- **Fault Indicator Name:** Name of the selected fault indicator.
- **Comment:** A comment, in free form.
- **Phase:** Select phases from (A, B, C, or N) to show the fault halo.
- **OK:** Click to add the event to the script.
- **Cancel:** Click to cancel adding the event.



4.8. Creating a Customer Call Event

You can create a customer call event using the Event Scripter's Customer Event form.

If you know the customer's account, name, and other details, you can access the form via the Event Scripter's Add a Call Event toolbar button  on the upper toolbar.

Note

Customer calls and AMI Power Out events cannot be created for customers that are connected to future equipment.

Note

Do not implement more than 20 customer call events per second using the Event Scripter. This causes performance issues.

CustomerEventForm

Time: 1.00

Customer Event:

- ☒ Call
- ☐ AMI Power Out Report
- ☐ AMI Power On Report

Customer Name: _____

Account Number: _____

Customer Key: _____

Placement X, Y: 0 0

Call Type: Dim/Bright Lights

Condition 1: None

Condition 2: None

Description: None

Operating Center: _____

Comment: _____

Call Back Number: _____

☐ Send Power Is On
☐ Keep Non-outage Separate

Meter Number: _____

Time Offset (sec): 0

OK Cancel

Figure 34. Event Scripter - Customer Event Form

The Customer Event form includes the following fields:

- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Customer Event:** Select the customer event type:
 - **Call:** Select to create the customer call event type.
 - **AMI Power Out Report:** Select to create the automated meter report about outage.
 - **AMI Power On Report:** Select to create the automated meter report about power restoration.
- **Customer Name:** The customer's name.
- **Account Number:** The customer's account number.

Note

For customers that have a non-unique account number, use the customer key to create customer call events.

- **Customer Key:** The customer's unique key number.
- **Placement X,Y:** The geographic coordinates of the customer location. This field is populated automatically.
- **Call Type:** Selects the type of the trouble (for example, Lights Out, Area Outage, Blinking Lights, and so on).
- **Condition 1:** Describes the first trouble code condition (for example, wires down, wires burning, and so on).
- **Condition 2:** Describes the second trouble code condition (for example, pole on fire, pole leaning, and so on).
- **Description:** Displays the trouble code description, if any (for example, pole/house, loud noise, and so on).
- **Operating Center:** Service center the customer belongs to.
- **Comment:** Enter a free-form description of the trouble.
- **Call Back Number:** Enter the customer's callback number.
 - **Send Power is On:** Sends a power on message via callback.
 - **Keep Non-Outage separate:** A separate non-outage incident is created for the same call type.

Note

This applies only to calls that result in non-outage incidents.

- **Meter Number:** The customer's meter ID. This field is populated automatically.
- **OK:** Click to add the customer call event to the script.
- **Cancel:** Click to cancel adding the event.

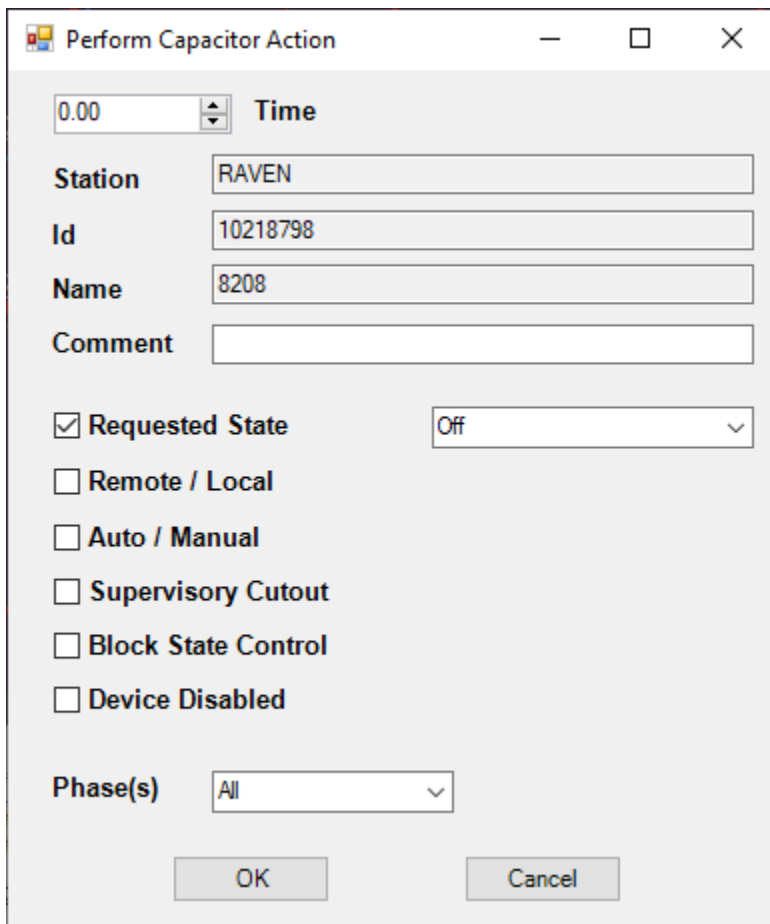
4.9. Creating a Device Input Event

The Event Scripter supports the creation of events to modify parameters of substations, power and step-up transformers, regulators, feeders, and capacitors.

4.9.1. Creating a Capacitor Input Event

You can set up a capacitor input event using the Perform Capacitor Action form. To access the form:

1. From the Event Scripter, open the geographic view by clicking the Call Up a Geographic Display  button
2. Select a capacitor, and then click the Add Capacitor Event button  on the upper toolbar.



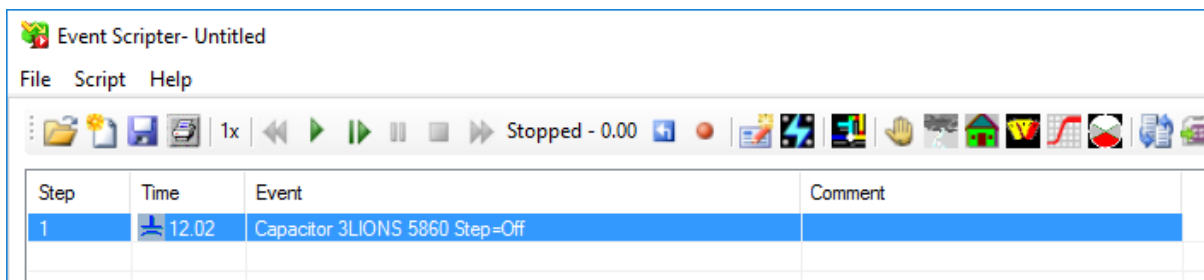
The image shows a screenshot of the 'Perform Capacitor Action' dialog box. It has a title bar with a standard Windows icon, a minus sign, a maximize button, and a close button. The dialog contains several input fields and checkboxes. At the top, there is a 'Time' field with a spinner set to '0.00'. Below this are four text input fields: 'Station' (containing 'RAVEN'), 'Id' (containing '10218798'), 'Name' (containing '8208'), and 'Comment' (empty). Below the text fields are five checkboxes: 'Requested State' (checked), 'Remote / Local', 'Auto / Manual', 'Supervisory Cutout', and 'Block State Control'. To the right of the 'Requested State' checkbox is a dropdown menu currently showing 'Off'. Below these checkboxes is another dropdown menu labeled 'Phase(s)' with 'All' selected. At the bottom of the dialog are two buttons: 'OK' and 'Cancel'.

Figure 35. Event Scripter - Perform Capacitor Action Form

The Perform Capacitor Action form includes the following fields:

- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Station:** The name of the substation.
- **ID:** Identification number of the selected capacitor.

- **Name:** Name of the selected capacitor.
- **Comment:** A comment, in free form.
- **Requested State:** When checked, the event changes the capacitor status to the selected value (On/Off).
- **Remote/Local:** When checked, you can select local or remote regulation.
- **Auto/Manual:** When checked, you can select to disable (Manual) or enable (Auto) regulation.
- **Supervisory Cutout:** When checked, you can turn on or off the supervisory cutout.
- **Block State Control:** Changes the Block State Control status to the selected value (Allow/Block). This defines whether device state controls are blocked.
- **Device Disabled:** Changes the Device Disabled status to the selected value (Enable/Disable). This defines whether the device is disabled for any controls except Enable. For example, the on/off and auto/manual controls are disabled.
- **OK:** Click to add the event to the script.
- **Cancel:** Click to cancel adding the event.



4.9.2. Creating a Transformer Input Event

You can set up a power transformer input event using the Perform Transformer Action form. To access the form:

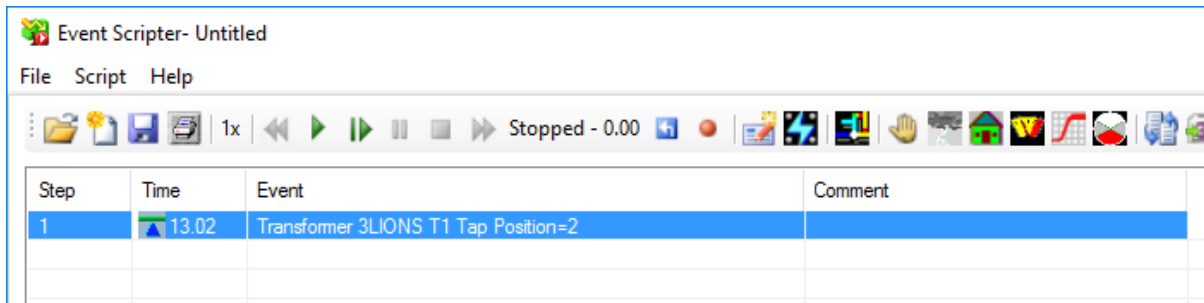
1. From the Event Scripter, open the geographic view by clicking the Call Up a Geographic Display  button.
2. Select a regulating, power, or step-up transformer, and then click the Add Transformer Input Event button  on the upper toolbar.

Figure 36. Event Scriptor - Perform Transformer Action Form

The Perform Transformer Action form includes the following fields:

- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Station:** The name of the substation.
- **ID:** Identification number of the selected transformer.
- **Name:** Name of the selected transformer.
- **Comment:** A comment, in free form.
- **Remote Enabled:** When selected, you can select local or remote regulation.
- **Regulation On (Auto/Manual):** When selected, you can select to disable (manual) or enable (auto) regulation.
- **Regulation Blocked:** When selected, you can select (True/False) whether to block the transformer from regulation.
- **Supervisory Cutout:** When selected, you can turn on or off the supervisory cutout.
- **Block State Control:** When selected, you can set the Block State Control status to the selected value (Allow/Block). This defines whether device state controls are blocked.

- **Disable Device:** When selected, you can set the Device Disabled status to the selected value (Enable/Disable). The Disable status specifies that the normal device functionality is disabled and the device is disabled for any controls except Enable. For example, the tap position raise/lower controls are disabled.
- **Tap Position:** When selected, you can enter a new tap position value.
- **OK:** Click to add the event to the script.
- **Cancel:** Click to cancel adding the event.



4.9.3. Creating a Substation and Feeder Input Event

A substation and feeder input event is created via the Substation Event form. This event is used to manually override substation and feeder input parameters used by DNAF. To access the form:

1. From the Event Scripter, open the geographic view by clicking the Call Up a Geographic Display  button.
2. Select a desired substation, and then click the Add a Substation/ Feeder Input Event button  on the upper toolbar.

Substation Event

Time: 1.00 DETAIL

Station Name: WINGED FOOT

Station ID: 8166757

Comment: [Empty text area]

☐ Regulators Blocked ☒ True ☐ False

☐ Voltage Reduction Enabled ☒ True ☐ False

☐ DNAF Enabled ☒ True ☐ False

☐ DSO Enabled ☒ True ☐ False

☐ DER SCADA Measurement Quality Valid ☒ True ☐ False

☐ Manual Temperature 0.00

☐ Manual Frequency 0.00

☐ Manual LVM Minimum Power Factor Target -1.00

☐ Manual LVM Maximum Power Factor Target 1.00

Feeder Parameters

Feeder Name: F7866 (selected), F7864, F7863, F7861, F7865

☐ Manual Load Scaling Factor 1.00

☐ Manual LVM Minimum Power Factor Target -1.00

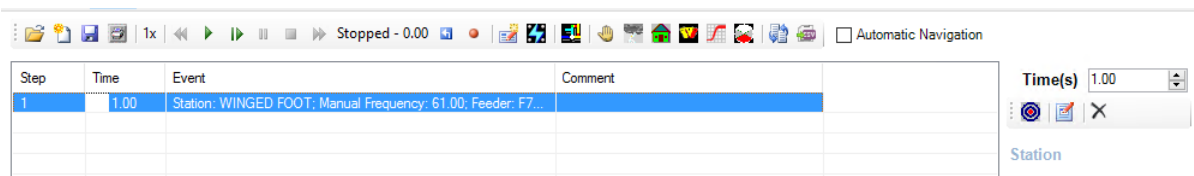
☐ Manual LVM Maximum Power Factor Target 1.00

OK Cancel

Figure 37. Event Scripter - Substation Event Form

The Substation Event form includes the following fields:

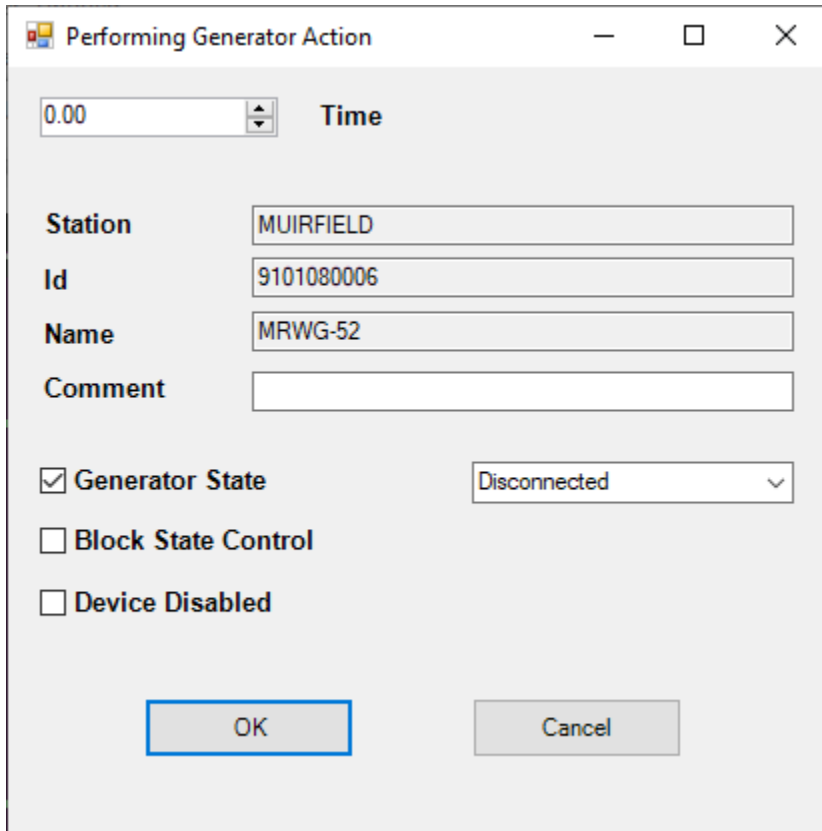
- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Geometry Name:** The name of the geographic geometry that is used in the display call-up URLs. In the example above, Geometry Name is "DETAIL".
- **Station Name:** Name of the selected substation.
- **Station ID:** Identification number of the selected substation.
- **Comment:** A comment, in free form.
- **Regulators Blocked:** When checked, voltage regulation for the substation is blocked.
- **Voltage Reduction Enabled:** When checked, voltage reduction for the substation is enabled.
- **DNAF Enabled:** When checked, basic DNAF functions (DPF, FL, and PRV) for the substation are enabled.
- **DSO Enabled:** When checked, advanced DNAF functions (AFR, FISR, and LVM) for the substation are enabled.
- **DER Scada Measurement Quality Valid:** When selected, the DER Scada measurement quality is valid.
- **Manual Temperature:** Manual override of the temperature (in °F) for the substation.
- **Manual Frequency:** Manual override of the frequency (in Hz) for the substation.
- **Manual LVM Minimum Power Factor Target:** Manual override of the LVM minimum power factor target for the substation. Power factor targets must be between -1.00 and +1.00.
- **Manual LVM Maximum Power Factor Target:** Manual override of the LVM maximum power factor target for the substation. Power factor targets must be between -1.00 and +1.00.
- **Feeder Parameters – Feeder Name:** Feeder name for the feeder parameters below. Only one feeder may be selected per event. To specify parameters for multiple feeders, create multiple events.
- **Feeder Parameters – Manual Load Scaling Factor:** Manual override of the per-unit load scaling factor for the selected feeder. A load scaling factor of 1.00 means no load scaling.
- **Feeder Parameters – Manual LVM Minimum Power Factor Target:** Manual override of the LVM minimum power factor target for the selected feeder. Power factor targets must be between -1.00 and +1.00.
- **Feeder Parameters – Manual LVM Maximum Power Factor Target:** Manual override of the LVM maximum power factor target for the selected feeder. Power factor targets must be between -1.00 and +1.00.
- **OK:** Click to add the event to the script.
- **Cancel:** Click to cancel adding the event.



4.9.4. Creating a Generator Input Event

You can set up a generator input event using the Performing Generator Action form. To access the form:

1. From the Event Scripter, open the geographic view by clicking the Call Up a Geographic Display  button.
2. Select a generator, and then click the Add Generator Event button  on the upper toolbar.



The image shows a dialog box titled "Performing Generator Action". It contains the following fields and controls:

- Time:** A numeric input field with the value "0.00" and a "Time" label.
- Station:** A text input field containing "MUIRFIELD".
- Id:** A text input field containing "9101080006".
- Name:** A text input field containing "MRWG-52".
- Comment:** An empty text input field.
- Generator State:** A checked checkbox followed by a dropdown menu currently showing "Disconnected".
- Block State Control:** An unchecked checkbox.
- Device Disabled:** An unchecked checkbox.
- Buttons:** "OK" and "Cancel" buttons at the bottom.

Figure 38. Event Scripter - Perform Capacitor Action Form

The Performing Generator Action form includes the following fields:

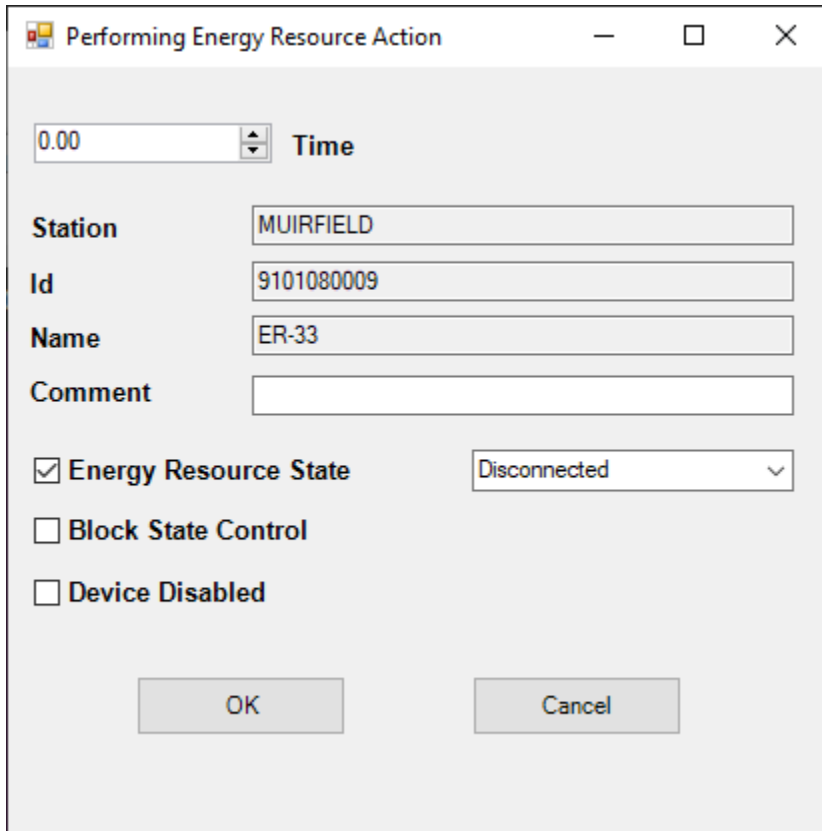
- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Station:** The name of the substation.
- **ID:** Identification number of the selected generator.
- **Name:** Name of the selected generator.
- **Comment:** A comment, in free form.
- **Generator State:** When checked, the event changes the generator status to the selected value (Connected/Disconnected).
- **Block State Control:** Changes the Block State Control status to the selected value (Allow/Block). This defines whether device state controls are blocked.

- **Device Disabled:** Changes the Device Disabled status to the selected value (Enable/Disable). This defines whether the device is disabled for any controls except Enable. For example, the connect/disconnect and power factor setpoint controls are disabled.
- **OK:** Click to add the event to the script.
- **Cancel:** Click to cancel adding the event.

4.9.5. Creating an Energy Resource Input Event

You can set up an energy resource input event using the Performing Energy Resource Action form. To access the form:

1. From the Event Scripter, open the geographic view by clicking the Call Up a Geographic Display  button.
2. Select an energy resource, and then click the Add Generator Event button  on the upper toolbar.



The image shows a screenshot of the 'Performing Energy Resource Action' dialog box. It has a title bar with a standard Windows icon and window controls. The form contains the following fields and controls:

- Time:** A numeric input field with '0.00' and a spinner control.
- Station:** A text input field containing 'MUIRFIELD'.
- Id:** A text input field containing '9101080009'.
- Name:** A text input field containing 'ER-33'.
- Comment:** An empty text input field.
- Energy Resource State:** A checked checkbox followed by a dropdown menu currently showing 'Disconnected'.
- Block State Control:** An unchecked checkbox.
- Device Disabled:** An unchecked checkbox.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom.

Figure 39. Event Scripter - Performing Energy Resource Action Form

The Performing Energy Resource Action form includes the following fields:

- **Time:** The time in seconds from the start of the script when the event is triggered.
- **Station:** The name of the substation.

- **ID:** Identification number of the selected energy resource.
- **Name:** Name of the selected energy resource.
- **Comment:** A comment, in free form.
- **Energy Resource State:** When checked, the event changes the energy resource status to the selected value (Connected/Disconnected).
- **Block State Control:** Changes the Block State Control status to the selected value (Allow/Block). This defines whether device state controls are blocked.
- **Device Disabled:** Changes the Device Disabled status to the selected value (Enable/Disable). This defines whether the device is disabled for any controls except Enable. For example, the connect/disconnect and power factor setpoint controls are disabled.
- **OK:** Click to add the event to the script.
- **Cancel:** Click to cancel adding the event.

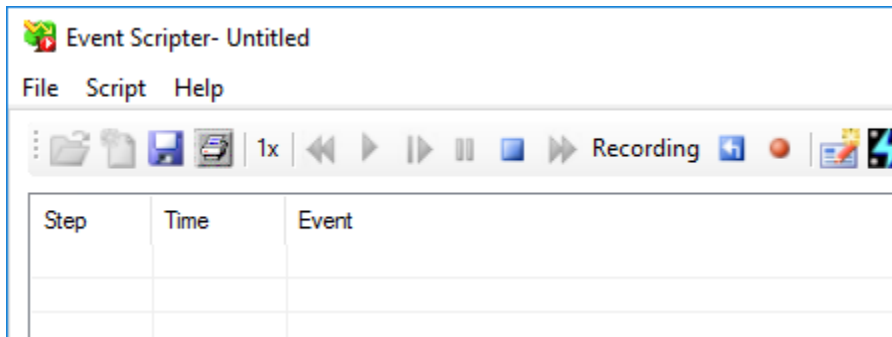
4.10. Recording Actions on the Event Scripter's Network View

Using the Record button  on the Script tab toolbar, you can record and add all your actions to a script on the Event Scripter's Network view.

To start recording:

1. Click the Record button  on the Script tab toolbar.

The status field on the main toolbar changes to "Recording". The Event Scripter is now in the recording mode.

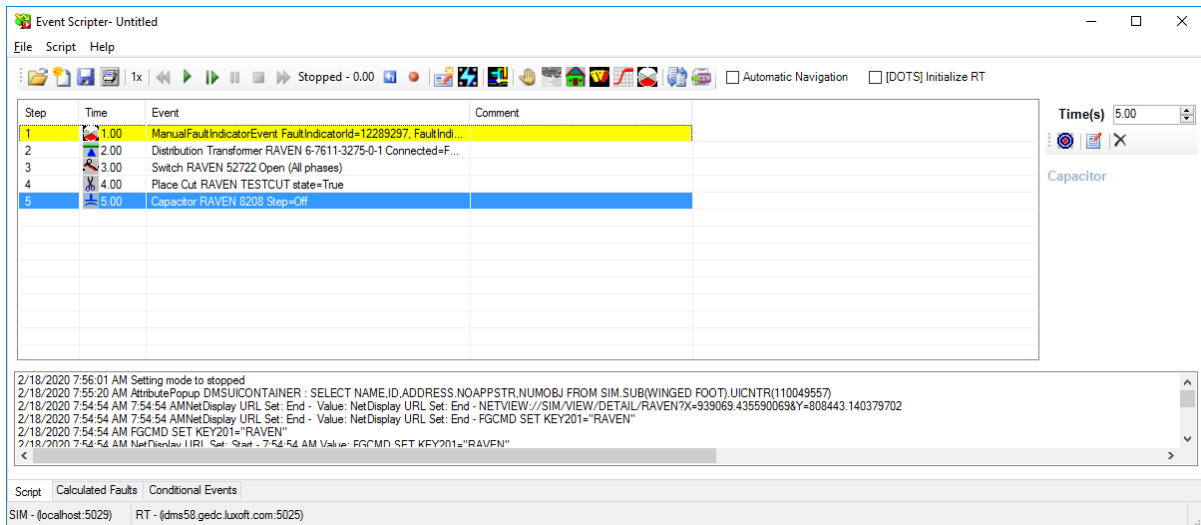


2. Click the Geographic View button  on the main toolbar.

The Event Scripter's Geographic View opens.

3. Select a substation.
4. Perform the actions that you need to record into the scripts (for example, open a switch, enter device parameters, place a cut, set a fault indicator, and so on).

All performed actions appear in the script. In the recording mode, you can also manually add steps using the event buttons on the Network View, or by using the Insert option of the right-click menu.



5. To stop recording, click the Stop  button.

4.11. Using the Fault Calculator for Creating Scripts

The fault calculator generates a list of faults for the various scenarios that can be used in the event scripts. For example, if you are creating a major storm training scenario, the tool generates a set of faults caused by the storm.

The Calculated Faults tab consists of three sections:

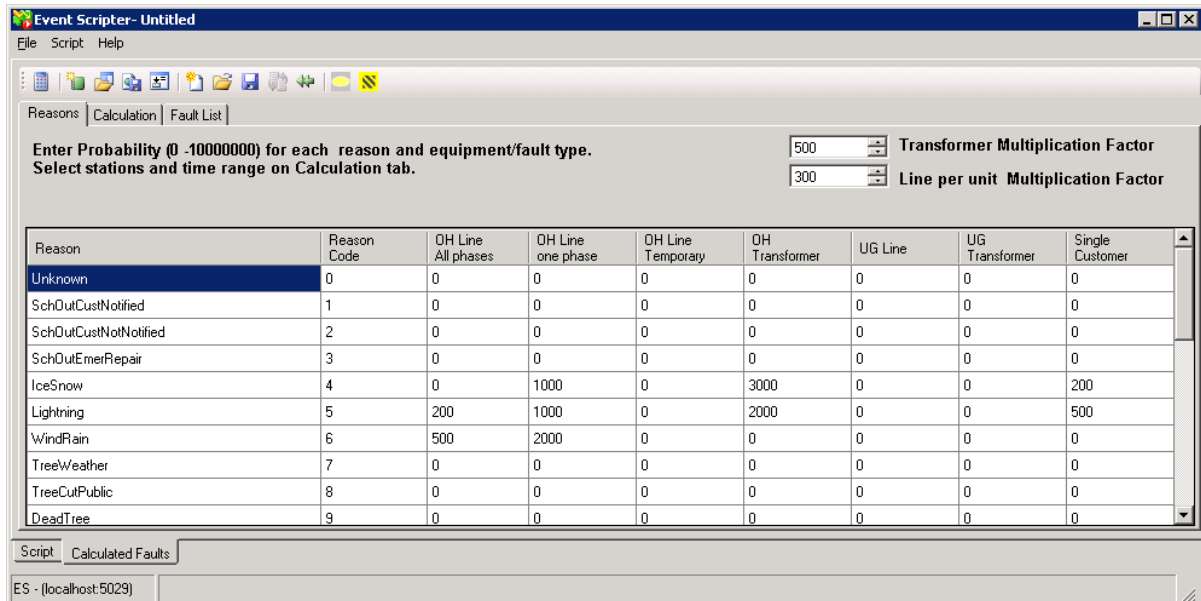
- **Reasons:** Allows you to enter the probability that a particular reason will cause faults on each equipment type. For example, an ice/snow storm (reason) may cause a much higher rate of faults on overhead line faults (equipment type) than on underground devices. A probability value of 100 roughly equates to one fault per hour in each of the selected stations.

Getting the desired result requires some model variations. Two multiplication factors affect the rate of faults applied in the probability table. These factors are shown at the top of the display: Transformer Multiplication Factor and Line per unit Multiplication Factor.

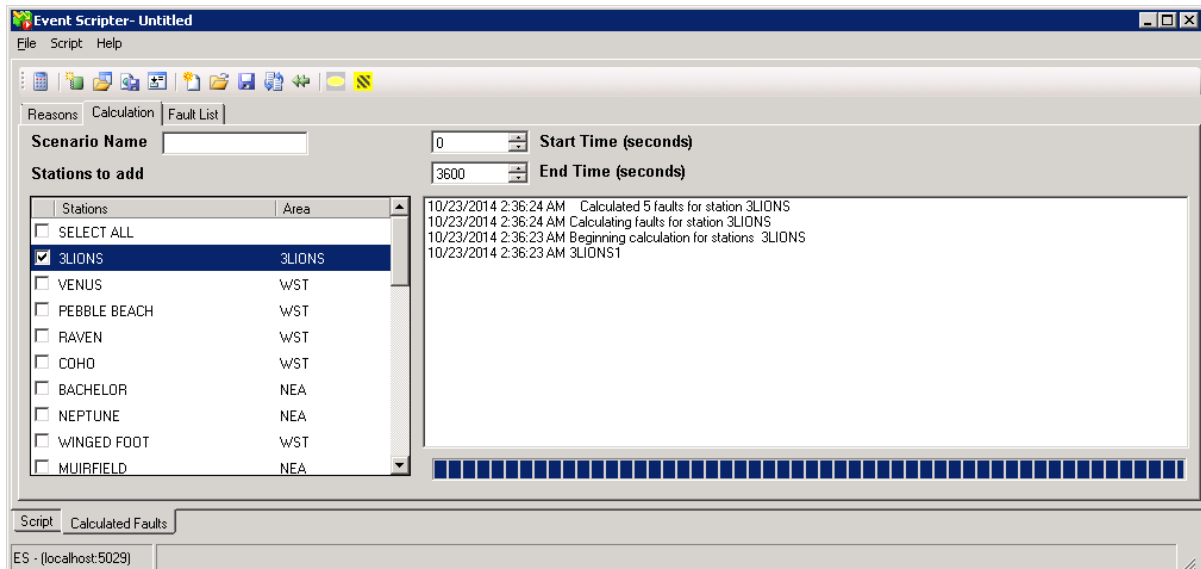
- The Transformer Multiplication Factor controls the rate at which faults are applied to a section of line according to the number of transformers. The factor is expressed as the number of transformers per fault. Therefore, the smaller the multiplication factor, the greater the number of faults applied.
- The Line per Unit Multiplication Factor represents the rate at which faults are applied per unit length of line. The factor is expressed as the number of units of line length per fault. Therefore, the smaller the multiplication factor, the greater the number of faults applied.

These factors are normally set by experimentation to match the overall configuration of the

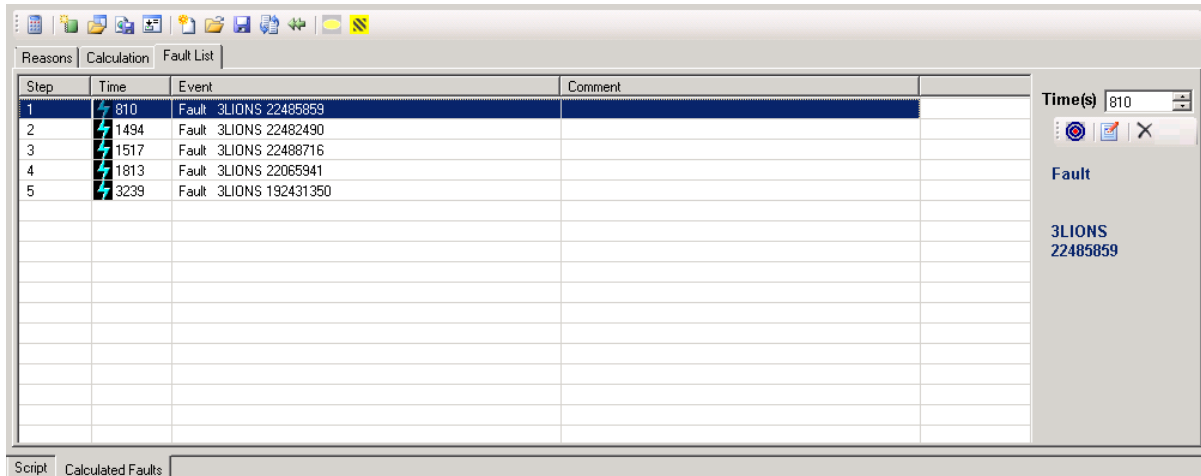
substation models and the desired fault rated in the probability table. Once appropriate multiplication factors have been set, further changes should not be required.



- **Calculation:** Allows you to select the stations at which the faults will be placed, and the time range of the scenario. It also shows a log of the calculation process. By setting different time ranges and/or stations, you can combine multiple runs of the Fault Calculator to generate a larger scenario such as a storm moving across a service area.



- **Fault List:** Shows the list of faults generated by the calculator and allows you to edit the faults using the editing toolbar of the Event Scriptor.



You can use the tabs below the Event Scriptor toolbar to switch among these sections.

After you finish calculating faults, you can save the fault list as an independent script, create a scenario, or merge it with the main script opened on the Script tab.

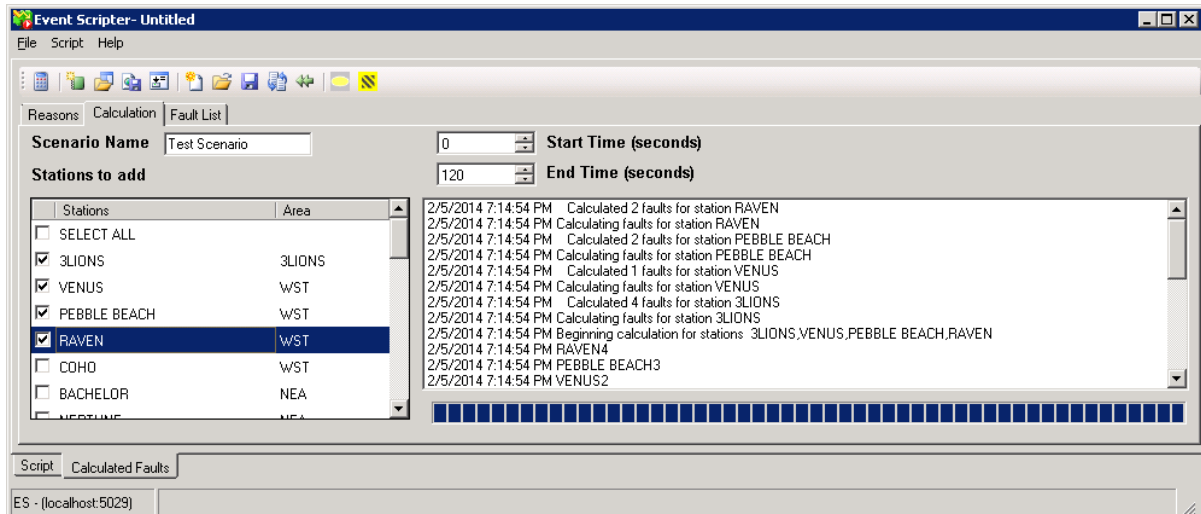
4.11.1. Creating a Fault Scenario

To calculate faults:

1. Open the Event Scriptor, and select the Calculated Faults tab at the bottom of the window.
2. Select the Calculation tab.
3. Enter a scenario name.
4. Check stations as needed.
5. Enter a start and/or end time in seconds.
6. Select the Reasons tab.
7. Enter the values in the Reasons table to show the probability of a particular reason causing faults on specific types of equipment.

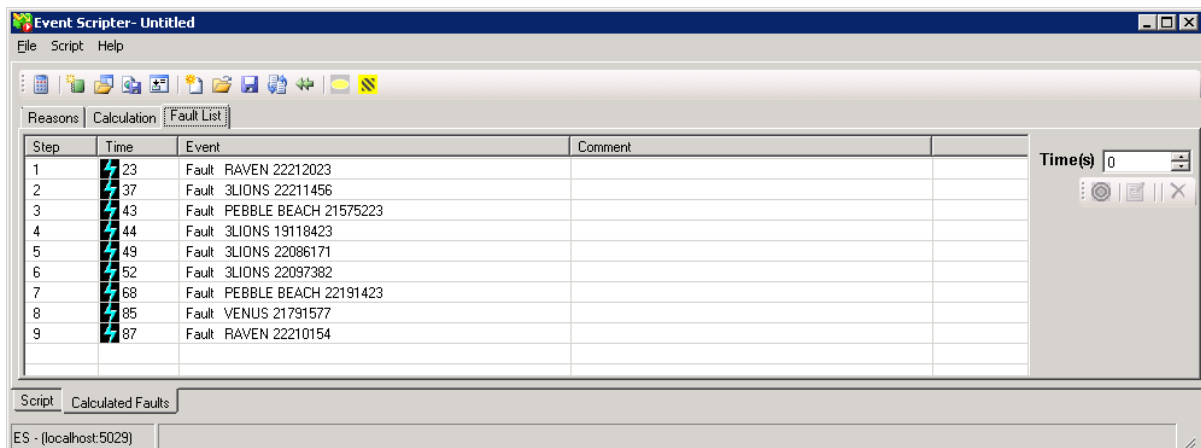
Enter values only for the fields applicable to the scenario you are creating. A probability of 100 is roughly equal to one fault event per hour randomly distributed for each selected station. Because of the differences in models and the station configurations, the results will vary. It may take some experimentation to get the desired result.

8. Save the scenario.
9. Click the Calculate Faults button on the toolbar to generate the faults.
10. Select the Calculation tab to see the log of the fault generation process. This gives a summary of the faults created for each station.



11. Select the Fault List tab.

This tab lists the faults and the times at which they are to be applied. The times are within the range in seconds set on the Calculations tab.



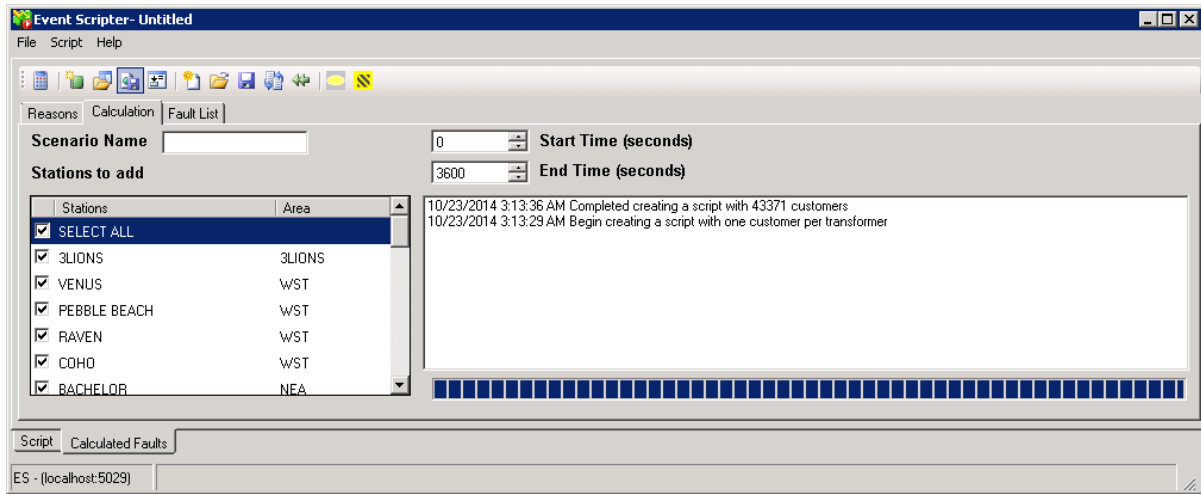
12. Save the Fault List, if required.

4.11.2. Creating a Customer Call Scenario

To create call events from all customers:

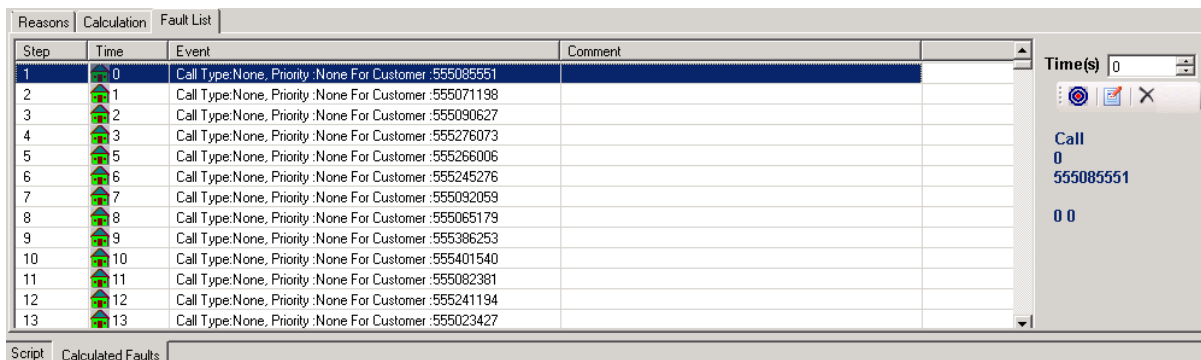
1. Open the Event Scriptor, and select the Calculated Faults tab at the bottom of the window.
2. Select the Calculation tab.
3. Do one of the following:
 - Click the  button to create call events from one customer for every transformer in the system.
 - Click the  button to create call events from all customers in the system.

4. Check the log of the call generation process.



5. Select the Fault List tab.

This tab lists the calls and the times at which they are to be applied.



4.11.3. Merging Customer Calls and Calculated Faults with a Script

To merge the faults list with a script, click the Merge button  on the calculated faults toolbar.

The calculated fault list is merged with the currently open main script.

4.12. Creating a Conditional Event

The conditional event functionality allows you to create scripts with event steps that trigger depending on certain network conditions. For example, you can create an event with a customer call that triggers if the customer's transformer is disconnected.

You can create scenarios using logical expressions (IF, ELSE IF, ELSE, END), logical operators (AND, OR) and comparison operators ("=", "!", ">", "<", ">=", and "<=").

Conditional event creation rules are as follows:

- A conditional event must begin with an IF statement.
- Optionally, the IF statement can be followed by one ELSE statements.
- Optionally, the IF statement can be followed by one or more ELSE IF statements.
- Optionally, the last ELSE IF statement can be followed by one ELSE statement.
- Optionally, step conditions can include multiple AND and OR logical operators. Conditions can be grouped using parenthesis. For example, (Condition_A AND Condition_B) OR Condition_C.
- A conditional event must end with an END statement.

For example:

```
IF (Condition_A AND Condition_B)
Event Step 1
ELSE IF (Condition_C OR Condition_D)
Event Step 2
ELSE
Event Step 3
END
```

Conditions are based on a value that can be queried from the server. Conditions have the following format:

```
RT.SUB(<Substation_Name>).<Query_Record_Name>(<Device_ID>):<Query_Field_Name_1>,<Query_Field_Name_N><C
omparison_Operator><Value_1><Value_N> <AND> <Condition_N1> <OR> Condition_N2>
```

For example:

```
(RT.SUB(RAVEN).SWT(22233017):CURROPEN,SWTCURPOSSTATUS=TrueOpen AND
RT.SUB(RAVEN).SWT(22233161):CURROPEN=RT.SUB(RAVEN).SWT(22233938):SWTCURPOSSTATUS) OR
RT.SUB(RAVEN).DISTTF(22233938):DYNAMICFEEDER=F8863
```

Note

You can configure comparison values to be queried from the RT server or SIM server using the netconfes.xml file.

For example, a conditional event can be modeled as follows:

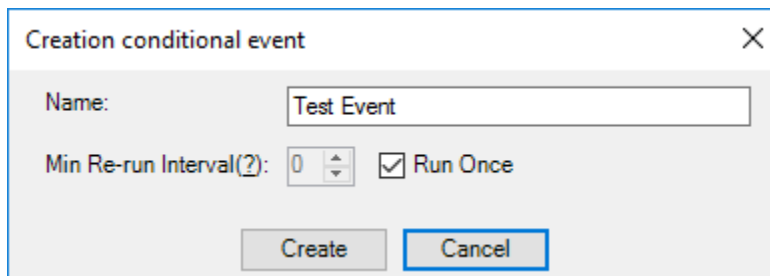
1. Step 1: IF RT.SUB(RAVEN).SWT(22239714):CURROPEN = True
2. Step 2: Change a measurement.
3. Step 4: ELSE
4. Step 5: Change a measurement.
5. Step 6: END

To create an example conditional event:

1. Open the Conditional Events tab.
2. On the Scripts toolbar, click Add New Event.

The "Creation conditional event" display appears.

3. Enter the event name, specify the event execution logic, and click Create. The event appears in the event list.
 - If "Run Once" is selected, an event is executed once when the specified event conditions are satisfied.
 - If "Min Rerun Interval" is specified, an event is executed continuously with the specified minimum interval between consequent executions (in seconds).

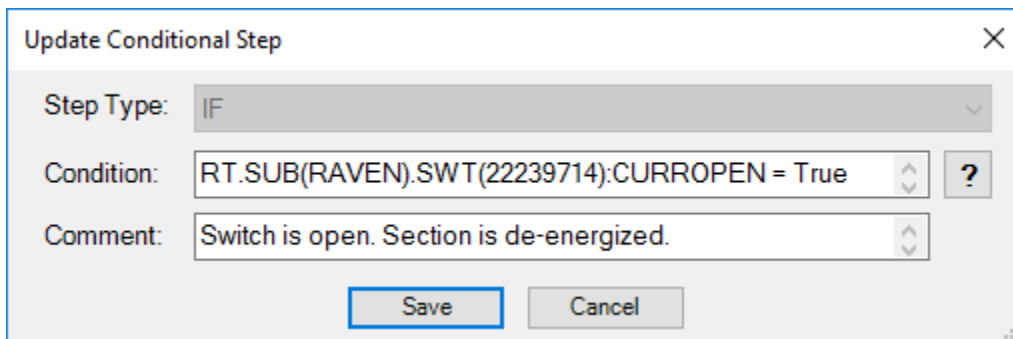


The "Creation conditional event" dialog box is shown. It has a title bar with a close button (X). The "Name:" field contains "Test Event". The "Min Re-run Interval(2):" field has a spinner set to 0. The "Run Once" checkbox is checked. At the bottom are "Create" and "Cancel" buttons.

4. On the Event Steps toolbar, click the Conditional Event Step button.

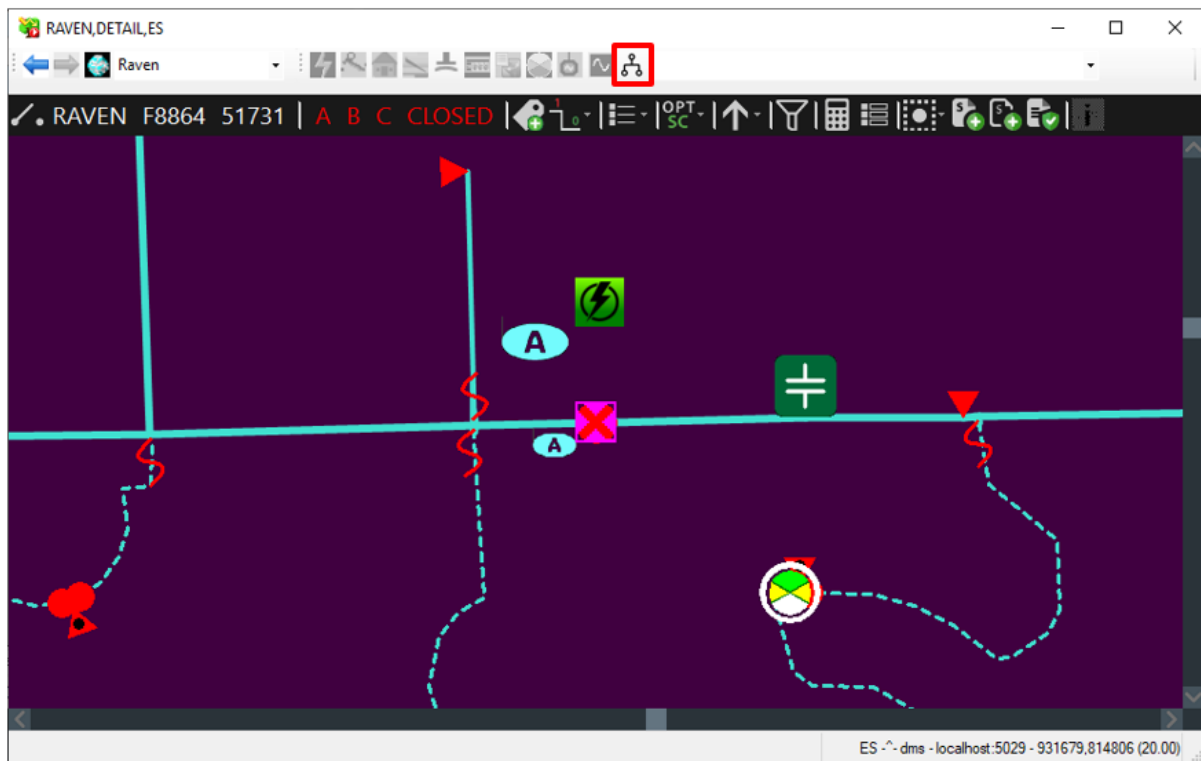
The Update Conditional Step display appears.

5. Select the IF step type. Enter a condition (for example, IF RT.SUB(RAVEN).SWT(22239714):CURROPEN = True) and click Save.



The "Update Conditional Step" dialog box is shown. It has a title bar with a close button (X). The "Step Type:" dropdown is set to "IF". The "Condition:" field contains "RT.SUB(RAVEN).SWT(22239714):CURROPEN = True". The "Comment:" field contains "Switch is open. Section is de-energized.". At the bottom are "Save" and "Cancel" buttons.

Alternatively, instead of step 4 and step 5, you can navigate to the device on the Event Scripter's network view and click the Add a Conditional Event Step button on the upper toolbar and enter a condition (for example, IF RT.SUB(RAVEN).SWT(22239714):CURROPEN = True) and click Create.



2: Create Conditional Step

Step Type: IF

ES.SUB(RAVEN).SWT(22239714) CURROPEN =

Condition: RT.SUB(RAVEN).SWT(22239714):CURROPEN = True

Comment: Switch is open. Section is de-energized.

Create Cancel

6. On the Event Steps toolbar, click Add Measurement Event Step.
7. Enter measurement details and click Save.

MeasurementEventForm

Time: 0.00

Station: RAVEN

Measurement type: Status

Measurement Id: RAVEN.DASW.50916.DAST

☐ Simulate Only

☐ Apply all in station

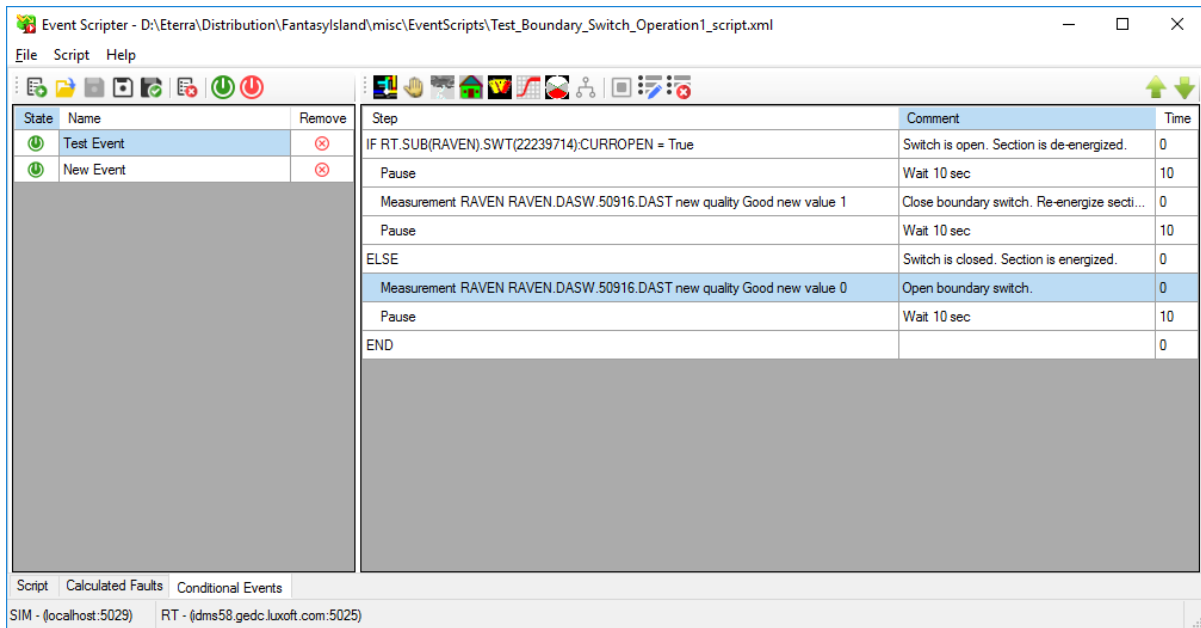
☒ Quality change: Good

☒ Value change: 1 - SDIS=1, XDIS=0

Comment: Close boundary switch. Re-energize section.

OK Cancel

8. Add ELSE IF and ELSE event steps, as appropriate.
9. Add an END event step.



10. Save the conditional event.

4.13. Creating a System Snapshot for the Current Time

Use the AdmsSystemSnapshot.ps1 script to collect all data needed to re-create the current system conditions. By default, this script is stored in the \server\bin folder.

The AdmsSystemSnapshot.ps1 script copies data from the model and dynamics folders (%IDMS_ROOT%\models\RT and %IDMS_ROOT%\dynamics) and compresses the data into a zip file <Unix_Timestamp>-dynamics.zip. The script includes only server dynamics and journals. Retired and backup dynamics and journals are excluded.

To collect all files needed to re-create the current system conditions, execute the AdmsSystemSnapshot.ps1 script from a PowerShell command line. The script prompts you for folders where to temporarily place copied files to be archived and where to place the output archive file with the current system data.

For example:

```
PS D:\Eterra\Distribution\Server\bin> .\AdmsSystemSnapshot.ps1
Which folder do you like to copy dynamics and model files to temporarily?: d:\temp\
Which folder do you like to see the final compressed file?:
d:\Eterra\Distribution\FantasyIsland\DOTS\SystemSnapshot\
```

```

Administrator: Windows PowerShell
PS D:\Eterra\Distribution\Server\bin> .\AdmsSystemSnapshot.ps1
which folder do you like to copy dynamics and model files to temporarily?: d:\temp\
which folder do you like to see the final compressed file?: d:\Eterra\Distribution\FantasyIsland\DOTS\SystemSnapshot\
69
34
d:\Eterra\Distribution\FantasyIsland\DOTS\SystemSnapshot\20200317062543-dynamics.zip

Directory: D:\temp\server\CallServer

Mode                LastWriteTime         Length Name
----                -
-a-----         3/17/2020   6:25 AM             0 CallServerJournal_2020_03_17_06_24_55.txt
D:\Eterra\Distribution\FantasyIsland\dynamics\server\CallServer\CallServerJournal_2020_03_17_06_24_55.txt

```

Note

The script folders must already exist before running the script.

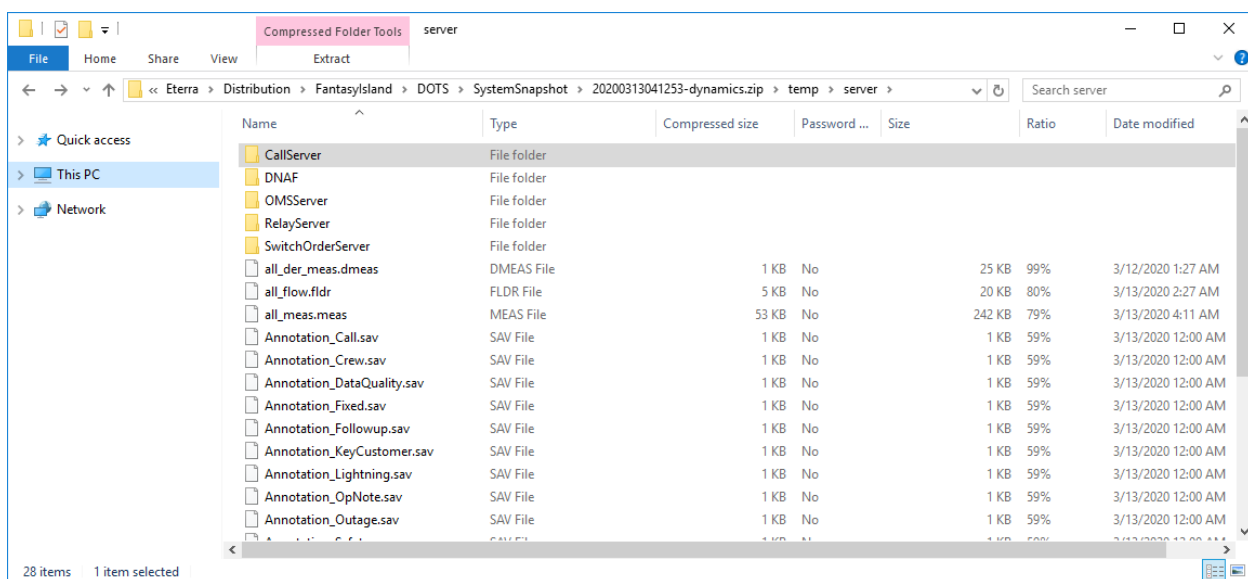


Figure 40. Creating a System Snapshot for the Current Time

4.14. Exporting Events from Dynamics and Call Journals

The Snapshot Converter application converts dynamics files and call journals into Event Scripter scripts. Use this tool to create scripts that accurately reflect network operations during a certain time period, including the actual event time and sequence.

To open the Snapshot Converter application, enter:

```
NETVIEW://RT/POPUP/SNAPSHOTCONVERTER
```

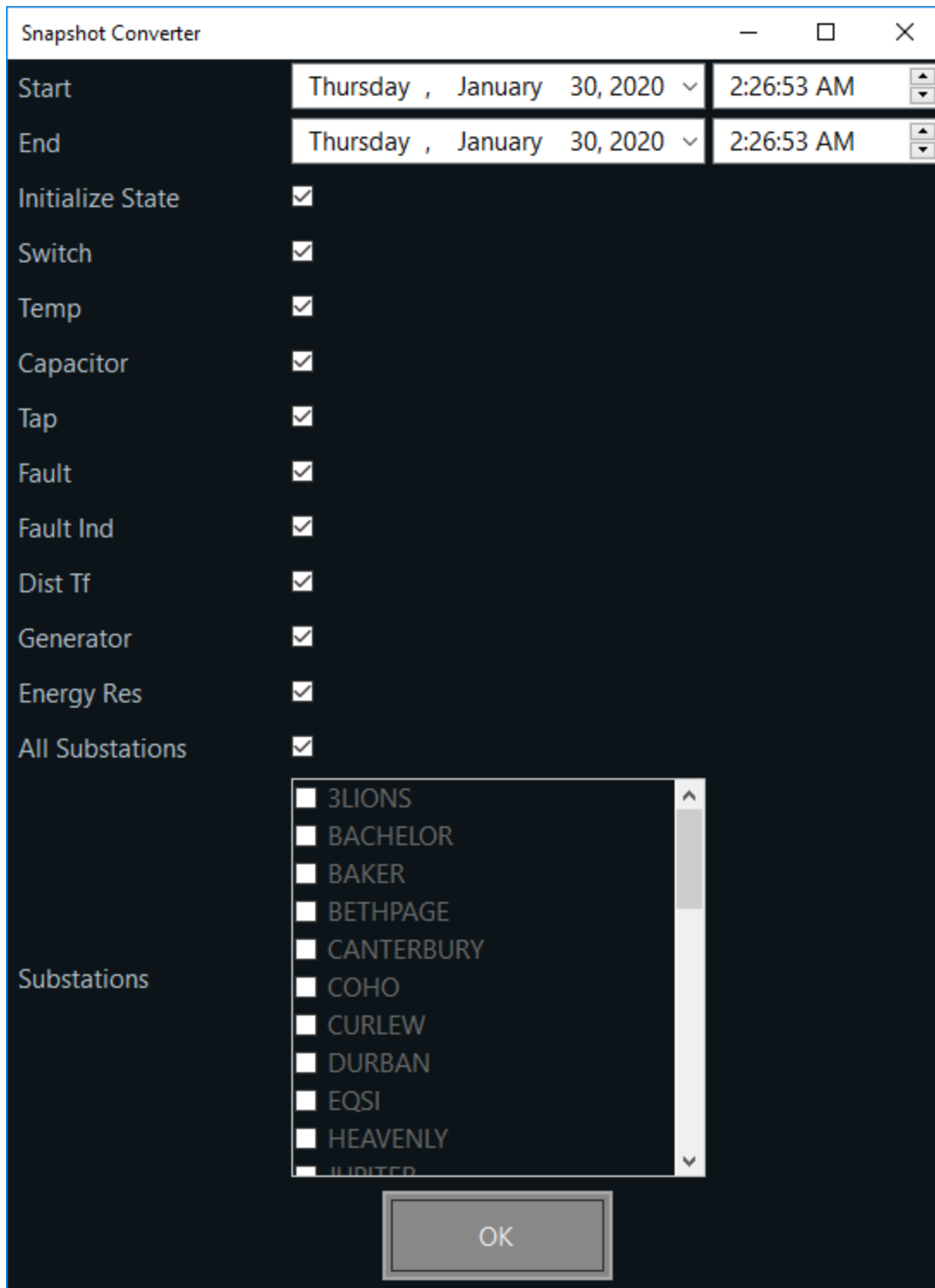


Figure 41. Snapshot Converter

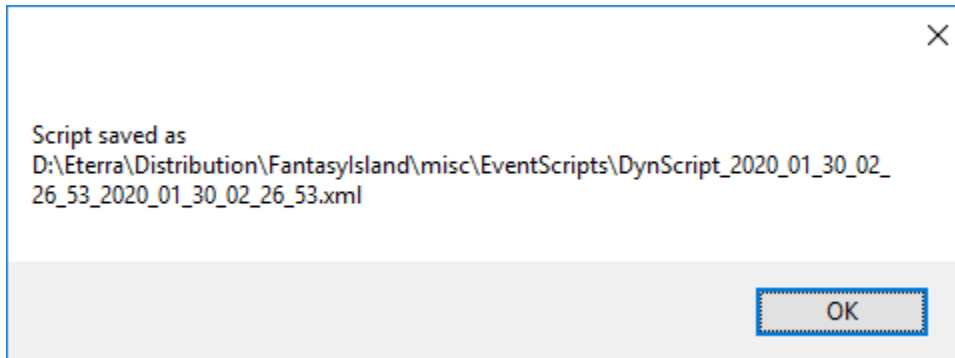
The Snapshot Converter application includes the following fields and controls:

- **Start/End Time:** The dynamics conversion start and end times. The tool extracts events for the specified time period from the system dynamics files.

- **Initialize State:** Specifies whether to reconstruct the system state that existed before the start time.

If this option is selected, all events that occurred in the system before the start time are included in a script. These events are inserted at the beginning of the script with the 0 time to reconstruct the existing system state.

- **Device Type:** Specifies device types to include events for.
- **Substation:** Specifies substations to include events for.
- **OK:** Creates an event script with the specified parameters. The script is placed in the %IDMS_ROOT%\misc\EventScripts\ folder with the name DynScript_<TimeStamp>.



Note

The Snapshot Converter command-line interface provides additional functions, such as including customer calls events in the generated script.

4.14.1. Snapshot Converter Command Line Operations

The Snapshot Converter command-line interface supports the following arguments.

`/DYNFILES <PATH TO FILE 1> <PATH TO FILE 2> <PATH TO FILE N>`

Specifies the path to the input SAV dynamics data files to export network events as event steps. This argument is required if JOURNALSPATH is not specified, and is optional if JOURNALSPATH is specified.

You can specify several dynamics files using the following format. This allows creating event scripts using both backup and new dynamics.

`/DYNFILES "DynamicsFile1" "DynamicsFile2" "DynamicsFile3"`

Note

When specifying several dynamics files, the events are loaded following the order of the files.

`/JOURNALSPATH <PATH TO FOLDER>`

Specifies the path to the folder with call journal files to export customer calls as event steps. This argument is required if DYNFILES is not specified, and is optional if DYNFILES is specified.

/OUTPUT [PATH TO FILE]

Specifies the path to a new Event Scripter's script file to be created from dynamics and/or journal files.

/STARTTIME [DATE AND TIME]

Specifies the event conversion start time in the format YYYY-MM-DDTHH:MM:SS (for example, "2009-06-15T13:45:30"). If not specified, all records before the event conversion end time are converted.

/ENDTIME [DATE AND TIME]

Specifies the event conversion end time in the format YYYY-MM-DDTHH:MM:SS (for example, "2009-06-15T13:45:30"). If not specified, all records after the event conversion start time are converted.

/GENINIT

This argument can only be used with STARTTIME. If specified, all records before the event conversion start time are also included in a script. These events are inserted at the beginning of the script with the 0 time to reconstruct a certain system state.

Exporting Events from Dynamics and Call Journals

You can export events from dynamics and call journals using the following command:

```
> SnapshotConverter.exe / DYNFILES "dynamics file path" /JOURNALSPATH "call journals folder path" /OUTPUT "export file path"
```

For example, to export events from dynamics.sav and call journal files, enter the following:

```
> D:\eterra\Distribution\Tools\bin\SnapshotConverter.exe /DYNFILES  
"%IDMS_ROOT%\dynamics\Server\dynamics.sav" /JOURNALSPATH  
"%IDMS_ROOT%\dynamics\Server\OMSServer\CallJournal" /OUTPUT  
"%IDMS_ROOT%\misc\EventScripts\ExportScript.xml"
```

Note

To create event scripts with all device status change events, not only those reflecting the current system state, set the PRESERVE_ALL_DYNAMICS and DISABLE_MID_NIGHT_DYNAMIC_COMPRESSION parameters to 1 in the NetServerReadOnlyConfig.txt configuration file.

5. Returning the Simulator to the Initial State

After running event scripts and training scenarios and manipulating the network view, you may need to clear the changes and return the dynamics to the initial state. To return the training simulator system's dynamics to the initial state, follow the procedures described in the following sections.

5.1. Clearing Changes via the Event Scriptor

To reset the simulator network view using the Event Scriptor tool:

1. Open the Event Scriptor tool.
2. On the toolbar, click the Go Back to Normal button .

The prompt dialog box appears.

3. Click OK.

All changes are now cleared.

Note

The "Go Back to Normal" function of the Event Scriptor can only reverse the actions that have been injected by the script or the instructor, such as the operation of a switch or the creation of a fault. It does not reverse the consequential operations that may have been caused by the event, such as the tripping of a breaker in response to the fault.

5.2. Clearing Changes via the Crew Simulator

To reset the simulator network view using the Crew Simulator tool:

1. Open the Crew Simulator (RoboCrew) tool.
2. From the File menu, select Clear All (Back to Nominal).

The Log tab displays the changes being cleared.

Note

When clearing the dynamics via the Crew Simulator Clear All option, no dialog box appears and all changes are reversed immediately.

5.3. Clearing Dynamics via the Batch Script

Running the batch script permanently clears all changes in the dynamics and deletes the logs on

both the server and the client. Note that this script does not remove any tags that are applied.

 Warning

Never use the `clear_dynamics.bat` script in a real-time system. The `clear_dynamics` batch script is intended for the training simulator only. If used in the real-time system, it permanently destroys all data.

To reset the Network View using the batch script:

1. Open the Process Manager and terminate all of the running processes.
2. Exit the Browser client.
3. Navigate to the location of the `clear_dynamics.bat` batch script location.

The default location is: `%IDMS_ROOT%\misc\demo\Scripts\`

4. Double-click the `clear_dynamics.bat` file to run the script.
5. Open the Process Manager and restart the processes.
6. Start the Browser client.
7. Navigate to the OMS overview displays to ensure that dynamics are cleared and the displays are empty.

The training simulator dynamics are reset to the initial state. All logs are deleted.

An example of the Clear Dynamics script is given in [Clear Dynamics Script](#).

6. System Setup

Integrating a Distribution Operator Training Simulator (DOTS) is similar to integrating a real-time network management system, since they use the same software components, interfaces, and user interfaces. You create a replica of the control room and the Scada system.

DOTS is less complex in the sense that it does not deal with redundancy, network security, external services, multiple locations, and so on. On the other hand, it is more complex because it requires configuring the servers for different modes of operations, it uses different datasets for different simulations, and it needs to integrate additional simulation tools. The Fantasy Island file provides the working example of these differences.

The installation and configuration aspects of the applications are documented in the *ADMS Software Installation and Maintenance Guide*.

The figure below illustrates the software modules and their interactions in the DOTS environment.

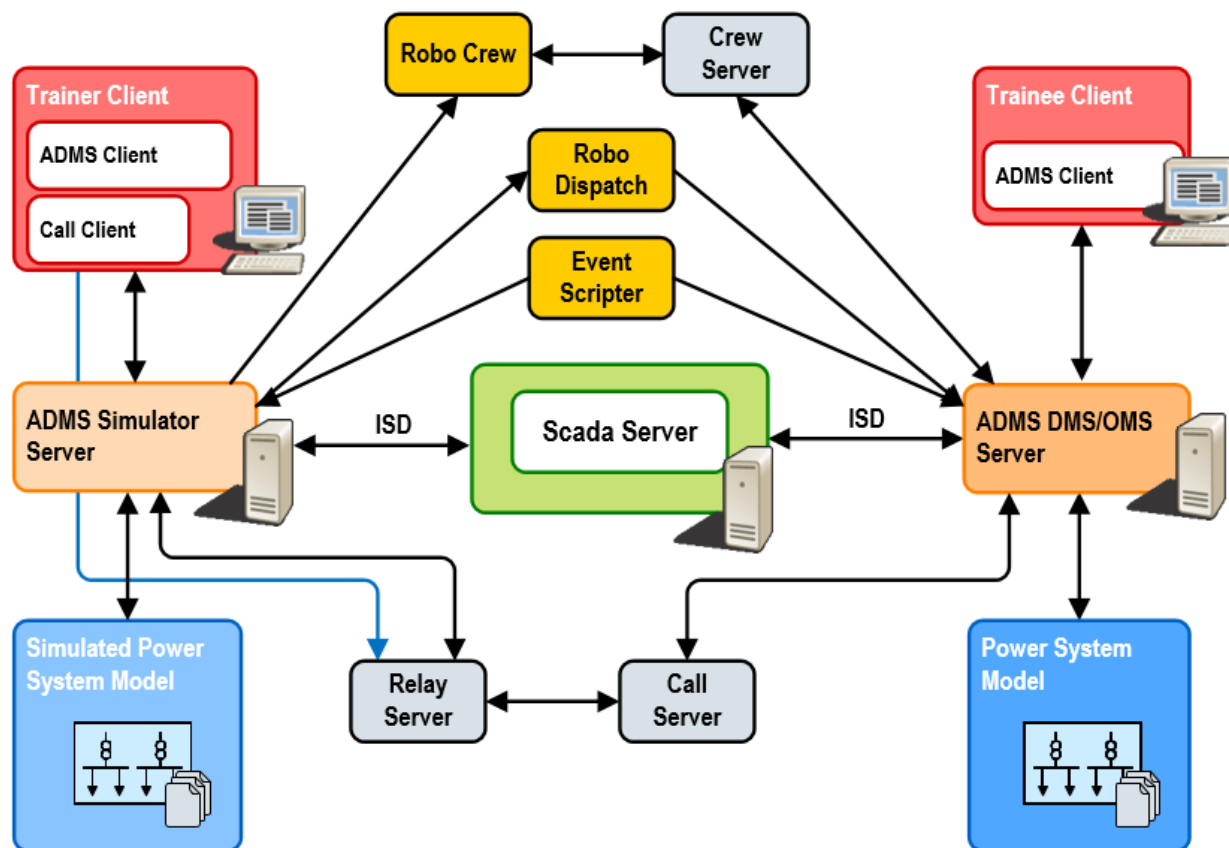


Figure 42. Distribution Operator Training Simulator Modules Interactions

- **ADMS DMS/OMS Server:** Hosts the NetSrv (DMS), NetServer (OMS), DnafSrv (DNAF), and switch order server applications.
- **ADMS Simulator Server:** Hosts the SimSrv (or NetServerSim) and switch order server applications. It also hosts the interface servers and simulation tools: RoboCrew,

RoboDispatcher, relay server, call server, and crew server.

- **Scada Server:** The host server for Scada applications and Habitat utilities. It hosts Scada applications and other shared Habitat applications such as ALARM, CFGMAN, PERMIT, TAGGING, SAFETY, etc.
- **Citrix Servers:** These servers host the ADMS clients for training non-control room dispatchers.
- **Trainee Consoles:** These consoles host the ADMS clients. The number of consoles depends on the training needed.

 Note

To operate the simulator on the instructor console, you need to install both the server and the client.

7. Clear Dynamics Script

The following is an example batch script to re-initialize the simulator. This script clears all of the dynamic data in the network models.

The "pushd" line in the script needs to be edited to point to the dynamics folder in each particular installation.

Warning

Never use the `clear_dynamics.bat` script in a real-time system. The `clear_dynamics` batch script is intended for the training simulator only. If used in a real-time system, it permanently destroys all data.

```
REM Clear Dynamics Batch Script
REM - Utilizing IDMS_ROOT
IF NOT EXIST %IDMS_ROOT% (
ECHO.
ECHO.
ECHO ERROR: The "IDMS_ROOT" Environment Variable is not defined for this context.
ECHO ERROR: Exiting Script...
ECHO.
EXIT /B
)
pushd %IDMS_ROOT%\dynamics
ECHO.
ECHO Begin Cleaning Dynamics...
ECHO.
del /q /s *journal*.txt
del /q /s *.sav
del /q /s sid.bin
del /q /s *.snapshot
del /q /s *.last
del /q /s *.cri
del /q /s *.mi
del /q /s *.qi
del /q /s *.lst
ECHO.
ECHO Dynamics Cleared.
ECHO.
pause
```

8. Server Configuration

The training simulator is a non-redundant system. It encompasses all of the real-time software as well as the simulation tools.

The server configuration includes the following important aspects:

- Creating a separate instance of the dynamics using the Net Dynamics Configuration Editor, as described in [SIM Dynamics Instance Configuration](#).
- Configuring the general server parameters and device operation settings using the `NetServerSimConfig.txt` configuration file as described in [NetServerSimConfig Default Parameters](#).
- Configuring the Outage Management System parameters. For example, customers' behavior when the outage occurs using the `NetServerSimConfig` configuration file as described in [OMS Server Configuration for the Simulator](#).
- Configuring the Relay Server parameters. For example, relay server communication with NetServer using the `RelayServerConfig` configuration file, as described in [Relay Server Configuration for the Simulator](#).
- Configuring the Crew Simulator parameters. For example, configuring the time for completing incidents by MDT crews using the `CrewSimulatorConfig` configuration file, as described in [Crew Simulator Configuration](#).

8.1. NetServerSimConfig Default Parameters

The ADMS Simulator server is configured with the `NetServerSimConfig` configuration file. The default parameters include:

- `SERVER_PORT` = 5029: The listening port of the Simulator server.
- `CALLSERVER_PORT` = 5033: This needs to match up the `<ListeningPort>5033</ListeningPort>` defined in `relayserverconfig.xml`.

Note

The simulator does not connect directly to the call server. Instead, it connects to the Relay server so the `CALLSERVER_PORT` and `CALLSERVER_HOST` are the relay server port and the relay server host.

- `CALLSERVER_HOST` = localhost: The host where the relay server runs. The default value is localhost.
- `CIS_IVR_INTERFACE_ENABLED` = 1: This flag enables (=1) or disables (=0) interface to the Call Server.
- `ISD_LOCAL_SITE` = NS: Set to match the `SYSTEM_SITE` field within SCADAMOM that represents the simulator site. Although not used directly by the ADMS server application, this can be a

good indication of which site represents the distribution system at the remote site. Therefore, using "NS" as the site name is recommended.

- `ISD_LOCAL_SCANNER_PATH = NS.SCADA_NS_ISD`: Set to match the `OBJECT_SITE` field in SCADAMOM that represents the ADMS site. It is used for InterSite Data.

If this parameter is not set, no communication on this path is attempted by the ADMS server application.

- `ISD_LOCAL_CONTROL_PATH = NS.SCADA_NS_CTRL`: Set to match the `CBJECT_SITE` field in SCADAMOM that represents the ADMS site. It is used for InterSite Control.

If this parameter is not set, no communication on this path is attempted by the ADMS server application.

- `ISD_LOCAL_TAGNOTES_PATH = EMS.TAGGING_NS_TGDATA`: Set to match the `TGOBJECT_SITE` field in SCADAMOM that represents the DMS site. It is used for InterSite Tagnotes.

If this parameter is not set, no communication on this path is attempted by the ADMS server application.

- `ISD_REMOTE_SITE = EMS`: Set to match the `SYSTEM_SITE` field within SCADAMOM that represents the remote Platform site. Although not used directly by the ADMS server application, this can be a useful indication of the remote site identifier.

- `ISD_REMOTE_SCANNER_PATH = EMS.SCADA_EMS_ISD`: Set to match the `OBJECT_SITE` field in SCADAMOM that represents the Platform site. It is used for InterSite Data.

If this parameter is not set, no communication on this path is attempted by the ADMS server application.

- `ISD_REMOTE_CONTROL_PATH = EMS.SCADA_EMS_CTRL`: Set to match the `CBJECT_SITE` field in SCADAMOM that represents the Platform site. It is used for InterSite Control.

If this parameter is not set, no communication on this path is attempted by the ADMS server application.

- `ISD_REMOTE_TAGNOTES_PATH = EMS.TAGGING_EMS_TGDATA`: Set to match the `TGOBJECT_SITE` field in SCADAMOM that represents the Platform site. It is used for InterSite Tagnotes.

If this parameter is not set, no communication on this path is attempted by the ADMS server application.

- `ISD_REMOTE_HOST1 = 127.0.0.1`: Set to match the first Scada host name or IP.
- `'ISD_REMOTE_HOST2 = EMSB`: Set to match the second Scada host name or IP.
- `'ISD_REMOTE_HOST3 =` Set to match the third (if applicable) Scada host name or IP.
- `'ISD_REMOTE_HOST4 =` Set to match the fourth (if applicable) Scada host name or IP.
- `ISD_REMOTE_PORT = 5001`: Set to match the port that is used for communication on the remote machines. This port should be set to match the `DEFPORT_ITEMS` field within the `PROXYSRV` database on the remote machine.

- ISD_REMOTE_POLLRATE = 2: Set to the desired period, in seconds, between periodic polls to the remote site. This parameter is only used for the InterSite Data protocol.
- ISD_REMOTE_INTEGRITYRATE = 900: Set to the desired period, in seconds, the ADMS server application uses to send all current data values to the remote site. This parameter is only used for the InterSite Data protocol.
- ISD_MAX_DOWNLOAD_TIME = 300: The number of seconds to wait for an ISD local or remote download. If the download is not completed, the server will disconnect the ISD path.
- ISD_KEEPALIVE_TIMEOUT = 300: Set to the desired period, in seconds, that the ADMS server application waits without receiving an appropriate message from the remote before disconnecting. Setting this value to zero disables keep-alive logic from NetSrv for InterSite protocols.
- ISD_DETAILED_MESSAGES = 1: If set to True ("1" and other non-zero values evaluate to True), verbose message logging relating to InterSite Data Protocol processing is produced. The default file is placed in the working directory of the ADMS server application and is called "isdlog.txt". Note that warning and error messages are sent to this log file regardless of the setting of the ISD_DETAILED_MESSAGES parameter.
- ISD_ANALOG_VALUE_MESSAGES = FALSE: If set to TRUE, ISD logs analog measurements received and their value and quality. It is logged to the file defined in ISD_LOG_FILE_NAME.
- ISD_DEFAULT_SITE = Set to match the DEFAULT_SITE field in SCADAMOM representing the ADMS site; it is used for InterSite Data.

If this parameter is not set, no communications on the path are attempted.

- USE_NEW_ISD_LOGIC = 1: A flag to enable (=1) or disable (=0) the new InterSite Data logic.
- ISD_RECONNECT_TIMEOUT = 5: Timeout (in seconds) for ISD reconnection.
- ISD_DUMP_SOCKET_TRAFFIC = 1: If set to True (=1), ISD logs the message size and other lower-level information to the file defined with the configuration parameter ISD_LOG_FILE_NAME.

8.1.1. Scada Control Parameters

The Simulator server parameters CONTROL_VALUE_0 and CONTROL_VALUE_1 included in the NetServerSimConfig configuration file are used to mimic the Control Function in Scada modeling for a specific control.

- CONTROL_VALUE_1 = RAISE, CLOSE, BLOCK, ON, ENABLE, AUTO, CPON, RESET, IN
- CONTROL_VALUE_0 = LOWER, TRIP, OPEN, UNBLOCK, DISABLE, OFF, MANUAL, CPOF, OUT
- MULTISTATE_MEAS_TRANSITION_CONFIG = %IDMS_ROOT%\config \server \MultistateMeasTransitionConfig.txt

Through the measurements mapped during modeling time, the simulator has the information of all the analogs, status points, and controls mapped to a device or a simulate-only point.

Control commands are passed from Scada to the simulator over ISD. Each command contains the composite ID of the device and the Control Name (ID_CTRL in Scada) to be performed.

- If the ID_CTRL matches with parameters of CONTROL_VALUE_1, the state will be changed to 1 if the current state of the device is 0.
- If the ID_CTRL matches with parameters of CONTROL_VALUE_0, the state will be changed to 0 if the current state of the device is 1.

Example: A breaker in the system is currently in the CLOSED state (value 1) in the simulator. A Scada control is received to open this breaker. The ID_CTRL in Scada for the control could be OPEN or TRIP or any other ID that meets the user's configuration. The ID_CTRL should be either in the server configuration CONTROL_VALUE_0 or CONTROL_VALUE_1.

Ideally in this example we would expect the string OPEN (or TRIP) to be part of CONTROL_VALUE_0. When simulator receives the control, it finds that ID_CTRL matches an entry in CONTROL_VALUE_0 and changes the device state to OPEN (value of 0). The new state value is sent to Scada.

Simulator has information about Tap Controls (RAISE/LOWER) based on the measurement mapping. Simulator also has information about the command increase or decrease for the tap position of the transformer or regulator.

Simulator also has information about simulate-only status controls (POPEN/PCLOSE). To reset the POPEN status to PCLOSE, the RESET value should be in the server configuration CONTROL_VALUE_1.

Simulator also has information about AVL switch controls (IN/OUT) based on the measurement mapping.

Switch State Transition Definitions

You can optionally specify state transition definitions for Scada switches using the MultistateMeasTransitionConfig.txt file. If specified, the SIM Server identifies the switch state based on the provided definitions.

For each multi-state switch, the file can contain possible state transition entries in the following format:

```
<New_Switch_State> <Current_SDIS ><Current_XDIS> <New_SDIS><New_XDIS>
```

For example:

```
AHL
OPEN 10 00
OPEN 00 01
CLOSE 01 00
CLOSE 00 10
```

```
3STS
OPEN 10 11
OPEN 11 00
CLOSE 00 01
CLOSE 01 10
OPEN 10 00
```


In the example above, the "OPEN 10 00" entry defines that the switch status measurement change from "SDIS = 1, XDIS = 0" to "SDIS = 0, XDIS = 0" results in the Open switch status. Similarly, the "CLOSE 01 00" entry sets the switch status to Close when the switch status measurement changes from "SDIS = 0, XDIS = 1" to "SDIS = 0, XDIS = 0".

8.1.2. Fault Indicator, Recloser, Breaker, and Meter Parameters

The following Simulator server parameters included in the NetServerSimConfig configuration file configure the fault indicator reset time, reclosing capability, reclosing schedule, and meter deadband.

- `NTP_FAULTIND_RESET_TIME = 300`: The system-wide telemetered fault indicator reset time, in seconds.
- `NTP_MANUAL_FAULTIND_RESET_TIME = 300`: The system-wide non-telemetered (manual) fault indicator reset time, in seconds.
- `ENABLE_RECLOSING = 1`: A flag to enable (=1) or disable (=0) the reclosing capability. For the instructor to create switch events that use a recloser or breaker, the breaker or recloser must be "reclosing enabled".

When reclosing is disabled, the lockout status measurement for a breaker or recloser upstream of a fault are set to True in the simulator. This allows a downstream fault to trigger FISR without waiting for device reclosing in the real-time environment.

- `BREAKER_RECLOSE_SCHEDULE = 1,30,60`: *VALUE1* is the time (in seconds) when the first reclose takes place after the initial fault, *VALUE2* is the second reclosing time (in seconds) counted after the first reclose, and *VALUE3* is the third reclosing time (in seconds) counted after the first reclose.
- `RECLOSER_RECLOSE_SCHEDULE = 1,30,60`: *VALUE1* is the time (in seconds) when the first reclose takes place after the initial fault, *VALUE2* is the second reclosing time (in seconds) counted after the first reclose, and *VALUE3* is the third reclosing time (in seconds) counted after the first reclose.
- `ANALOG_MEASUREMENT_DEADBAND = 0.1`: The deadband for analog measurements.
- `ENABLE_BREAKER_OVERCURRENT_TRIP = 0`: A flag to enable (=1) or disable (=0) the feeder breaker overcurrent tripping when the feeder loading exceeds the overcurrent relay settings (phase trip or ground trip) in simulator mode.
- `ENABLE_MANUAL_TRIP_LOCKOUT = 1`: If enabled (=1) the lockout status is set to "Locked Out" for a breaker or recloser when the device is manually tripped in simulator mode.

This option should be disabled (=0) when changing the lockout status using the Event Scripter.

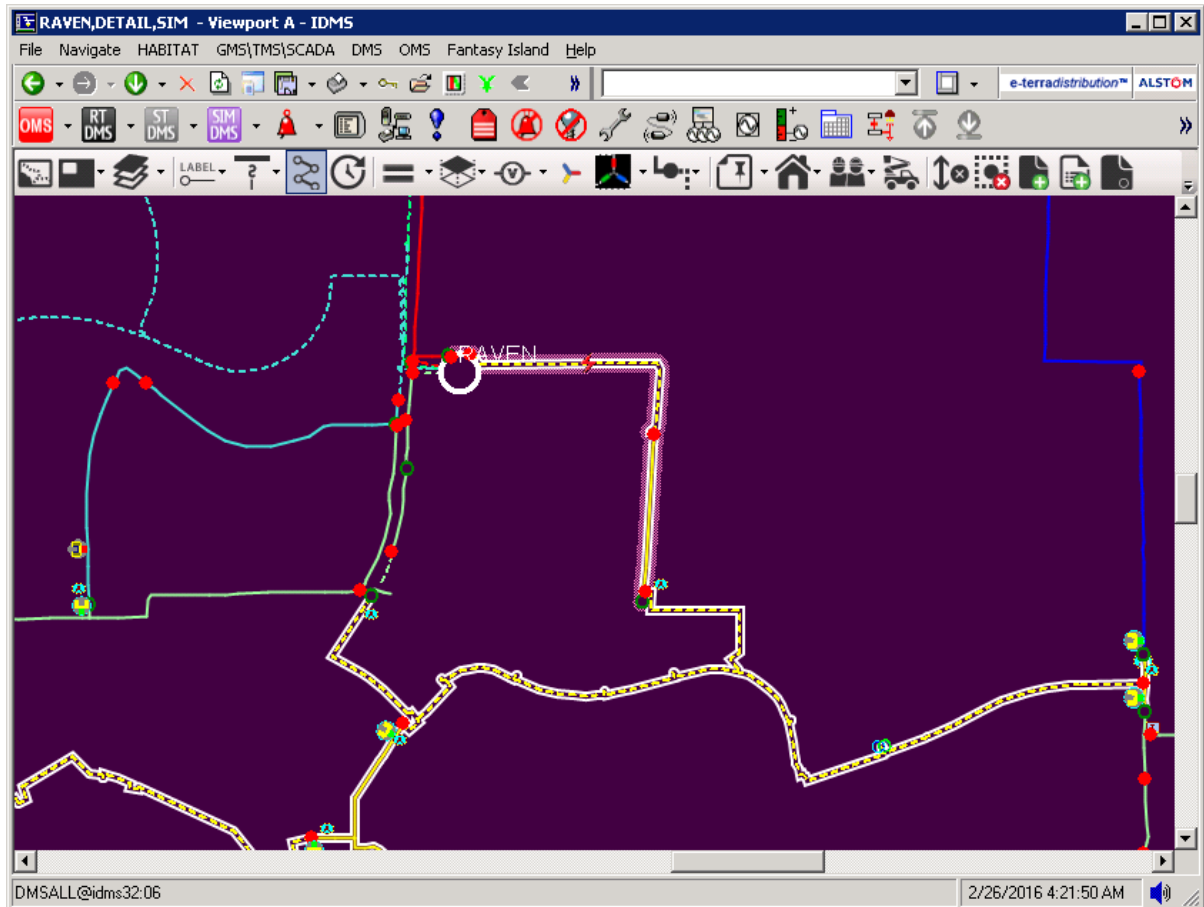
Note

In real-time mode, the lockout status is always set to "Locked Out" for a breaker or recloser

when the device is manually tripped by the operator.

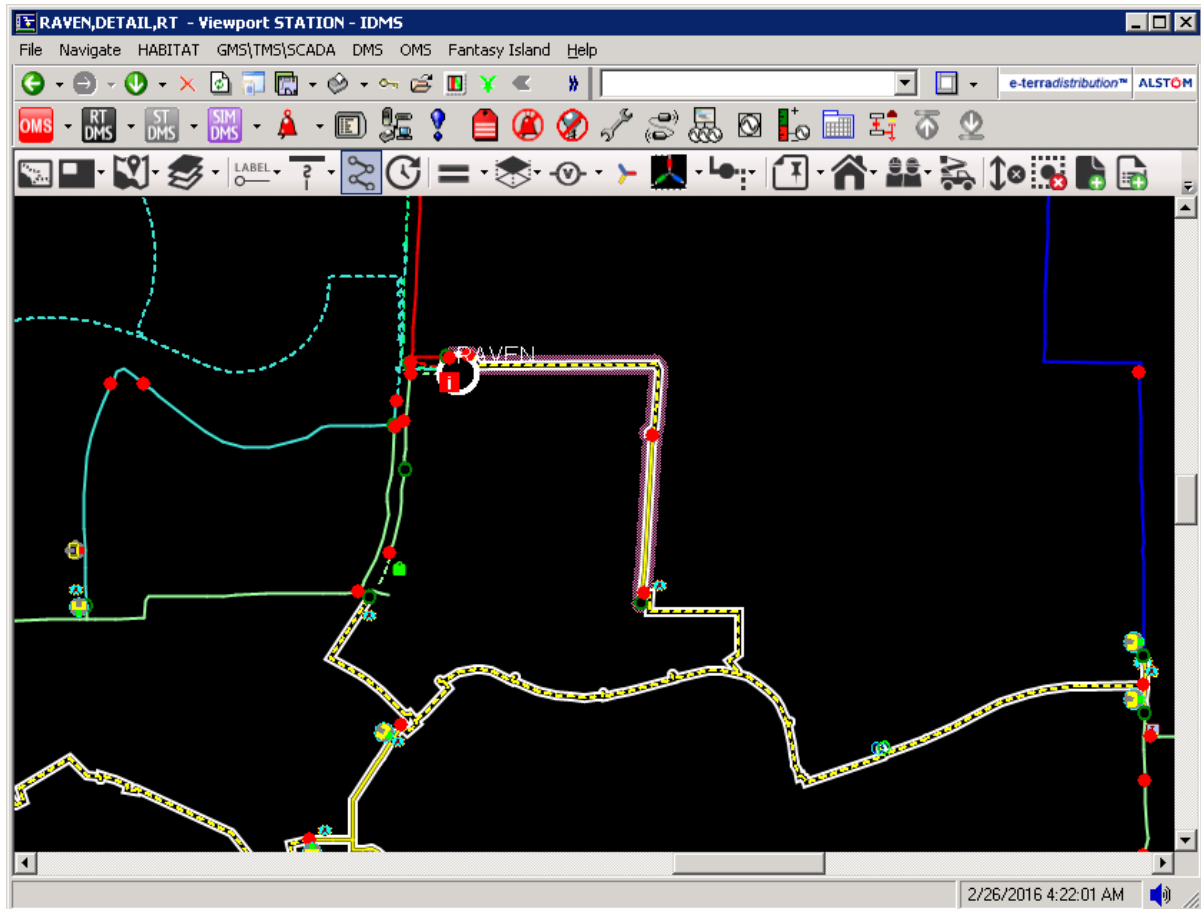
The example below demonstrates operation of the reclose schedules and fault indicators.

1. From the Event Scripter, click the Open Geographic Display button to open the network view.
2. Create a temporary fault event with the duration of 40 seconds on a feeder backbone downstream from a station recloser.
3. Execute the script.
4. Open the simulator network view.



The fault is located to the right of the Raven station.

5. Open the real-time network view.



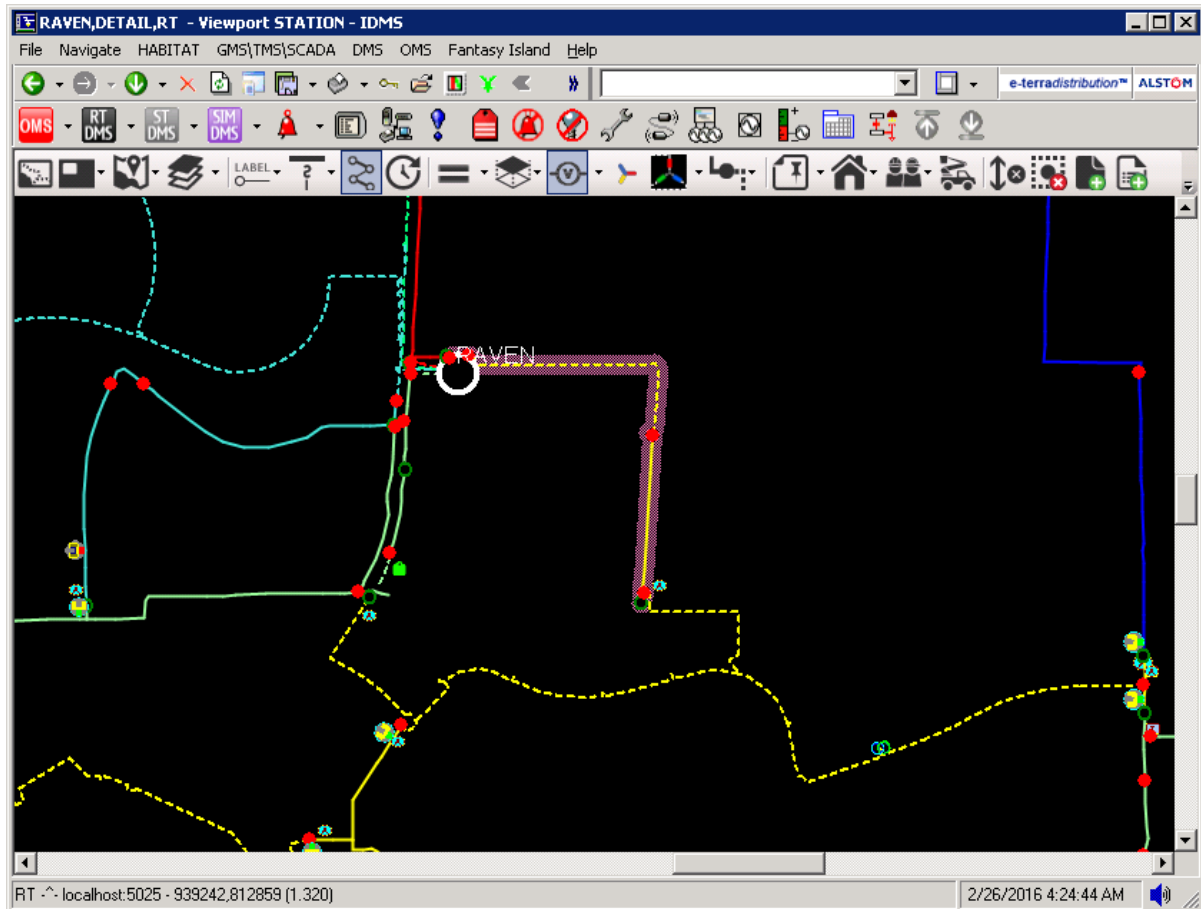
The feeder is de-energized. The pink halo shows the possible fault location based on the state of telemetered (Scada) fault indicators. The fault halo around the lines is shown between the furthest downstream set fault indicator and the next set of fault indicators that are unset.

The first and second reclosing attempts do not succeed as the fault still exists on a line.

6. Wait until the fault is cleared from the simulator network view (40 seconds after the fault event start).

The fault is removed, but the line is still de-energized.

7. When the third reclosing attempt is performed (60 seconds after the fault is placed), the line is re-energized from the Raven station.



The feeder is energized. However, the pink halo continues showing the possible fault location.

8. The pink halo disappears as per the fault indicator reset time (300 seconds after the fault is placed).

8.2. OMS Server Configuration for the Simulator

You can modify the Outage Management System parameters using the `OmsServerSimConfig_localhost.txt` configuration file, where `localhost` is a server name. The default location of the configuration file is: `%IDMS_ROOT%\config\server\`

After changing the default parameters, save the changes and restart the client. The configuration file's default parameters include:

- `CALL_LIKELIHOOD = 0.25`: Percent of probability for customer calls when an outage occurs.
- `CALL_AGGREGATE_LIKELIHOOD = 0.05`: Percent of customer calls that are aggregated with the specified call conditions.
- `CALL_DELAY = 10`: Delay time in seconds after an outage occurs and before the first customer call is made.
- `CALL_WINDOW = 300`: Time period in seconds during which all customer calls are made after an

outage.

- `CALL_PEAK_TIME = 30`: Time in seconds at which most customers make calls after an outage, for example, most customers call 30 seconds after an outage occurs.
- `CALL_BATCH_TIMER = 60`: The time period in seconds during which customers calls are sent.
- `CALLBACK_DELAY = 30`: Delay time in seconds for a customer callback.
- `CALLBACK_FAIL_LIKELIHOOD = 0.0`: Percent of probability that a callback fails. Every failed callback creates an isolated incident for a customer.
- `AMR_MODE = 1`: "1" enables the simulation of automatic meter reporting (AMR), "0" disables the simulation of the AMR reporting.
- `AMR_REPORT_OUT_LIKELIHOOD = 0.5`: Percent of likelihood that a meter will report a power outage.
- `AMR_REPORT_IN_LIKELIHOOD = 1.0`: Percent of likelihood that a meter will report when power is restored.
- `AMR_POWER_OUT_DELAY = 30`: Delay in seconds after power is out before AMR reports.
- `AMR_POWER_ON_DELAY = 60`: Delay in seconds after power is restored before AMR reports.
- `AMR_EXCEPTION_MODE = 1`: TBD.
- `AMR_POLL_MODE_PERIOD = 120`: TBD.
- `AMR_BATCH_TIMER = 1`: Time period in seconds when AMR calls are sent.
- `AMR_FAIL_LIKELIHOOD = 0.02`: Percent of likelihood that an AMR call fails.
- `AMR_CALL_PEAK_TIME = 30`: Time in seconds after power is out during which most AMR calls are made.
- `AMR_CALL_WINDOW = 60`: Time period in seconds after power is out during which all AMR calls are made.
- `METER_BASE_SECONDARY_VOLTAGE = 110.0`: Meter voltage base.
- `METER_SECONDARY_VOLTAGE_VIOLATION_THRESHOLD = 0.8`: Meter voltage violation threshold that is used to determine the meter brown out and high voltage conditions.
- `VOLTAGE_LEVEL_REQUIRED_TIME = 10.0`: Time period in seconds to delay sending a meter voltage event notification after an event occurs (for example, nominal voltage or high voltage event).
- `MOMENTARY_OUTAGE_CALL_TYPE = 1`: Call type that is applied for momentary outages. The value is the enumeration of the `CallType` entry in the coded translation configuration (for example, 1 = Lights Out, 8 = Blinking Lights).
- `DEFAULT_OUTAGE_CALL_TYPE = 1`: Call type that is applied for non-momentary outages. The value is the enumeration of the `CallType` entry in the coded translation configuration.
- `CALL_AGGREGATE_KEY_NAME_1 = Condition_1`: The first of the three customer call conditions that are randomly applied to aggregated calls. The value is the name of an entry in the coded translation configuration.
- `CALL_AGGREGATE_KEY_VALUE_1 = 1`: The value of the first condition. The value is the

enumeration of the entry in the coded translation configuration (for example, 1 = Wires Down).

- **CALL_AGGREGATE_KEY_NAME_2 = Condition_2:** The second of the three customer call conditions that are randomly applied to aggregated calls. The value is the name of an entry in the coded translation configuration.
- **CALL_AGGREGATE_KEY_VALUE_2 = 2:** The value of the second condition. The value is the enumeration of the entry in the coded translation configuration (for example, 2 = Pole Down).
- **CALL_AGGREGATE_KEY_NAME_3 = Description:** The third of the three customer call conditions that are randomly applied to aggregated calls. The value is the name of an entry in the coded translation configuration.
- **CALL_AGGREGATE_KEY_VALUE_3 = 2:** The value of the third condition. The value is the enumeration of the entry in the coded translation configuration (for example, 2 = Pole/House).

In the example above, one of the three defined conditions (Condition 1 = Wires Down, Condition 2 = Pole Down, and Description = Pole/House) is randomly applied to an aggregated call.

8.3. Relay Server Configuration for the Simulator

The RelayServerConfig.xml configuration file's default parameters include:

- **ListeningHost:** localhost: The host where the application is installed and is listening to other interfaces.
- **ListeningPort:** 5033: The relay server listening port.
- **UseCredentialFile:** false: If set to true, this parameter enables the use of the Credential file that defines the private key and user SIDs.
- **CallServerPort:** 5032: TCP/IP listening port number of the Call Server.
- **Relay** – true: Set to true only for the Relay Server. For the Call Server, this parameter always needs to be false; do not change it.

Warning

XML files are case-sensitive; use all lower-case letters when changing the Boolean settings.

8.4. Crew Simulator Configuration

You can modify the Crew Simulator (RoboCrew) parameters using the CrewSimulatorConfig.xml configuration file. The default location of the configuration file is %IDMS_ROOT%\config\client\.

After changing the default parameters, save your changes and restart the Crew Simulator. The configuration file's default parameters include:

- **ETRSimulatorElement:** Parameters for simulating an incident reason and ETR when an incident is cleared by a crew.

- **CrewRadio:** Crew radio number.
- **Reason:** Reason that is applied to the incident.
- **ETR:** ETR that is applied to the incident.
- **CrewDatabaseFile:** Path to the crew model file.
- **NetConfFile:** Path to the client configuration file specific to the crew simulator.
- **BitMapPath:** The path to the folder with bitmaps. Used for the crew modeling functionality of the Crew Simulator.
- **GridTabularPath:** The path to the folder with grid tabular displays. Used for the crew modeling functionality of the Crew Simulator.
- **ModelMessagesUri:** URL for WFM modeling messages.
- **CrewServerHost:** Host name on which the Crew Server runs.
- **CrewServerPort:** TCP/IP port used by the Crew Server.
- **EnableCrewServerTcpip:** Specifies whether communication with the Crew Server via TCP is enabled. Set to False when the Crew Simulator web service client is enabled and HTTP message exchange with the Crew Server is supported.
- **EnableTcpipTls:** If set to true, the TLS protocol is enabled for the TCP/IP connection.
- **CertSubjectName:** The TLS certificate subject name's CN value.
- **LogPath:** Path to the folder where the crew simulator log is recorded.
- **CodedTranslationFile:** Path to the Coded Translation file.
- **CheckMessageSecurity:** If set to true, checking the security token for authenticity and expiration is enabled.
- **CheckMessageSequence:** If set to true, checking the sequence number of server messages is enabled.
- **UseCredentialFile:** If set to true, using the credential file for defining the private key and user SIDs is enabled.
- **AllowUnregisteredSids:** If set to true, any user SID is considered valid.
- **CredentialInfoFile:** Path to the Credential Info file that stores valid SIDs and the secret shared key.
- **TimeToLive:** The time period in seconds to sustain IDDS messages on the network.
- **ConnectionDebug:** Controls whether debug files are created for TCP connections. Debug files are created in the ".\Server\log" folder and start with "ConnectionDebug_".
- **FaultClearTimerDelay:** Time period in minutes for RoboCrew to wait before starting the automatic incident clearance process.
- **IsSyncCapable:** If set to true, this parameter enables synchronization queries for RoboCrew.
- **SendVehicleLocationsAsLatitudeLongitude:** If set to True, vehicle locations are specified with latitude and longitude. If set to False, vehicle locations are specified with X/Y coordinates.
- **CoordinatesConverterAssemblyName:** Name of the assembly (DLL) that contains a class that

converts between latitude/longitude and system X/Y coordinates.

- **CoordinatesConverterClassName:** Name of the class that converts between latitude/longitude and X/Y coordinates.

The coordinate converter options allow you to configure navigation from the network view to a URL with a certain latitude and longitude (for example, Google Maps).

- **DefaultWaitOnSiteTime:** Time period, in seconds, for the assigned MDT crew to wait on site before completing the incident. This time period can be used to un-assign a crew or send a damage assessment report.
- **AutomaticCrewUpdates:** If enabled, crew behavior is automatically simulated. If disabled, MDT crews do not change position and status; new MDT crew statuses need to be manually sent by the operator from the Crew Config tab of the Robo Crew.
- **CompressMessage:** If set to True, Crew Simulator messages are compressed before sending. This decreases message size and improves performance. This CompressMessage parameter must be set with the same value as the CompressMessage, OMS_SEND_COMPRESSED_MESSAGE, and OMS_COMPRESSED_MESSAGE_FOR_MANAGED_CONNECTION parameters in other server and client configuration files.
- **Plugin URI Parameters:** This group includes URI parameters that apply to all Crew Simulator web service plugins.
 - **UriScheme:** URI scheme, HTTP or HTTPS. Applies to all Crew Simulator web service plugins.
 - **UriHost:** URI host. Applies to all Crew Simulator web service plugins.
 - **UriPort:** URI port. Applies to all Crew Simulator web service plugins.

Note

Use different ports for different web services.

- **UriPath:** URI path. Applies to all Crew Simulator web service plugins.
- **FullMessages:** If set to true, entire message payloads are written to the log file.
- **Log Service Plugin Parameters:** This group includes parameters for the Crew Simulator logging service plugin.
 - **Enabled:** If set to true, the Crew Simulator logging service is enabled.
 - **Type:** The type that implements the Crew Simulator logging service plugin.
 - **Name:** Crew Simulator logging service plugin name.
 - **LogMessageTypes:**
 - **Type:** Application for which the log is created.
 - **SeparateLogToDisk:** If set to true, a separate log is created for this type.
 - **Color:** Color of log messages in the Crew Simulator log (for the list of available colors, refer to <https://docs.microsoft.com/en-us/dotnet/api/system.drawing.color?>

`view=netframework-4.8).`

- **LogToDisk:**
 - **Name:** Name of the log file.
 - **EnableAppendLog:** If set to true, a log appender is added.
 - **EnableDailyLog:** If set to true, a separate log file is created daily.
 - **MaxEntries:** Specifies the maximum number of entries in the log file.
- **Swagger Docs Plugin Parameters:** This group includes Swagger Docs plugin parameters.
 - **Enabled:** If set to true, the Swagger Docs plugin is enabled.
 - **Type:** The type that implements the Swagger Docs plugin.
 - **Name:** Swagger Docs plugin name.
- **Crew Simulator Switching Order Web Service Plugin Parameters:** This group includes parameters for the Workforce Management System (WFM) Switching Order web service plugin.
 - **Enabled:** If set to true, the plugin is enabled and HTTP message exchange is supported. If set to False, the legacy TCP message exchange is used.
 - **Type:** The type that implements the plugin.
 - **Name:** The plugin name.
 - **BasePath:** The plugin root URL.
 - **BaseUri:** The plugin root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
 - **Name:** The plugin alternate name.
 - **Version:** The plugin message version.
 - **Description:** The plugin description.
 - **DateTimeFormatString:** Date and time format. By default, the value is "O", which is a custom "round-trip" DateTime format specifier that is equal to "yyyy-MM-ddTHH:mm:ss.fffffffK".
 - **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
 - **PW:** The password value that is used to hash and place in the request authorization header. Use the Encryption tool for encrypting the password.
 - **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.
 - **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **Crew Simulator Crew Web Service Plugin Parameters:** This group includes parameters for the WFM Crew simulator web service plugin.
 - **Enabled:** If set to true, the plugin is enabled, and HTTP message exchange is supported. If

set to False, the legacy TCP message exchange is used.

- **Type:** The type that implements the plugin.
- **Name:** The plugin name.
- **BasePath:** The plugin root URL.
- **BaseUri:** The plugin root URL. If populated, BaseUri takes precedence over the URL constructed by the UriScheme, UriHost, UriPort, UriPath, BasePath, and Version parameters.
- **Name:** The plugin alternate name.
- **Version:** The plugin message version.
- **Description:** The plugin description.
- **DateTimeFormatString:** Date and time format. By default, the value is "O", which is a custom "round-trip" DateTime format specifier that is equal to "yyyy-MM-ddTHH:mm:ss.fffffffK".
- **UseHeaderAuthorization:** If set to true, "authorization" header checking is enabled for incoming web service calls.
- **PW:** The password value that is hashed and then placed in the request authorization header. Use the Encryption tool for encrypting the password.
- **JsonFormattingIgnoreNull:** If set to true, properties with null values in the class are ignored for the web service plugin.
- **UseUAAToken:** Specifies whether UAA authentication is enabled. Enables validating a UAA authentication token (for servers and web services) or enables retrieving a valid UAA authentication token (for clients).
- **UaaIntrospectEndPoint:** Endpoint for validating a UAA authentication token.
- **SecureWebService:** Configures whether a secure web service connection is used (the web service runs in HTTPS mode). The default value is false.

9. Event Scripter Configuration

When the Event Scripter is started from the shortcut, it applies the client configuration file `netconfes.xml`.

To change the configuration file:

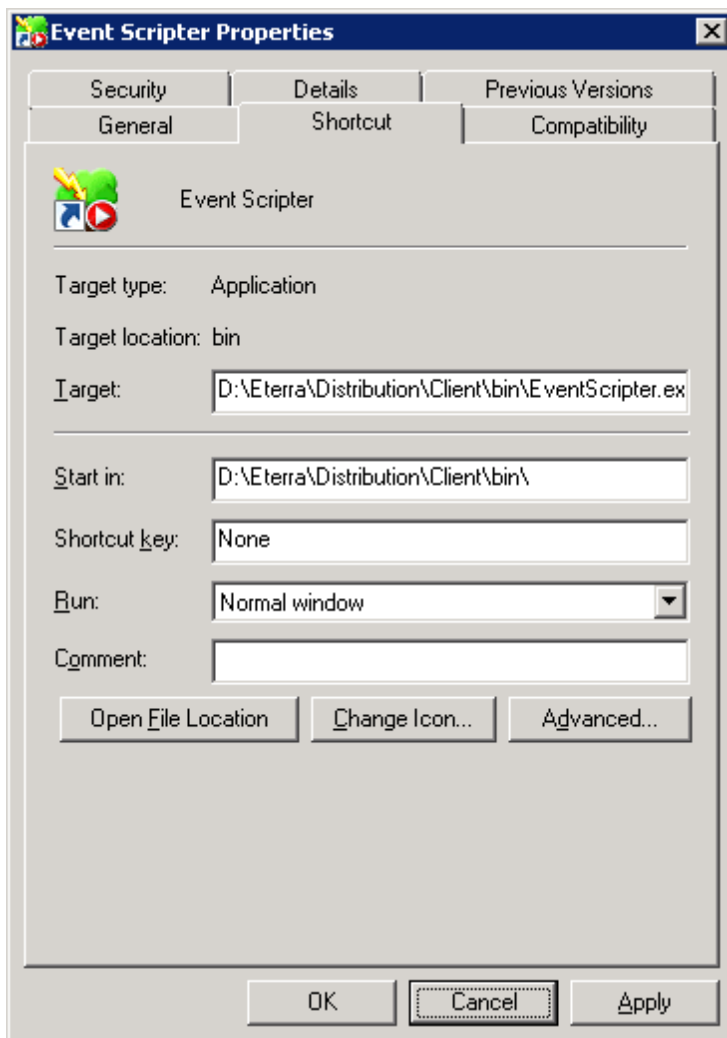
1. Locate the Event Scripter shortcut.
2. Right-click the shortcut and select the Properties option.

The Event Scripter Properties window appears.

3. On the Shortcut tab, modify the path to the configuration file in the Target field.

By the default, the Target field contains the following:

```
D:\Eterra\Distribution\Client\bin\EventScripter.exe /NETCONF  
"%IDMS_ROOT%\config\client\netconfes.xml"
```



4. Click OK.

Event Scriptor can run on a computer separate from the DOTS server in a client/server configuration. To achieve this configuration, in the `netconfes.xml` file, specify the server IP and port:

- **ServerIPs:** This is the IP address of the server(s) hosting the Simulator (or Real-Time) server.
- **Port:** This is the listening port of the Simulator (or Real-Time) server.

i Note

The `netconfes.xml` file can be modified using the Net Dynamics Configuration Editor tool. For more details, refer to [SIM Dynamics Instance Configuration](#).

In the lower pane of its user interface, the Event Scriptor indicates whether it is connected with the Simulator and Real-Time servers. If it is connected, the Event Scriptor indicates the host name and listening port of the servers.

10. SIM Dynamics Instance Configuration

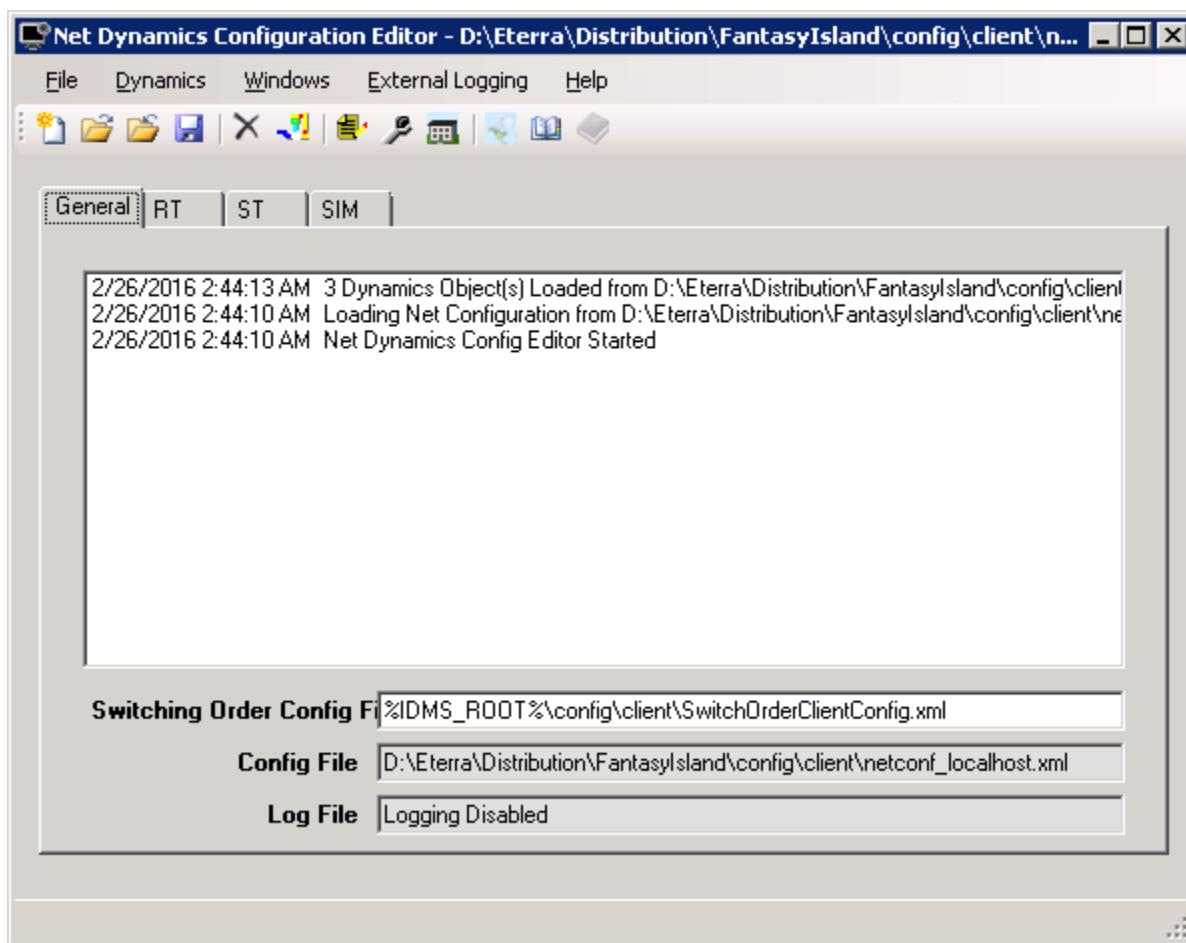
To modify the configuration of the SIM dynamic:

1. Using Microsoft Windows Explorer, locate the `Net DynamicsConfigEditor.exe` file in the `bin` directory of the local installation.
2. To start the Net Dynamics Configuration Editor application, double-click the `Net DynamicsConfigEditor.exe` file.

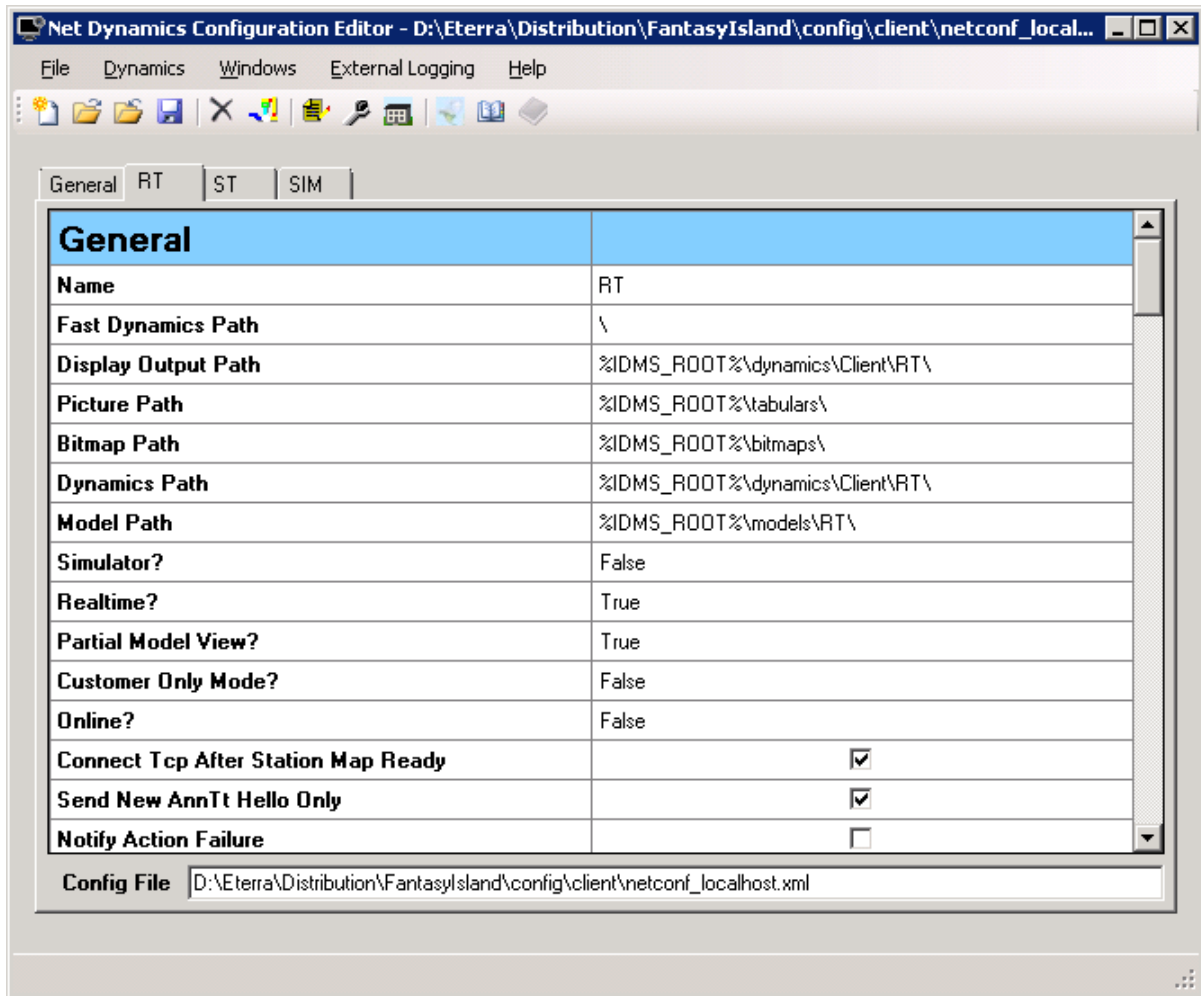
The Net Dynamics Configuration Editor application starts. A message appears in the General tab log area with the date, timestamp, and the messages "Input File Not Found" and "No Dynamics Loaded".

3. Select **File > Open**, and select the `netconf_localhost.xml` file from the list on the Open pop-up display, where `<localhost>` is a server name. The default path to the configuration file is:
`%IDMS_ROOT%\config\client\`

The network configuration is uploaded, and three new tabs named "RT"(real-time), "ST"(study), and "SIM"(training simulator) appear, as shown below:



4. Select the SIM tab.



The dynamics parameters associated with a specific configuration are shown on the SIM tab. Now you can modify the dynamics parameters, as described in the next step.

Note

All the path fields can be defined using absolute or relative paths. The product sample files use relative paths, so that the same configuration files can be used even if the installer chooses a non-default location.

5. Modify or enter the fields on the selected dynamics view, as follows:

The following settings relate to the general setup:

- **Name:** The Dynamics tab header name (for example, "SIM" for simulator mode). Enter this name, and save the configuration file for the changes to take effect.

A good practice is to keep the name short but descriptive because it is used in every URL call-up.

For example, to call up the simulator mode (SIM) detailed geographic display for the

Kirkland substation, the URL is as follows: NETVIEW://SIM/view/detail/Kirkland

- **Fast Dynamics Path:** Enter the full path for the folder location for the client, to store the fast dynamics files it receives from the server.
- **Display Output Path:** Enter the full path to the folder for tabular results output. The default path may look like the following: D:\Eterra\Distribution\Client\netview\dynamics\SIM\
- **Picture Path:** Enter the full path to the folder containing the picture files used for toolbars and tabular displays. This can be an absolute path or a relative path, but it must end with a backslash.
- **Bitmap Path:** Enter the full path to the folder containing the bitmap files. This can be an absolute path or a relative path, but it must end with a backslash.
- **Dynamics Path:** Specifies the folder where dynamics files should be stored after receiving them from the server. This can be an absolute path or a relative path, but it must end with a backslash. For example: D:\Eterra\Distribution\Client\netview\dynamics\SIM\
- **Model Path:** Sets the path to the station model files. This can be an absolute path or a relative path, but it must end with a backslash. For example: D:\Eterra\Distribution\Client\netview\models\RT\
- **Simulator?:** For an SIM client configuration, this field must be set to True. Set to False for ST and RT modes.
- **Real-time?:** Set to True for a real-time client configuration. In real-time mode, the use of Scada energization information is enabled.

Set to False to set to study mode where pseudo-energy sources are programmatically inserted into the model.

For a SIM client configuration, set the "Real-time?" flag to False.

- **Partial Model View?:** Set to True to enable the partial model view feature.

For distribution systems with more than 15 substations, the performance impacts to the client are such that it is better to only look at portions of the network at one time. By default, the partial model view (PMV) loads the requested station and two levels of electrical adjacency. This way, only a handful of stations are loaded into memory, which greatly improves UI performance. The default value is True.

Note

Setting PMV to False is not supported and not recommended.

- **Customer Only Mode?:** If set to True, the Client runs in the Customer Only mode.
- **Online?:** Set to True to allow connections to server(s). Set to False to run as an isolated client.
- **Allow to Load Fatal Model:** If checked, allow loading the station with fatal error. This feature should be disabled by default.
- **Temp. Ground Change Event Enable:** This is used for development purposes only. Leave

unchecked.

- **Dynamic Query State:** This is for internal use only. Leave unchecked.
- **Transmission Adjacent Enable:** If checked, transmission lines are taken into account in the adjacency algorithm.
- **Distribution Adjacent Enable:** If checked, distribution lines are taken into account in the electrical adjacency algorithm.
- **Resync on Connect:** If checked, the client cleans its dynamics and downloads a completely fresh copy from the server.

i Note

Checking the Resync on Connect box significantly impacts performance. It is recommended that you leave this unchecked.

- **Value Placement Query Enable:** Enables/disables querying value placement data from the server.
- **Potential Fault Locations Query Enable:** If checked, fast dynamics querying is enabled, so the client downloads the fault location dynamic results.
- **Measurement Quality Query Enable:** If checked, fast dynamics querying is enabled, so the client downloads the measurement quality information.

By default, this should be enabled.

- **DNAF Input Query Enable:** If checked, the clients can modify values in the DNAF manual entry pop-ups.

By default, this should be enabled.

- **Measurement Query Enable:** By default, this should be enabled.
- **Trouble Ticket Query Enable:** This feature is not enabled. Leave unchecked.
- **Power Flow Query Enable:** If checked, enables power flow violation data, as calculated on the server, to be passed to the client and presented as halos on the geographic display.

By default, this should be enabled.

The server must also have the corresponding parameter FAST_DYNAMIC_ENABLED set in its configuration file.

- **Use Dynamics:** This parameter is for internal use only. Always set it to True (checked).
- **Is Admin or DB Builder:** If checked, allows users with Administrator credentials to modify fixed annotations.
- **Read-Only Color:** This is only useful for an environment where permissions are being used. It can be set to a color different from the background color, to indicate that the current operator does not have permission to control devices within that colored region. Shades of gray are good to use for this feature.

GE does not recommend using a color that is being used as one of the feeder colors.

- **Pop-Up Background Color:** Defines the background color used behind pop-ups and grid tabular displays. This allows pop-ups to be configured to match the background color of a read-only or study mode background.

By default, the color is gray for RT and SIM modes, and yellow for ST mode.

- **Background Color:** Defines the main background color used behind the electrical lines and components drawn by the software. You can change the color by clicking on the color square.

Black is the default and most optimal choice for real-time mode. The default color is gray for ST mode, and purple for SIM mode.

The following settings relate to the grid tabular setup:

- **Grid Tabular Grid Line Style:** Defines the grid tabular border style.
- **Grid Tabular Grid Lines Color:** Defines the color for grid lines.
- **Grid Tabular Even Rows Color:** Defines the color for even rows. If this parameter is not defined in the configuration file, the grid tabular display (.xml) settings are used by default.
- **Grid Tabular Odd Rows Color:** Defines the color for odd rows. If this parameter is not defined in the configuration file, the grid tabular display (.xml) settings are used by default.
- **Grid Tabular Header Cells Color:** Defines the color for header cells. If this parameter is not defined in the configuration file, the grid tabular display (.xml) settings are used by default.
- **Grid Tabular Title Text Color:** Defines the display's title color (black by default).
- **Grid Tabular Title Color End:** Sets the end color for the display's title background gradient. The title gradient is painted left-to-right using the specified colors. If this parameter is not defined in the configuration file, the background is blue by default.
- **Grid Tabular Title Color Start:** Sets the start color for the display's title background gradient. The title gradient is painted left-to-right using the specified colors. If this parameter is not defined in the configuration file, the background is blue by default.

The following settings relate to study case setup:

- **Savecase Model File Limit:** Specified as an integer, this limits the number of model files allowed to be included in a single savecase.
- **Study Server Model Path:** To allow for the automated features of the CopyFromRealTime and savecase capabilities, the client needs to know about the study server's model path. Set the same as the folder defined in the study server's configuration file.
- **Study Server Dynamics Path:** To allow for the automated features of the CopyFromRealTime and savecase capabilities, the client needs to know about the study server's dynamics path. Set the same as the folder defined in the study server's configuration file.
- **Savecase Path:** Set to the folder where savecase information should be stored.

- **Copy from Real-Time Temp Path:** Set to the folder where temporary data should be stored.
- **Study Case Setup Station Adjacency:** Specified as an integer, this number defines the level of adjacency when the stations are copied from the real-time server. To enable the retrieval of all station model files, set this parameter to a number greater than the total number of model files. For example, if there are 52 stations, set this value to 60. This will retrieve all of the model files in the folder regardless of geographical or electrical adjacency and account for future model additions.

The following settings relate to the Outage Management System(OMS) case setup:

- **OMS Coded Translation Config File:** Sets the path to the coded translation configuration file.
- **OMS Client Config Name:** Sets the full path to the OMS client configuration file.
- **OMS Customer File Name:** Sets the full path to the OMS customer file.
- **OMS Timer Period:** Enter the time in seconds for the client to send a periodic query to the server (for example, every five seconds).
- **OMS Online?:** If set to True, only the client starts sending a query to the server.
- **OMS Customer Cache Path:** Path to the location where the mapping of customer data to the station model files is cached. These are transient files that allow for faster call-up of the client.
- **Client Side Dynamic Whole Saving Frequency:** Number of times the dynamics content has been changed. Whenever the dynamics content changes this value of times, a clear copy of the whole dynamics is saved and appended at the end of the dynamics.sav file.
- **Safety RUSER Command (%X %Y %ID Proxies):** Enter the RUSER command to move the safety document on the geographic network view.
- **Message Version:** This should always be set to 0.
- **Cache Size Limit:** Enter the maximum size for cached data in megabytes.
- **Viewer Configuration Name:** Use this combo box to select from the list of defined viewer configurations. Click the Viewer Config Name Editor button on the toolbar to switch to the viewer configuration mode and define viewer configurations for use with this combo box. For more information, refer to the *NetDynamics Configuration Editor User Guide*.
- **WebMap Name:** WebMap provider name.
- **Background Name:** Use this combo box to select from the list of defined background configurations. Click the Backgrounds Editor button on the toolbar to switch to the background view and define background configurations for use with this combo box. For more information, refer to the *NetDynamics Configuration Editor User Guide*.
- **Port:** The TCP/IP port of the ADMS server(s). Set this the same as the listening port on the desired server.
- **Server IPs:** Set to the host(s) name(s) that the client is connecting to for this particular set of dynamics data. Multiple hosts should be separated with a comma. When connecting to a

server running on the same machine (for example, the default local study server), set this field to "localhost".

- **Model Extension:** Defines the Microsoft Windows file extension to associate with model files. The default is `.mod`.